# 第八章 多元函数积分学

## §1. 重积分的概念及其性质

### 习题 P78.

1. 设平面闭区域 $D=\{(x,y)\mid x^2+y^2\le r^2\}$，求 $\iint_D\sqrt{r^2-x^2-y^2}\,d\sigma$.




解：$z=f(x,y)=\sqrt{r^2-x^2-y^2}$ 表示半径为 $r$ 的半球面

在 $D$ 的边界上：$x^2+y^2=r^2$ 即 $z=0$. 所求重积分即为半球体积.

$\therefore \iint_D\sqrt{r^2-x^2-y^2}\,dxdy=\frac{2}{3}\pi r^3$.

（二重积分的几何意义：曲顶柱体的体积. 所求曲顶柱体为半球体.）

$$\iint_D\sqrt{r^2-x^2-y^2}\,dxdy=\iint_{D'}\sqrt{r^2-\rho^2}\cdot\rho\,d\rho d\theta=\int_0^{2\pi}d\theta\cdot\int_0^r\sqrt{r^2-\rho^2}\cdot\frac{1}{2}d\rho^2=\frac{2}{3}\pi r^3$$

2. 设有平面区域 $D_1,D_2$，$D_1\supset D_2$，$f$ 是 $D_1$ 上的非负连续函数，证明：

$$\iint_{D_1}f(x,y)\,d\sigma\ge\iint_{D_2}f(x,y)\,d\sigma.$$

证明：$\because D_1\supset D_2$. $\therefore D_1=D_2\cup D_2$. $D_2$ 为 $D_1$ 中挖去 $D_2$ 后所余的区域.

由重积分的可加性：$\iint_{D_1}f(x,y)\,d\sigma=\iint_{D_2}f(x,y)\,d\sigma+\iint_{D_2}f(x,y)\,d\sigma$

$\because f$ 在 $D_1$ 上非负连续，$\therefore f$ 在 $D_2$ 上非负连续. 即 $\iint_{D_2}f(x,y)\,d\sigma\ge0$.

$\therefore \iint_{D_1}f(x,y)\,d\sigma\ge\iint_{D_2}f(x,y)\,d\sigma$.

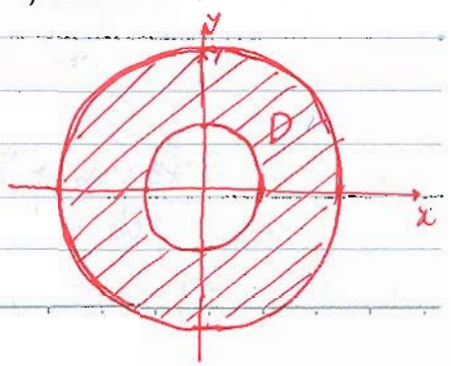
3. 设 $D$ 为一空间区域，比较重积分 $\iiint_D\sin^2(x+2y+3z)\,d\sigma$ 和 $\iiint_D(x+2y+3z)^2\,d\sigma$.

解：当 $\theta\ge0$ 时恒有 $\sin\theta\le\theta$. 令 $\theta=(x+2y+3z)^2$. 则 $\sin^2(x+2y+3z)\le(x+2y+3z)^2$.

由重积分的保序性：$\iiint_D\sin^2(x+2y+3z)\,d\sigma\le\iiint_D(x+2y+3z)^2\,d\sigma$

4. 利用重积分的性质，估计积分值：

1) $\iint_D-\sin(x^2+y^2)\,d\sigma$，其中 $D=\{(x,y)\mid\frac{\pi}{4}\le x^2+y^2\le\frac{5\pi}{4}\}$.

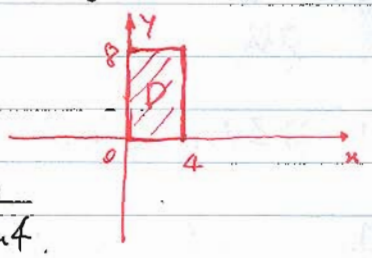



解：在 $D$ 上，$\frac{\pi}{4}\le x^2+y^2\le\frac{5\pi}{4}$. 有：$\frac{\sqrt{2}}{2}\le\sin(x^2+y^2)\le1$.

$$\iint_D d\sigma = S(D) = \pi\cdot\frac{5\pi}{4} - \pi\cdot\frac{\pi}{4} = \frac{\pi^2}{2}.$$

$$\therefore \quad \frac{\sqrt{2}}{2}\cdot S(D) \le \iint_D \sin(x^2+y^2)\,d\sigma \le 1\cdot S(D). \qquad \text{即: } \frac{\sqrt{2}\pi^2}{4} \le \iint_D \sin(x^2+y^2)\,d\sigma \le \frac{\pi^2}{2}.$$

(2) $\iint_D \frac{dxdy}{\ln(4+x+y)}$, 其中 $D=\{(x,y)\mid 0\le x\le 4,\ 0\le y\le 8\}$

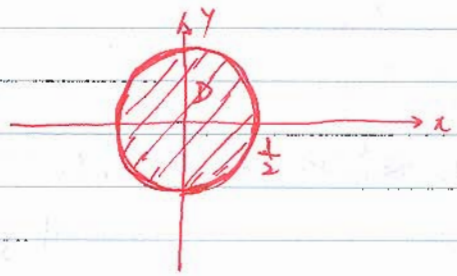



解: 在D上: $4\le 4+x+y\le 16 \Rightarrow \frac{1}{\ln 16} \le \frac{1}{\ln(4+x+y)} \le \frac{1}{\ln 4}$.

即: $\frac{1}{4\ln 2}\cdot S(D) \le \iint_D \frac{dxdy}{\ln(4+x+y)} \le \frac{1}{2\ln 2}\cdot S(D)$. $\quad S(D)=\iint_D dxdy = 32$.

$$\therefore \quad \frac{8}{\ln 2} \le \iint_D \frac{dxdy}{\ln(4+x+y)} \le \frac{16}{\ln 2}.$$

(3) $\iint_D e^{x^2+y^2}\,d\sigma$. 其中 $D=\{(x,y)\mid x^2+y^2\le \frac{1}{4}\}$.




解: 在D上: $1 \le e^{x^2+y^2} \le e^{\frac{1}{4}}$. $\quad S(D)=\iint_D d\sigma = \frac{\pi}{4}$

$$\therefore \quad \frac{\pi}{4} \le \iint_D e^{x^2+y^2}\,d\sigma \le \frac{\pi}{4}e^{\frac{1}{4}}.$$

5. 设f是三元连续函数, 试求极限: $\lim_{r\to 0}\frac{1}{r^3}\iiint_{\Omega_r} f(x,y,z)\,d\sigma$.

其中 $\Omega_r=\{(x,y,z)\mid (x-a)^2+(y-b)^2+(z-c)^2\le r^2\}$

解: 由积分中值定理, 存在 $(\xi,\eta,\zeta)\in\Omega_r$, 使得:

$$\iiint_{\Omega_r} f(x,y,z)\,d\sigma = f(\xi,\eta,\zeta)\iiint_{\Omega_r} d\sigma = f(\xi,\eta,\zeta)\cdot\frac{4}{3}\pi r^3.$$

$$\lim_{r\to 0}\frac{1}{r^3}\iiint_{\Omega_r} f(x,y,z)\,d\sigma = \lim_{r\to 0}\frac{1}{r^3} f(\xi,\eta,\zeta)\cdot\frac{4}{3}\pi r^3 = \frac{4}{3}\pi\lim_{r\to 0} f(\xi,\eta,\zeta).$$

∵ f是三元连续函数, ∴ 当 $r\to 0$ 时, $\Omega_r\to 0$, $f(\xi,\eta,\zeta)\to f(a,b,c)$.

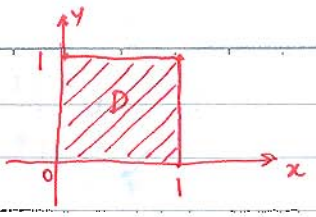
$$\therefore \quad \lim_{r\to 0}\frac{1}{r^3}\iiint_{\Omega_r} f(x,y,z)\,d\sigma = \frac{4}{3}\pi f(a,b,c).$$

## §2. 二重积分的计算

P.87 习题

1. 计算二重积分:

(1) $\iint_D xy e^{xy^2} d\sigma$，其中 $D=\{(x,y) \mid 0\le x\le 1,\ 0\le y\le 1\}$.

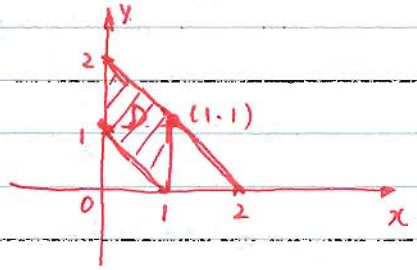



解：$\iint_D xy e^{xy^2} = \int_0^1 dx \int_0^1 xy e^{xy^2} dy$

$= \int_0^1 dx \int_0^1 e^{xy^2} \cdot \frac{1}{2} d(xy^2) = \frac{1}{2}\int_0^1 dx \cdot e^{xy^2}\Big|_0^1$

$= \frac{1}{2}\int_0^1 (e^x - 1)\,dx = \frac{1}{2}(e^x - x)\Big|_0^1 = \frac{e}{2} - 1.$

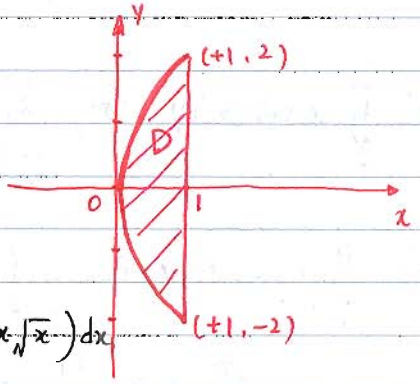
(2) $\iint_D \frac{1}{x+y} d\sigma$，其中 $D=\{(x,y) \mid 0\le x\le 1,\ 1\le x+y\le 2\}$.




解：$\iint_D \frac{1}{x+y} d\sigma = \int_0^1 dx \int_{1-x}^{2-x} \frac{1}{x+y} dy$

$= \int_0^1 dx \cdot \ln(x+y)\Big|_{1-x}^{2-x} = \int_0^1 \ln 2\, dx = \ln 2.$

(3) $\iint_D xy^2 d\sigma$，其中 $D=\{(x,y) \mid 4x\ge y^2,\ x\le 1\}$.

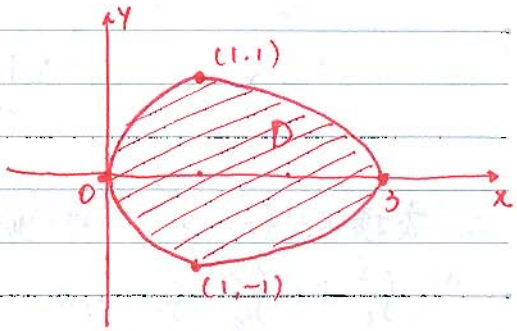



解：$\iint_D xy^2 d\sigma = \int_0^1 dx \int_{-2\sqrt{x}}^{2\sqrt{x}} xy^2 dy$

$= \int_0^1 dx \cdot \frac{1}{3}xy^3\Big|_{-2\sqrt{x}}^{2\sqrt{x}} = \int_0^1 \frac{1}{3}x(8x\sqrt{x} + 8x\sqrt{x})\,dx$

$= \frac{16}{3}\int_0^1 x^2\sqrt{x}\,dx = \frac{16}{3}\int_0^1 x^{\frac{5}{2}} dx = \frac{16}{3}\cdot\frac{2}{7}x^{\frac{7}{2}}\Big|_0^1 = \frac{32}{21}.$

(4) $\iint_D (y^2 - y)\,d\sigma$，其中 D 由 $x = y^2$ 与 $x = 3 - 2y^2$ 围成.




解：$\iint_D (y^2 - y)\,d\sigma = \int_{-1}^1 dy \int_{y^2}^{3-2y^2} (y^2 - y)\,dx$

$= \int_{-1}^1 (y^2 - y)\,dy \cdot \int_{y^2}^{3-2y^2} dx = \int_{-1}^1 (y^2 - y)\cdot(3 - 3y^2)\,dy$

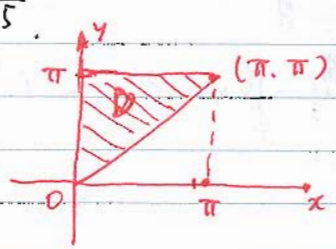
$= 3\int_{-1}^1 (-y^4 + y^3 + y^2 - y)\,dy = 3\cdot\left(-\frac{1}{5}y^5 + \frac{1}{4}y^4 + \frac{1}{3}y^3 - \frac{1}{2}y^2\right)\Big|_{-1}^1 = 3\cdot\left(-\frac{2}{5} + \frac{2}{3}\right) = \frac{4}{5}.$

$\iint_D (y^2 - y)\,d\sigma = \int_0^1 dx \int_{-\sqrt{x}}^{\sqrt{x}} (y^2 - y)\,dy + \int_1^3 dx \int_{-\sqrt{\frac{3-x}{2}}}^{\sqrt{\frac{3-x}{2}}} (y^2 - y)\,dy$

$= \int_0^1 dx \cdot \left(\frac{1}{3}y^3 - \frac{1}{2}y^2\right)\Big|_{-\sqrt{x}}^{\sqrt{x}} + \int_1^3 dx \cdot \left(\frac{1}{3}y^3 - \frac{1}{2}y^2\right)\Big|_{-\sqrt{\frac{3-x}{2}}}^{\sqrt{\frac{3-x}{2}}} = \int_0^1 \frac{2}{3}x\sqrt{x}\,dx + \int_1^3 \frac{2}{3}\cdot\frac{3-x}{2}\sqrt{\frac{3-x}{2}}\,dx$

$$= \frac{2}{3}\cdot\frac{2}{5}x^{\frac{5}{2}}\Big|_0^1 - 2\cdot\frac{2}{3}\cdot\frac{2}{5}\left(\frac{3-x}{2}\right)^{\frac{5}{2}}\Big|_1^3 = \frac{4}{15}+\frac{8}{15} = \frac{4}{5}.$$

(5) $\iint_D \cos(x+y)\,d\sigma$，其中 $D=\{(x,y)\mid x\geq 0,\ x\leq y\leq \pi\}$.




解：$\iint_D \cos(x+y)\,d\sigma = \int_0^\pi dx\int_x^\pi \cos(x+y)\,dy$

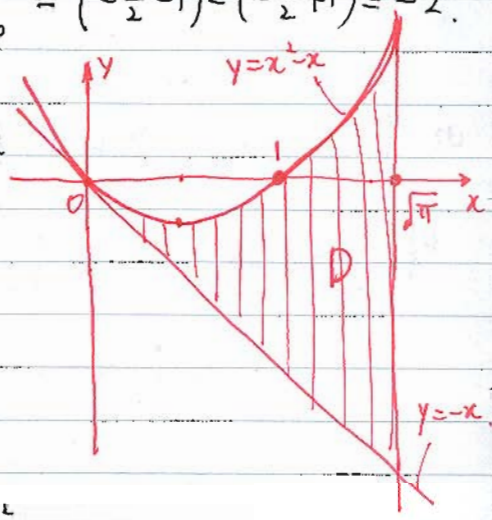
$= \int_0^\pi dx\cdot\left[+\sin(x+y)\right]\Big|_{y=x}^{y=\pi} = \int_0^\pi(-\sin x-\sin 2x)\,dx$

$= \left[\cos x+\frac{1}{2}\cos 2x\right]\Big|_0^\pi = \left(-1+\frac{1}{2}\right)-\left(1+\frac{1}{2}\right) = -2.$

$\iint_D \cos(x+y)\,d\sigma = \int_0^\pi dy\cdot\int_0^y \cos(x+y)\,dx = \int_0^\pi dy\cdot\left[\sin(x+y)\right]\Big|_{x=0}^{x=y}$

$= \int_0^\pi(\sin 2y-\sin y)\,dy = \left(-\frac{1}{2}\cos 2y+\cos y\right)\Big|_0^\pi = \left(-\frac{1}{2}-1\right)-\left(-\frac{1}{2}+1\right) = -2.$

(6) $\iint_D x\sin(x+y)\,d\sigma$，其中 D 由直线 $x=\sqrt{\pi}$，抛物线 $y=x^2-x$ 及点 (0,0) 处的切线围成.




解：$\iint_D x\sin(x+y)\,d\sigma = \int_0^{\sqrt{\pi}} dx\int_{-x}^{x^2-x} x\sin(x+y)\,dy$

$= \int_0^{\sqrt{\pi}} x\,dx\cdot\left[-\cos(x+y)\right]\Big|_{y=-x}^{y=x^2-x}$

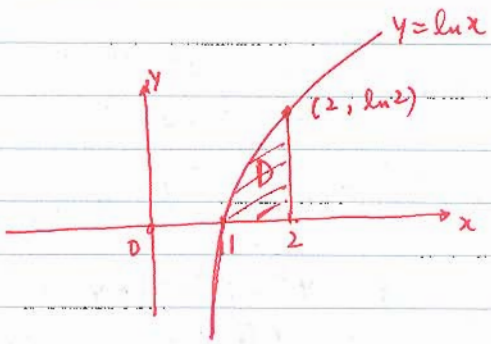
$= \int_0^{\sqrt{\pi}} x\cdot(-\cos x^2+1)\,dx = \frac{1}{2}\int_0^{\sqrt{\pi}}(1-\cos x^2)\,dx^2$

$= \left(\frac{1}{2}x^2-\frac{1}{2}\sin x^2\right)\Big|_0^{\sqrt{\pi}} = \frac{\pi}{2}.$

2. 交换下列二次积分的积分顺序：

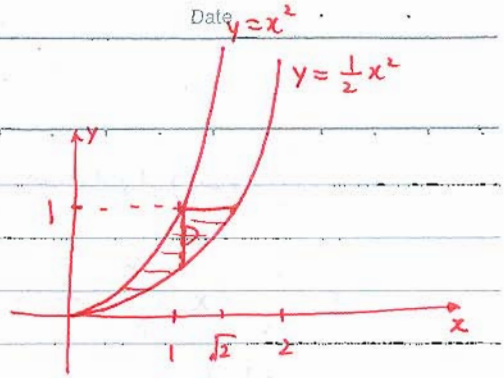
(1) $\int_1^2 dx\int_0^{\ln x} f(x,y)\,dy$

$= \int_0^{\ln 2} dy\int_{e^y}^2 f(x,y)\,dx$



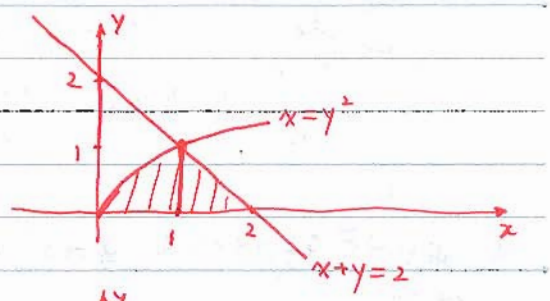
(2) $\int_0^1 dy \int_{\sqrt{y}}^{\sqrt{2y}} f(x-y)\,dx$.

$= \int_0^1 dx \int_{\frac{1}{2}x^2}^{x^2} f(x,y)\,dy + \int_1^{\sqrt{2}} dx \int_{\frac{1}{2}x^2}^{1} f(x-y)\,dy$.



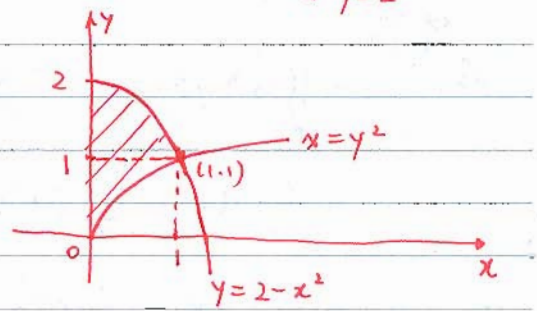

(3) $\int_0^1 dy \int_{y^2}^{2-y} f(x-y)\,dx$.

$= \int_0^1 dx \int_0^{\sqrt{x}} f(x-y)\,dy + \int_1^2 dx \int_0^{2-x} f(x-y)\,dy$.




(4) $\int_0^1 dy \int_0^{y^2} f(x-y)\,dx + \int_1^2 dy \int_0^{\sqrt{2-y}} f(x-y)\,dx$

$= \int_0^1 dx \int_{\sqrt{x}}^{2-x^2} f(x-y)\,dy$.



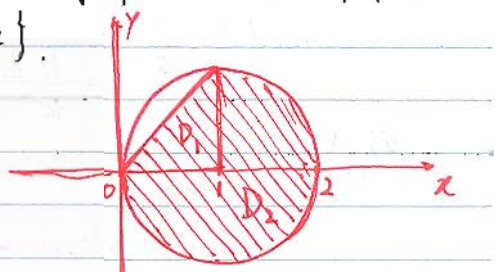

3. 求由曲面 $z=x-y$, $z=0$ 及圆柱面 $x^2+y^2=2x$ 在 $z\geq 0$ 中所围成的空间体的体积.

解: (I). $f(x,y)=x-y$. $D=\{(x,y)\mid x^2+y^2\leq 2x,\ y\leq x\}$.

$V=\iint_D f(x,y)\,d\sigma = \iint_D (x-y)\,d\sigma$

$= \int_0^1 dx \int_{-\sqrt{2x-x^2}}^{x} (x-y)\,dy + \int_1^2 dx \int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} (x-y)\,dy$

$= \int_0^1 dx \cdot (xy-\frac{1}{2}y^2)\Big|_{y=-\sqrt{2x-x^2}}^{y=x} + \int_1^2 dx \cdot (xy-\frac{1}{2}y^2)\Big|_{y=-\sqrt{2x-x^2}}^{y=\sqrt{2x-x^2}}$




$= \int_0^1 (x+x\sqrt{2x-x^2})\,dx + \int_1^2 2x\sqrt{2x-x^2}\,dx$. $\left(\text{令 } 1-x=\sin\theta \text{ 则 } x=1-\sin\theta\right)$

$= \int_{\frac{\pi}{2}}^0 (1-\sin\theta)\cdot\cos\theta\cdot(-\cos\theta)\,d\theta + \frac{1}{2}x^2\Big|_0^1 + 2\int_0^{-\frac{\pi}{2}} (1-\sin\theta)\cdot\cos\theta\cdot(-\cos\theta)\,d\theta$

$= \frac{1}{2} + \int_0^{\frac{\pi}{2}} (\cos^2\theta - \sin\theta\cos^2\theta)\,d\theta + 2\int_0^{-\frac{\pi}{2}} (\sin\theta\cdot\cos^2\theta - \cos^2\theta)\,d\theta$

$= \frac{1}{2} + \left(\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta + \frac{1}{3}\cos^3\theta\right)\Big|_0^{\frac{\pi}{2}} + 2\left(-\frac{1}{3}\cos^3\theta - \frac{1}{2}\theta - \frac{1}{4}\sin 2\theta\right)\Big|_0^{-\frac{\pi}{2}}$

$= \frac{1}{2} + \frac{\pi}{4} - \frac{1}{3} + \frac{\pi}{2} + \frac{2}{3} = \frac{5}{6} + \frac{3\pi}{4}$.

(II). $V = \iint_D (x-y)\,d\sigma = \iint_{D_1} (x-y)\,d\sigma + \iint_{D_2} (x-y)\,d\sigma$.

在 $D_2$ 中令: $\begin{cases} x-1=r\cos\theta \\ y=r\sin\theta \end{cases}$

$d\sigma = r\,dr\,d\theta$.

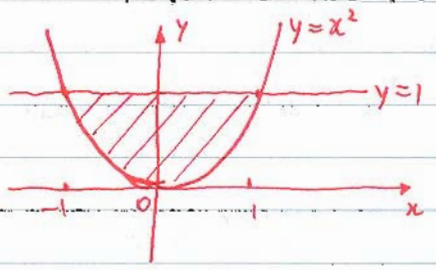
$$V=\iint_{D_1}(x-y)\,dxdy+\iint_{D_2}(x-y)\,d\sigma$$

$$=\int_0^1 dx\int_0^x (x-y)\,dy+\int_0^1 dr\int_{-\pi}^{\frac{\pi}{2}}(1+r\cos\theta-r\sin\theta)\,r\,d\theta$$

$$=\int_0^1 dx\cdot\left(xy-\frac{1}{2}y^2\right)\Big|_{y=0}^{y=x}+\int_0^1 r\,dr\cdot(\theta+r\sin\theta+r\cos\theta)\Big|_{-\pi}^{\frac{\pi}{2}}$$

$$=\int_0^1\frac{1}{2}x^2\,dx+\int_0^1\left(\frac{3}{2}\pi r+2r^2\right)dr=\frac{1}{6}+\left(\frac{3}{4}\pi r^2+\frac{2}{3}r^3\right)\Big|_0^1=\frac{5}{6}+\frac{3\pi}{4}.$$

4. 求由旋转抛物面 $z=x^2+y^2$，柱面 $y=x^2$ 及平面 $y=1$ 和 $z=0$ 所围成的空间立域的体积.




解：$V=\iint_D(x^2+y^2)\,d\sigma$，$D=\{(x,y)\mid x^2\le y\le 1\}$.

$$=\int_0^1 dy\int_{-\sqrt{y}}^{\sqrt{y}}(x^2+y^2)\,dx$$

$$=\int_0^1 dy\cdot\left(\frac{1}{3}x^3+y^2x\right)\Big|_{-\sqrt{y}}^{\sqrt{y}}=\int_0^1\left(\frac{2}{3}y^{\frac{3}{2}}+2y^{\frac{5}{2}}\right)dy$$

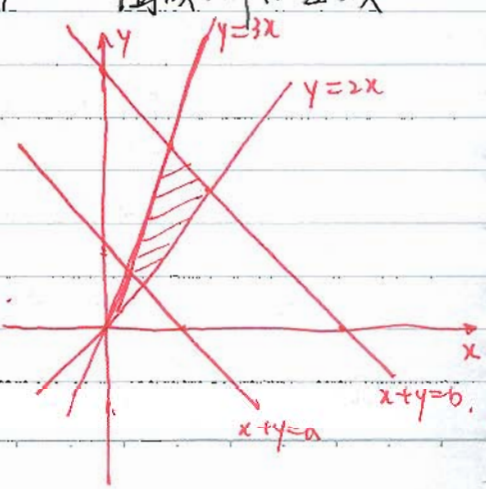
$$=\left(\frac{2}{3}\cdot\frac{2}{5}y^{\frac{5}{2}}+2\cdot\frac{2}{7}y^{\frac{7}{2}}\right)\Big|_0^1=\frac{88}{105}.$$

或：$V=\int_{-1}^1 dx\int_{x^2}^1(x^2+y^2)\,dy=\int_{-1}^1 dx\cdot\left(x^2y+\frac{1}{3}y^3\right)\Big|_{y=x^2}^{y=1}$

$$=\int_{-1}^1\left(x^2+\frac{1}{3}-x^4-\frac{1}{3}x^6\right)dx=\left(\frac{1}{3}x^3+\frac{1}{3}x-\frac{1}{5}x^5-\frac{1}{21}x^7\right)\Big|_{-1}^1$$

$$=2\left(\frac{1}{3}+\frac{1}{3}-\frac{1}{5}-\frac{1}{21}\right)=2\cdot\frac{70-21-5}{105}=\frac{88}{105}.$$

5. 作适当的变量代换，求由 $x+y=a$，$x+y=b$，$y=2x$，$y=3x$ 围成的平面立域的面积，其中 $b>a>0$.




解：作变换 $\begin{cases}u=x+y\\ v=\frac{y}{x}\end{cases}$ 则 $J=\begin{bmatrix}x'_u & x'_v\\ y'_u & y'_v\end{bmatrix}=\begin{bmatrix}u'_x & u'_y\\ v'_x & v'_y\end{bmatrix}^{-1}$.

$$\therefore J=\begin{bmatrix}1 & 1\\ -\frac{y}{x^2} & \frac{1}{x}\end{bmatrix}^{-1}=\frac{1}{\frac{x+y}{x^2}}\begin{bmatrix}\frac{1}{x} & -1\\ \frac{y}{x^2} & 1\end{bmatrix}=\frac{x^2}{x+y}\begin{bmatrix}\frac{1}{x} & -1\\ \frac{y}{x^2} & 1\end{bmatrix}.$$

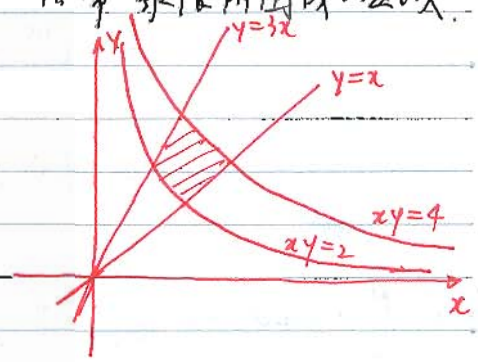
$$|J|=\frac{x^4}{(x+y)^2}\cdot\left(\frac{1}{x}+\frac{y}{x^2}\right)=\frac{x^2}{x+y}=\frac{u}{(v+1)^2}.$$

$$\therefore S(D)=\iint_D dx\,dy=\iint_{D'}|J|\,du\,dv=\iint_{D'}\frac{u}{(v+1)^2}\,du\,dv$$

$$=\int_a^b du\int_2^3\frac{u}{(v+1)^2}dv=\int_a^b u\,du\cdot\int_2^3\frac{1}{(v+1)^2}dv$$

$$=\frac{1}{2}u^2\Big|_a^b\cdot\left(-\frac{1}{v+1}\right)\Big|_2^3=\frac{1}{2}(b^2-a^2)\cdot\left(-\frac{1}{4}+\frac{1}{3}\right)=\frac{1}{24}(b^2-a^2).$$

6. 计算 $\iint_D x^2y^2\,d\sigma$，其中 D 是由 $xy=2$，$xy=4$，$y=x$，$y=3x$ 在第一象限所围成的区域.




解：作变换 $\begin{cases}u=xy\\ v=\dfrac{y}{x}\end{cases}$，则 $J=\begin{bmatrix}x'_u & x'_v\\ y'_u & y'_v\end{bmatrix}=\begin{bmatrix}u'_x & u'_y\\ v'_x & v'_y\end{bmatrix}^{-1}$

$$\therefore J=\begin{bmatrix}y & x\\ -\frac{y}{x^2} & \frac{1}{x}\end{bmatrix}^{-1}=\frac{x}{2y}\begin{bmatrix}\frac{1}{x} & -x\\ \frac{y}{x^2} & y\end{bmatrix}$$

$$|J|=\left(\frac{x}{2y}\right)^2\cdot\left(\frac{y}{x}+\frac{y}{x}\right)=\frac{x}{2y}=\frac{1}{2v}.$$

$$\therefore S(D)=\iint_D dx\,dy=\iint_{D'}|J|\,du\,dv=\iint_{D'}\frac{1}{2v}\,du\,dv$$

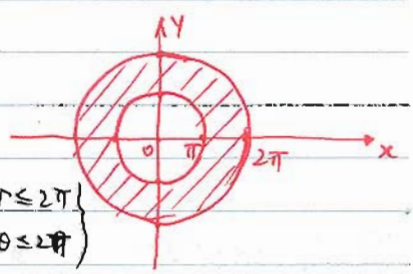
$$=\int_2^4 du\int_1^3\frac{1}{2v}dv=$$

$$\iint_D x^2y^2\,d\sigma=\iint_{D'}u^2|J|\,du\,dv=\iint_{D'}u^2\cdot\frac{1}{2v}\,du\,dv=\int_2^4 u^2du\cdot\int_1^3\frac{1}{2v}dv$$

$$=\frac{1}{3}u^3\Big|_2^4\cdot\frac{1}{2}\ln v\Big|_1^3=\frac{1}{3}(64-8)\cdot\frac{1}{2}(\ln 3)=\frac{28}{3}\ln 3.$$

7. 在极坐标系下计算下列二重积分：

(1) $\iint_D \sin\sqrt{x^2+y^2}\,d\sigma$，其中 $D=\{(x,y)\mid \pi^2\le x^2+y^2\le 4\pi^2\}$.

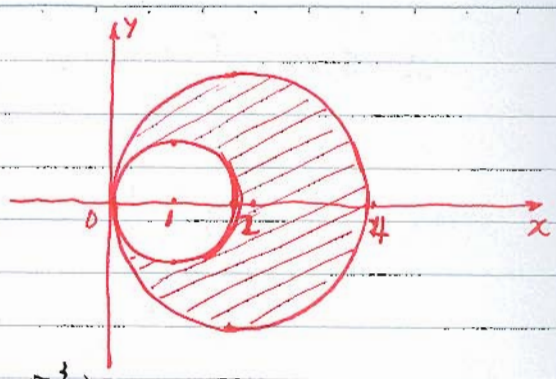



解：令 $\begin{cases}x=r\cos\theta\\ y=r\sin\theta\end{cases}$，则 $d\sigma=dx\,dy=r\,dr\,d\theta$. $D'=\left\{(r,\theta)\,\middle|\,\begin{matrix}\pi\le r\le 2\pi\\ 0\le\theta\le 2\pi\end{matrix}\right\}$

$$\iint_D\sin\sqrt{x^2+y^2}\,d\sigma=\int_0^{2\pi}d\theta\int_\pi^{2\pi}\sin r\cdot r\,dr=2\pi\int_\pi^{2\pi}r\,d\cos r$$

$$=2\pi\left(-r\cos r\Big|_\pi^{2\pi}+\int_\pi^{2\pi}\cos r\,dr\right)=2\pi\left(-2\pi-\pi\right)=-6\pi^2.$$

(2) $\iint_D (x^2+y^2)\,d\sigma$, 其中 $D=\{(x,y)\mid 2x\le x^2+y^2\le 4x\}$




解: 令 $\begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases}$ $d\sigma=dxdy=r\,dr\,d\theta$

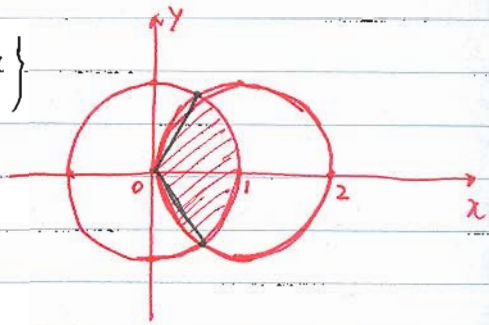
$$D'=\left\{(r,\theta)\,\middle|\,\begin{matrix} 2\cos\theta\le r\le 4\cos\theta \\ 0\le\theta<2\pi \end{matrix}\right\}$$

$$\iint_D (x^2+y^2)\,d\sigma=\iint_{D'} r^2\cdot r\,dr\,d\theta=\int_0^{2\pi}d\theta\cdot\int_{2\cos\theta}^{4\cos\theta} r^3\,dr$$

$$=\int_0^{2\pi}\frac{1}{4}r^4\Big|_{2\cos\theta}^{4\cos\theta}d\theta=60\int_0^{2\pi}\cos^4\theta\,d\theta$$

$$=60\cdot\int_0^{2\pi}\left(\frac{3}{8}+\frac{1}{2}\cos 2\theta+\frac{1}{8}\cos 4\theta\right)d\theta=\frac{45}{2}\pi.$$

(3) $\iint_D (x^2+y^2)^{3/2}\,d\sigma$, 其中 $D=\{(x,y)\mid x^2+y^2\le 1,\ x^2+y^2\le 2x\}$




解: 令 $\begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases}$ $d\sigma=dxdy=r\,dr\,d\theta$.

$$D'=\left\{(r,\theta)\,\middle|\,\begin{matrix} 0\le r\le 1 \\ 0\le r\le 2\cos\theta \end{matrix}\right\}$$

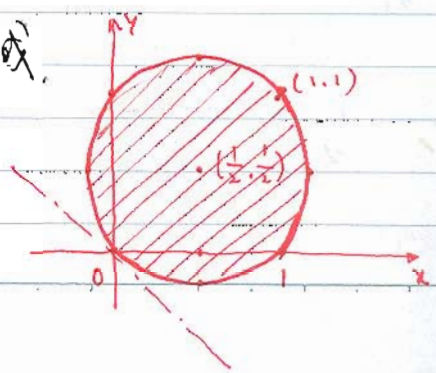
$$\iint_D (x^2+y^2)^{3/2}d\sigma=\iint_D r^3\cdot r\,dr\,d\theta=\iint_D r^4\,dr\,d\theta$$

$$=\int_{-\frac{\pi}{2}}^{-\frac{\pi}{3}}d\theta\int_0^{2\cos\theta}r^4dr+\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}}d\theta\int_0^1 r^4dr+\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}d\theta\int_0^{2\cos\theta}r^4dr$$

$$=\int_{-\frac{\pi}{2}}^{-\frac{\pi}{3}}\frac{32}{5}\cos^5\theta\,d\theta+\frac{2}{3}\pi\cdot\frac{1}{5}+\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\frac{32}{5}\cos^5\theta\,d\theta$$

$$=\frac{2\pi}{15}+\frac{512}{75}-\frac{98}{25}\sqrt{3}.$$

(4) $\iint_D (x+y)\,d\sigma$, 其中 $D$ 是由曲线 $x^2+y^2=x+y$ 所包围的平面区域.




解: 令 $\begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases}$ $d\sigma=dxdy=r\,dr\,d\theta$.

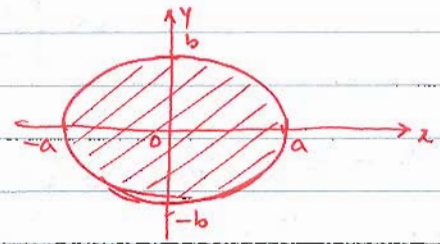
$$D'=\left\{(r,\theta)\,\middle|\,\begin{matrix} 0\le r\le\cos\theta+\sin\theta \\ -\frac{\pi}{4}\le\theta\le\frac{3\pi}{4} \end{matrix}\right\}$$

$$\iint_D (x+y)\,d\sigma = \iint_{D'} r(\cos\theta+\sin\theta)\,r\,dr\,d\theta = \int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} (\cos\theta+\sin\theta)\,d\theta \cdot \int_0^{\cos\theta+\sin\theta} r^2\,dr$$

$$= \int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} (\cos\theta+\sin\theta)\,d\theta \cdot \frac{1}{3}(\cos\theta+\sin\theta)^3 = \frac{1}{3}\int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} (\cos\theta+\sin\theta)^4\,d\theta$$

$$= \frac{1}{3}\int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} (1+\sin 2\theta)^2\,d\theta = \frac{1}{3}\int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} \left(1+2\sin 2\theta+\frac{1}{2}-\frac{1}{2}\cos 4\theta\right)d\theta = \frac{\pi}{2}.$$

8. 计算 $\iint_D \left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)d\sigma$，其中 $D$ 是由椭圆 $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ 所围成的区域.




解：令 $\begin{cases} x = ar\cos\theta \\ y = br\sin\theta \end{cases}$ $d\sigma = dx\,dy = \left|\begin{matrix} x'_r & x'_\theta \\ y'_r & y'_\theta \end{matrix}\right| dr\,d\theta$

$= ab\,r\,dr\,d\theta.$

$D' = \{(r,\theta) \mid 0\le r\le 1,\ 0\le\theta\le 2\pi\}.$

$$\iint_D \left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)d\sigma = \iint_{D'} r^2 \cdot ab\,r\,dr\,d\theta = \int_0^{2\pi} ab\,d\theta \int_0^1 r^3\,dr$$

$$= ab\cdot 2\pi\cdot \frac{1}{4}r^4\Big|_0^1 = \frac{\pi ab}{2}.$$

9. 求由曲线 $\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)^2 = \frac{xy}{c^2}$ $(a,b,c>0)$ 在第一象限中所围图形面积.

解：令 $\begin{cases} x = ar\cos\theta \\ y = br\sin\theta \end{cases}$ $d\sigma = dx\,dy = ab\,r\,dr\,d\theta.$

曲线方程：$r^4 = \frac{ab}{c^2}r^2\cos\theta\sin\theta$

即：$r = \sqrt{\frac{ab}{c^2}\sin\theta\cos\theta}$

$$\therefore S(D) = \iint_D d\sigma = \int_0^{\frac{\pi}{2}} d\theta \int_0^{\sqrt{\frac{ab}{c^2}\sin\theta\cos\theta}} ab\,r\,dr$$

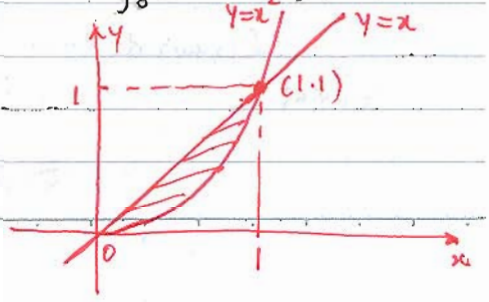
$$= \int_0^{\frac{\pi}{2}} d\theta\,\frac{1}{2}ab\left(\frac{ab}{c^2}\sin\theta\cos\theta\right) = \frac{a^2b^2}{4c^2}\int_0^{\frac{\pi}{2}} \sin 2\theta\,d\theta = \frac{a^2b^2}{4c^2}.$$

10. 设一元函数 $f$ 在 $[0,1]$ 上连续，证明：$\int_0^1 dy\int_y^{\sqrt{y}} e^y f(x)\,dx = \int_0^1 (e^x-e^{x^2})f(x)\,dx$.

证明：∵ $f$ 在 $[0,1]$ 连续，∴ 可积.

$$\int_0^1 dy\int_y^{\sqrt{y}} e^y f(x)\,dx = \int_0^1 dx\int_{x^2}^{x} e^y f(x)\,dy$$

$$= \int_0^1 f(x)\,dx\int_{x^2}^{x} e^y\,dy = \int_0^1 (e^x-e^{x^2})f(x)\,dx.$$



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11、设一元非负函数 $f$ 在 $[a,b]$ 上连续，证明 $\iint\limits_{[a,b]\times[a,b]}\left[f(x)+\frac{1}{f(y)}\right]dxdy\geqslant 2(b-a)^2$.

证明：$\iint\limits_{[a,b]\times[a,b]}\left[f(x)+\frac{1}{f(y)}\right]dxdy=\iint\limits_{[a,b]\times[a,b]}f(x)\,dxdy+\iint\limits_{[a,b]\times[a,b]}\frac{1}{f(y)}\,dxdy$

$$=\int_a^b dy\int_a^b f(x)dx+\int_a^b dx\int_a^b\frac{1}{f(y)}dy$$

$$=(b-a)\int_a^b f(x)\,dx+(b-a)\int_a^b\frac{1}{f(x)}dx$$

$$=(b-a)\int_a^b\left[f(x)+\frac{1}{f(x)}\right]dx\geqslant(b-a)\int_a^b 2\,dx=2(b-a)^2.$$

12、设 $D[0,1)\times[0,1]$，证明：$1\leq\iint\limits_D\left[\sin(x^2)+\cos(y^2)\right]dxdy\leq\sqrt{2}$.

证明：$I=\iint\limits_D\left[\sin(x^2)+\cos(y^2)\right]dxdy=\iint\limits_D\sin(x^2)dxdy+\iint\limits_D\cos(y^2)dxdy$

$$=\int_0^1 dy\int_0^1\sin(x^2)dx+\int_0^1 dx\int_0^1\cos(y^2)dy$$

$$=\int_0^1\sin(x^2)dx+\int_0^1\cos(x^2)dx=\int_0^1\left[\sin(x^2)+\cos(x^2)\right]dx$$

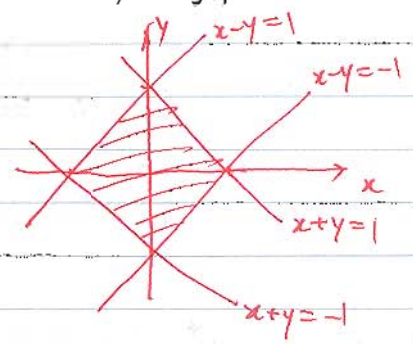
$$=\int_0^1\sqrt{2}\sin\left(x^2+\frac{\pi}{4}\right)dx$$

当 $0\leq x\leq 1$ 时，$\frac{1}{\sqrt{2}}\leq\sin\left(x^2+\frac{\pi}{4}\right)\leq 1$. $\therefore\quad 1\leq I\leq\sqrt{2}$.

13、设一元函数 $f(u)$ 在 $[-1,1]$ 上连续，证明：$\iint\limits_{|x|+|y|\leq 1}f(x+y)\,dxdy=\int_{-1}^1 f(u)\,du$.

证明：作变换 $\begin{cases}u=x+y\\ v=-(x-y)\end{cases}$ $J=\begin{bmatrix}x'_u & x'_v\\ y'_u & y'_v\end{bmatrix}=\begin{bmatrix}u'_x & u'_y\\ v'_x & v'_y\end{bmatrix}^{-1}$

$$J=\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}^{-1}=\frac{1}{2}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}.\qquad |J|=\frac{1}{4}\cdot 2=\frac{1}{2}.$$




$$d\sigma=dxdy=|J|\cdot dudv=\frac{1}{2}dudv.$$

$$D'=\left\{(u,v)\,\middle|\,\begin{matrix}-1\leq u\leq 1\\ -1\leq v\leq 1\end{matrix}\right\}$$

$$\iint\limits_{|x|+|y|\leq 1}f(x+y)\,dxdy=\iint\limits_{D'}f(u)\cdot\frac{1}{2}dudv.$$

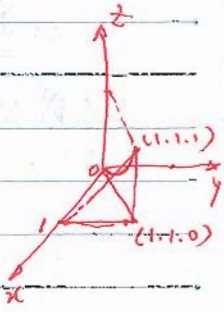
$$=\frac{1}{2}\int_{-1}^1 dv\int_{-1}^1 f(u)\,du=\int_{-1}^1 f(u)\,du.$$

## §3. 三重积分的计算和应用

P. 98. 习题

1. 计算下列三重积分：

(1) $\iiint_\Omega xy^2z^3\,dv$，其中 $\Omega$ 是由曲面 $z=xy$ 和平面 $y=x$，$x=1$，$z=0$ 所围成的区域。

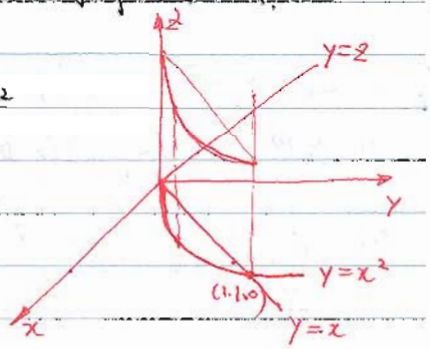



解：$\iiint_\Omega xy^2z^3\,dv = \int_0^1 dx\int_0^x dy\int_0^{xy} xy^2z^3\,dz = \int_0^1 x\,dx\int_0^x y^2dy\int_0^{xy} z^3\,dz$

$= \int_0^1 x\,dx\int_0^x y^2dy\cdot\frac{1}{4}x^4y^4 = \int_0^1 \frac{1}{4}x^5dx\int_0^x y^6dy$

$= \frac{1}{4}\int_0^1 x^5\cdot\frac{1}{7}x^7dx = \frac{1}{28}\int_0^1 x^{12}dx = \frac{1}{28}\cdot\frac{1}{13} = \frac{1}{364}.$

(2) $\iiint_\Omega xz\,dv$，其中 $\Omega$ 是由平面 $z=0$，$x=y$，$y=z$ 以及抛物柱面 $y=x^2$ 所围成的闭区域。

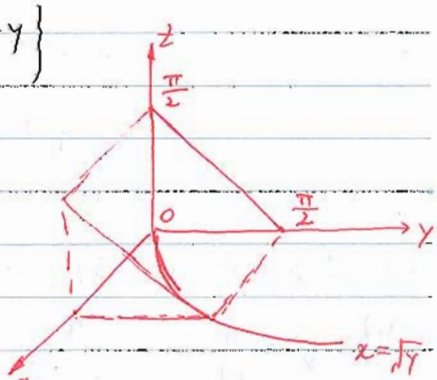



解：$\iiint_\Omega xz\,dv = \int_0^1 dx\int_{x^2}^x dy\int_0^y xz\,dz = \int_0^1 dx\int_{x^2}^x dy\cdot x\cdot\frac{1}{2}y^2$

$= \int_0^1 \frac{1}{2}x\,dx\int_{x^2}^x y^2dy = \frac{1}{2}\int_0^1 x\,dx\cdot\frac{1}{3}(x^3-x^6)$

$= \frac{1}{6}\int_0^1 (x^4-x^7)\,dx = \frac{1}{6}\left(\frac{1}{5}-\frac{1}{8}\right) = \frac{1}{80}.$

(3) $\iiint_\Omega x\sin(y+z)\,dv$，其中 $\Omega = \left\{(x,y,z)\,\middle|\,0\le x\le\sqrt{y},\ 0\le z\le\frac{\pi}{2}-y\right\}$




解：$\iiint_\Omega x\sin(y+z)\,dv = \int_0^{\frac{\pi}{2}} dy\int_0^{\sqrt{y}} dx\int_0^{\frac{\pi}{2}-y} x\sin(y+z)\,dz$

$= \int_0^{\frac{\pi}{2}} dy\int_0^{\sqrt{y}} x\,dx\cdot\left[-\cos(y+z)\right]_{z=0}^{z=\frac{\pi}{2}-y}$

$= \int_0^{\frac{\pi}{2}} dy\int_0^{\sqrt{y}} x\,dx\cdot\cos y = \int_0^{\frac{\pi}{2}} \cos y\cdot\frac{1}{2}x^2\Big|_0^{\sqrt{y}}dy$

$= \int_0^{\frac{\pi}{2}} \frac{1}{2}y\cos y\,dy = \frac{1}{2}\int_0^{\frac{\pi}{2}} y\,d\sin y = \frac{1}{2}y\sin y\Big|_0^{\frac{\pi}{2}} - \frac{1}{2}\int_0^{\frac{\pi}{2}}\sin y\,dy$

$= \frac{\pi}{4} + \frac{1}{2}\cos y\Big|_0^{\frac{\pi}{2}} = \frac{\pi}{4} + \frac{1}{2}(-1) = \frac{\pi}{4} - \frac{1}{2}.$

(4) $\iiint_\Omega \frac{xyz}{1+x^2+y^2+z^2}\,dv$，其中 $\Omega = \left\{(x,y,z)\,\middle|\,x\ge0,\ z\ge0,\ x^2+y^2+z^2\le1\right\}$

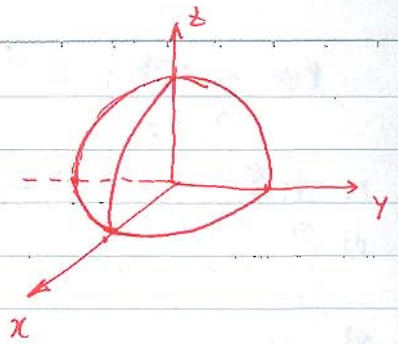
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解: $\iiint_\Omega \frac{xyz}{1+x^2+y^2+z^2}\,d\sigma = \int_0^1 dx \int_0^{\sqrt{1-x^2}} dz \int_{-\sqrt{1-x^2-z^2}}^{\sqrt{1-x^2-z^2}} \frac{xyz}{1+x^2+y^2+z^2}\,dy$

$= \int_0^1 dx \int_0^{\sqrt{1-x^2}} dz \cdot \frac{1}{2} xz \ln(1+x^2+y^2+z^2) \Big|_{y=-\sqrt{1-x^2-z^2}}^{y=\sqrt{1-x^2-z^2}}$

$= \frac{1}{2}\int_0^1 x\,dx \int_0^{\sqrt{1-x^2}} z\,dz \cdot 0 = 0.$




(或做球坐标: $\begin{cases} x = r\sin\varphi\cos\theta \\ y = r\sin\varphi\sin\theta \\ z = r\cos\varphi \end{cases}$ $\Omega = \left\{(r,\varphi,\theta) \,\middle|\, 0\le r\le 1,\ 0\le\varphi\le\frac{\pi}{2},\ 0\le\theta\le\pi\right\}$

$d\sigma = dx\,dy\,dz = r^2\sin\varphi\,dr\,d\varphi\,d\theta$

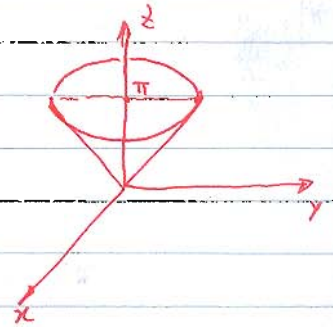
$\iiint_\Omega \frac{xyz}{1+x^2+y^2+z^2}\,d\sigma = \iiint_\Omega \frac{r^3\sin^2\varphi\cos\varphi\sin\theta\cos\theta}{1+r^2}\cdot r^2\sin\varphi\,dr\,d\varphi\,d\theta$

$= \int_0^1 \frac{r^5}{1+r^2}dr \cdot \int_0^{\pi}\sin\theta\cos\theta\,d\theta \cdot \int_0^{\frac{\pi}{2}}\sin^3\varphi\cos\varphi\,d\varphi = 0.$ )

(其中 $\int_0^{\pi}\sin\theta\cos\theta\,d\theta = 0$)

2. 解下列三重积分问题:

(1) 求 $\iiint_\Omega \sin z\,d\sigma$, 其中 $\Omega$ 由锥面 $z=\sqrt{x^2+y^2}$ 和平面 $z=\pi$ 围成.



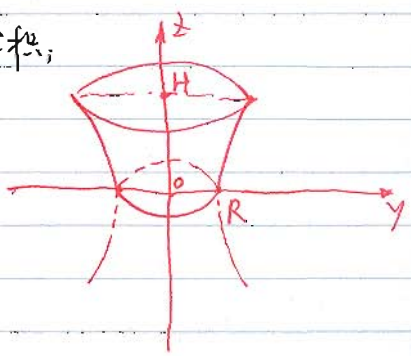

解: $\iiint_\Omega \sin z\,d\sigma = \int_0^{\pi} dz \iint_{x^2+y^2\le z^2} \sin z\,dx\,dy = \int_0^{\pi}\sin z\,dz\cdot\pi z^2$

$= \pi\int_0^{\pi} z^2\sin z\,dz = \pi\int_0^{\pi} z^2\,d(-\cos z)$

$= -\pi z^2\cos z\Big|_0^{\pi} + \pi\int_0^{\pi} 2z\cos z\,dz = \pi^3 + 2\pi\int_0^{\pi} z\,d\sin z$

$= \pi^3 + 2\pi z\sin z\Big|_0^{\pi} - 2\pi\int_0^{\pi}\sin z\,dz = \pi^3 + 2\pi\cos z\Big|_0^{\pi} = \pi^3 - 4\pi.$

(2) 设 $\Omega$ 由单叶双曲面 $x^2+y^2-z^2=R^2$ 和平面 $z=0$, $z=H$ 围成, 求 $\Omega$ 的体积;




解: $V = \iiint_\Omega d\sigma = \int_0^H dz \iint_{0\le x^2+y^2\le z^2+R^2} dx\,dy = \int_0^H dz\cdot\pi r^2 \Big|_{r^2=0}^{r^2=z^2+R^2}$

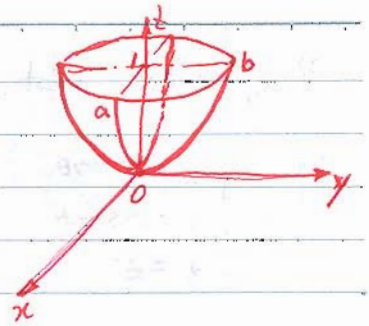
$= \int_0^H dz\cdot\pi(z^2+R^2) = \pi\left(\frac{1}{3}z^3+R^2z\right)\Big|_0^H = \pi\left(\frac{1}{3}H^3+R^2H\right).$

(3) 求由 $\Omega = \left\{(x,y,z) \,\middle|\, \frac{x^2}{a^2}+\frac{y^2}{b^2} \le z \le 1\right\}$ 确定的立体的质量, 其中密度 $\rho$ 为常数.

解: $M = \rho V = \rho\iiint_\Omega d\sigma = \rho\int_0^1 dz \iint_{0\le\frac{x^2}{a^2}+\frac{y^2}{b^2}\le z} dx\,dy = \rho\int_0^1 dz \cdot \iint_{0\le\frac{x^2}{za^2}+\frac{y^2}{zb^2}\le 1} dx\,dy$

$= \rho \int_0^1 dz \cdot \pi ab z = \pi\rho ab \cdot \frac{1}{2} z^2 \Big|_0^1 = \frac{\pi\rho ab}{2}$.

$\therefore \frac{M}{\rho} = \frac{\pi ab}{2}$. 设质心坐标 $(\bar{x}, \bar{y}, \bar{z})$.




则: $\bar{x} = \frac{1}{M} \iiint_\Omega x\rho \, d\sigma = \frac{\rho}{M} \iiint_\Omega x \, d\sigma = \frac{\rho}{M} \int_0^1 dz \cdot \iint_{0 \le \frac{x^2}{a^2} + \frac{y^2}{b^2} \le z} x \, dx \, dy$

令 $\begin{cases} x = ar\cos\theta \\ y = br\sin\theta \end{cases}$ $D = \{(r, \theta) \mid 0 \le r \le \sqrt{z},\ 0 \le \theta \le 2\pi\}$. $dx\,dy = abr\,dr\,d\theta$.

$\bar{x} = \frac{\rho}{M} \int_0^1 dz \cdot \int_0^{2\pi} d\theta \int_0^{\sqrt{z}} ar\cos\theta \cdot abr\,dr = \frac{\rho}{M} \int_0^1 dz \cdot \underbrace{\int_0^{2\pi} \cos\theta \, d\theta}_{=0} \int_0^{\sqrt{z}} a^2 b r^2 dr$

$= \frac{\rho}{M} \cdot \int_0^1 dz \cdot 0 \cdot \int_0^{\sqrt{z}} a^2 b r^2 dr = 0$.

$\bar{y} = \frac{\rho}{M} \iiint_\Omega y \, d\sigma = \frac{\rho}{M} \int_0^1 dz \int_0^{2\pi} d\theta \int_0^{\sqrt{z}} br\sin\theta \cdot abr\,dr$

$= \frac{\rho}{M} \cdot \int_0^1 dz \cdot \underbrace{\int_0^{2\pi} \sin\theta \, d\theta}_{=0} \cdot \int_0^{\sqrt{z}} ab^2 r^2 dr = 0$.

$\bar{z} = \frac{\rho}{M} \iiint_\Omega z \, d\sigma = \frac{\rho}{M} \int_0^1 dz \cdot \int_0^{2\pi} d\theta \cdot \int_0^{\sqrt{z}} z \cdot abr\,dr$

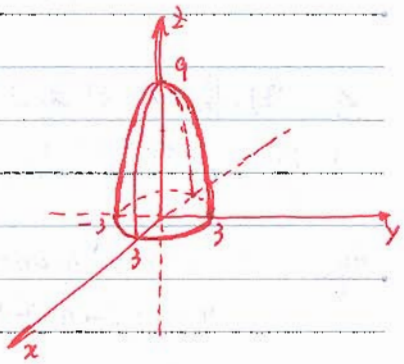
$= \frac{\rho}{M} \int_0^1 z\,dz \int_0^{2\pi} d\theta \int_0^{\sqrt{z}} abr\,dr = \frac{\rho}{M} \cdot 2\pi \cdot \int_0^1 z\,dz \cdot \frac{1}{2} abz$

$= \frac{\pi\rho}{M} ab \int_0^1 z^2 dz = \frac{\rho}{M} \cdot \pi ab \cdot \frac{1}{3} = \frac{2}{\pi ab} \pi ab \frac{1}{3} = \frac{2}{3}$.

$\therefore$ 质心坐标为 $(0, 0, \frac{2}{3})$.

3. 用柱面坐标计算三重积分:

(1) $\iiint_\Omega \sqrt{x^2+y^2} \, d\sigma$. 其中 $\Omega = \{(x, y, z) \mid 0 \le z \le 9 - x^2 - y^2\}$.




解: 令 $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases}$ $\Omega = \{(x, y, z) \mid 0 \le z \le 9 - x^2 - y^2\}$

$= \{(r, \theta, z) \mid \begin{matrix} 0 \le z \le 9 - r^2 \\ 0 \le \theta \le 2\pi \end{matrix}\}$.

$d\sigma = dx\,dy\,dz = r\,dr\,d\theta\,dz$.

$\iiint_\Omega \sqrt{x^2+y^2} \, d\sigma = \int_0^{2\pi} d\theta \int_0^3 dr \int_0^{9-r^2} r^2 \, dz = 2\pi \cdot \int_0^3 r^2(9 - r^2) dr = 2\pi \left(3r^3 - \frac{1}{5} r^5\right) \Big|_0^3$.

$= 2\pi \cdot \left(81 - \frac{243}{5}\right) = \frac{324}{5}\pi$.

(2) $\iiint_\Omega y\,d\sigma$, 其中 $\Omega=\{(x,y,z)\mid 1\le x^2+y^2\le 4,\ 0\le z\le x+2\}$.

解: $\begin{cases} x=r\cos\theta \\ y=r\sin\theta \\ z=z \end{cases}$ $\Omega=\{(x,y,z)\mid 1\le x^2+y^2\le 4,\ 0\le z\le x+2\}$

$=\{(r,\theta,z)\mid 1\le r\le 2,\ \begin{matrix}0\le z\le r\cos\theta+2\\ 0\le\theta\le 2\pi\end{matrix}\}$.

$d\sigma=r\,dr\,d\theta\,dz$.

$$\iiint_\Omega y\,d\sigma=\int_0^{2\pi}d\theta\int_1^2 dr\int_0^{r\cos\theta+2} r\sin\theta\cdot r\,dz=\int_0^{2\pi}\sin\theta\,d\theta\int_1^2 r^2dr\cdot(r\cos\theta+2)$$

$$=\int_0^{2\pi}\sin\theta\,d\theta\int_1^2(r^3\cos\theta+2r^2)dr=\int_0^{2\pi}\sin\theta\,d\theta\left(\frac14 r^4\cos\theta+\frac23 r^3\right)\Big|_1^2$$

$$=\int_0^{2\pi}\sin\theta\,d\theta\left(4\cos\theta+\frac{16}{3}-\frac14\cos\theta-\frac23\right)=\int_0^{2\pi}\left(\frac{15}{4}\cos\theta+\frac{14}{3}\right)\sin\theta\,d\theta$$

$$=-\int_0^{2\pi}\left(\frac{15}{4}\cos\theta+\frac{14}{3}\right)d\cos\theta=\left(-\frac{15}{8}\cos^2\theta-\frac73\cos\theta\right)\Big|_0^{2\pi}=0.$$

(3) $\iiint_\Omega x^2\,d\sigma$, 其中 $\Omega=\{(x,y,z)\mid x^2+y^2\le 1,\ 0\le z\le 2\sqrt{x^2+y^2}\}$.

解: $\begin{cases} x=r\cos\theta \\ y=r\sin\theta \\ z=z \end{cases}$ $\Omega=\{(x,y,z)\mid x^2+y^2\le 1,\ 0\le z\le 2\sqrt{x^2+y^2}\}$

$=\{(r,\theta,z)\mid \begin{matrix}0\le r\le 1\\ 0\le\theta\le 2\pi\end{matrix},\ 0\le z\le 2r\}$. $d\sigma=r\,dr\,d\theta\,dz$.

$$\iiint_\Omega x^2\,d\sigma=\int_0^{2\pi}d\theta\int_0^1 dr\int_0^{2r} r\,dz\cdot r^2\cos^2\theta=\int_0^{2\pi}\cos^2\theta\,d\theta\int_0^1 r^3dr\cdot\int_0^{2r}dz$$

$$=\int_0^{2\pi}\frac12(1+\cos2\theta)\,d\theta\cdot\int_0^1 2r^4dr=\frac12\cdot2\pi\cdot2\cdot\frac15=\frac{2\pi}{5}.$$

4. 用球坐标计算三重积分:

(1) $\iiint_\Omega xe^{(x^2+y^2+z^2)^2}d\sigma$, 其中 $\Omega$ 是第一卦限中球面 $x^2+y^2+z^2=1$ 与 $x^2+y^2+z^2=4$ 之间的部分

解: 令 $\begin{cases} x=r\sin\varphi\cos\theta \\ y=r\sin\varphi\sin\theta \\ z=r\cos\varphi \end{cases}$ $\Omega=\{(x,y,z)\mid 1\le x^2+y^2+z^2\le 4,\ x\ge0,\ y\ge0,\ z\ge0\}$

$=\{(r,\varphi,\theta)\mid 1\le r\le 2,\ 0\le\theta\le\frac{\pi}{2},\ 0\le\varphi\le\frac{\pi}{2}\}$

$d\sigma=dx\,dy\,dz=r^2\sin\varphi\,dr\,d\varphi\,d\theta$.

$$\iiint_\Omega xe^{(x^2+y^2+z^2)^2}d\sigma=\int_0^{\frac{\pi}{2}}d\theta\int_0^{\frac{\pi}{2}}d\varphi\int_1^2 r\sin\varphi\cos\theta\cdot e^{r^4}\cdot r^2\sin\varphi\,dr$$

$$= \int_0^{\frac{\pi}{2}} \cos\theta \, d\theta \cdot \int_0^{\frac{\pi}{2}} \sin^2\varphi \, d\varphi \cdot \int_1^2 r^3 e^{r^4} dr$$

$$= \sin\theta \Big|_0^{\frac{\pi}{2}} \cdot \int_0^{\frac{\pi}{2}} \frac{1}{2}(1-\cos 2\varphi) d\varphi \cdot \frac{1}{4} e^{r^4} \Big|_1^2 = 1 \cdot \frac{1}{2} \cdot \frac{\pi}{2} \cdot \frac{1}{4} (e^{16} - e)$$

$$= \frac{\pi}{16}(e^{16} - e).$$

2) $\iiint_\Omega \frac{z \ln(1+x^2+y^2+z^2)}{1+x^2+y^2+z^2} dv$, 其中 $\Omega$ 是由单位球体，即 $\Omega = \left\{ (x,y,z) \,\middle|\, 0 \le z \le \sqrt{1-x^2-y^2} \right\}$.

解：
$$\begin{cases} x = r\sin\varphi\cos\theta \\ y = r\sin\varphi\sin\theta \\ z = r\cos\varphi \end{cases}$$

$$\Omega = \left\{ (x,y,z) \,\middle|\, 0 \le z \le \sqrt{1-x^2-y^2} \right\} = \left\{ (r,\theta,\varphi) \,\middle|\, 0 \le r \le 1, \ 0 \le \varphi \le \frac{\pi}{2}, \ 0 \le \theta \le 2\pi \right\}.$$

$$dv = r^2 \sin\varphi \, dr \, d\varphi \, d\theta.$$

$$\iiint_\Omega \frac{z \ln(1+x^2+y^2+z^2)}{1+x^2+y^2+z^2} dv = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^1 \frac{r\cos\varphi \cdot \ln(1+r^2)}{1+r^2} r^2 \sin\varphi \, dr$$

$$= 2\pi \cdot \int_0^{\frac{\pi}{2}} \sin\varphi\cos\varphi \, d\varphi \cdot \int_0^1 \frac{r^3 \ln(1+r^2)}{1+r^2} dr$$

$$= 2\pi \cdot \frac{1}{2}\sin^2\varphi \Big|_0^{\frac{\pi}{2}} \cdot \frac{1}{2} \int_0^1 \frac{(1+r^2-1)\ln(1+r^2)}{1+r^2} d(1+r^2)$$

$$= \frac{\pi}{2}\left[ \int_0^1 \frac{-\ln(1+r^2)}{1+r^2} d(1+r^2) + \int_0^1 \ln(1+r^2) \cdot d(1+r^2) \right]$$

$$= \frac{\pi}{2}\left[ (1+r^2)\ln(1+r^2)\Big|_0^1 - \int_0^1 d(1+r^2) - \int_0^1 \ln(1+r^2) \, d\ln(1+r^2) \right]$$

$$= \frac{\pi}{2}\left[ 2\ln 2 - (1+r^2)\Big|_0^1 - \frac{1}{2}\ln^2(1+r^2)\Big|_0^1 \right] = \frac{\pi}{2}\left[ 2\ln 2 - 1 - \frac{1}{2}\ln^2 2 \right]$$

$$= \pi\left[ \ln 2 - \frac{1}{2} - \frac{1}{4}\ln^2 2 \right].$$

3) $\iiint_\Omega (x+y) dv$, 其中 $\Omega = \left\{ (x,y,z) \,\middle|\, 1 \le z \le 1 + \sqrt{1-x^2-y^2} \right\}$

解：
$$\begin{cases} x = r\sin\varphi\cos\theta \\ y = r\sin\varphi\sin\theta \\ z = r\cos\varphi \end{cases}$$

$$\Omega = \left\{ (x,y,z) \,\middle|\, 1 \le z \le 1 + \sqrt{1-x^2-y^2} \right\} = \left\{ (r,\theta,\varphi) \,\middle|\, \begin{array}{l} 1 \le r\cos\varphi \le 1 + \sqrt{1-r^2\sin^2\varphi} \\ 0 \le \theta \le 2\pi \end{array} \right\}.$$

$$dv = r^2 \sin\varphi \, dr \, d\varphi \, d\theta.$$

$$\iiint_\Omega (x+y)\,d\sigma = \int_0^{2\pi} d\theta \iint_D r\sin\varphi(\sin\theta+\cos\theta)\cdot r^2\sin\varphi\,dr\,d\varphi \qquad D=\left\{(r,\varphi)\,\middle|\, 1\le r\cos\varphi \le 1+\sqrt{1+r^2\sin^2\varphi}\right\}$$

$$= \int_0^{2\pi}(\sin\theta+\cos\theta)\,d\theta \cdot \iint_D r^3\sin^2\varphi\,dr\,d\varphi = 0.$$

$\int_0^{2\pi}(\sin\theta+\cos\theta)\,d\theta = 0$

(4) $\iiint_\Omega \sqrt{x^2+y^2+z^2}\,d\sigma$, 其中 $\Omega=\left\{(x,y,z)\,\middle|\, x^2+y^2+z^2\le z\right\}$.

解: $\begin{cases} x=r\sin\varphi\cos\theta \\ y=r\sin\varphi\sin\theta \\ z=r\cos\varphi \end{cases}$

$$\Omega=\left\{(x,y,z)\,\middle|\, x^2+y^2+z^2\le z\right\} = \left\{(r,\varphi,\theta)\,\middle|\, 0\le r\le\cos\varphi,\ 0\le\varphi\le\frac{\pi}{2},\ 0\le\theta\le 2\pi\right\}$$

$d\sigma = r^2\sin\varphi\,dr\,d\varphi\,d\theta$.

$$\iiint_\Omega \sqrt{x^2+y^2+z^2}\,d\sigma = \int_0^{2\pi}d\theta\int_0^{\frac{\pi}{2}}d\varphi\int_0^{\cos\varphi} r\cdot r^2\sin\varphi\,dr$$

$$= 2\pi\cdot\int_0^{\frac{\pi}{2}}\sin\varphi\,d\varphi\cdot\int_0^{\cos\varphi} r^3\,dr = 2\pi\cdot\int_0^{\frac{\pi}{2}}\sin\varphi\cdot\frac{1}{4}\cos^4\varphi\,d\varphi$$

$$= -\frac{\pi}{2}\int_0^{\frac{\pi}{2}}\cos^4\varphi\,d\cos\varphi = -\frac{\pi}{2}\cdot\frac{1}{5}\cos^5\varphi\Big|_0^{\frac{\pi}{2}} = \frac{\pi}{10}.$$

5. 求半径为 a 的半球体的质心，该球体上任何一点的密度与它到底面的距离成正比.

解: $\rho(x,y,z)=kz$.

$$M=\iiint_\Omega \rho\,d\sigma = \iiint_\Omega kz\,d\sigma$$

$$= \int_0^{2\pi}d\theta\int_0^{\frac{\pi}{2}}d\varphi\int_0^a kr^3\sin\varphi\cos\varphi\,dr$$

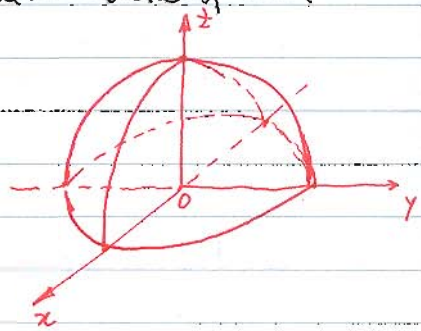
$$= 2\pi k\int_0^{\frac{\pi}{2}}\sin\varphi\cos\varphi\,d\varphi\int_0^a r^3\,dr$$

$$= 2\pi k\cdot\frac{1}{2}\sin^2\varphi\Big|_0^{\frac{\pi}{2}}\cdot\frac{1}{4}r^4\Big|_0^a = \frac{\pi k a^4}{4}$$

$\begin{cases} x=r\sin\varphi\cos\theta \\ y=r\sin\varphi\sin\theta \\ z=r\cos\varphi \end{cases}$

$\Omega=\left\{(r,\varphi,\theta)\,\middle|\, \begin{array}{l} 0\le r\le a, \\ 0\le\varphi\le\frac{\pi}{2} \\ 0\le\theta\le 2\pi \end{array}\right\}$

$d\sigma = r^2\sin\varphi\,dr\,d\varphi\,d\theta$




由对称性可知: $\bar{x}=\bar{y}=0$.

$$\bar{z} = \frac{1}{M}\iiint_\Omega z\rho\,d\sigma = \frac{1}{M}\iiint_\Omega kz^2\,d\sigma = \frac{1}{M}\cdot\int_0^{2\pi}d\theta\cdot\int_0^{\frac{\pi}{2}}d\varphi\cdot\int_0^a kr^2\cos^2\varphi\cdot r^2\sin\varphi\,dr$$

$$= \frac{1}{M}\cdot 2\pi\cdot\int_0^{\frac{\pi}{2}}\cos^2\varphi\sin\varphi\,d\varphi\cdot\int_0^a kr^4\,dr = \frac{2\pi}{M}\cdot\left(-\frac{1}{3}\cos^3\varphi\Big|_{\frac{\pi}{2}}^0\right)\cdot\frac{1}{5}kr^5\Big|_0^a$$

$$= \frac{2\pi}{M}\cdot\frac{ka^5}{15} = 2\pi\cdot\frac{4}{\pi ka^4}\cdot\frac{k}{15}a^5 = \frac{8}{15}a. \quad \therefore \text{质心}\left(0,0,\frac{8}{15}a\right)$$

6. 设一物体占有空间区域 $\Omega=\{(x,y,z)\mid x^2+y^2\le R^2,\ |z|\le H\}$，其密度为常值，已知它关于 $x$ 轴及 $z$ 轴的转动惯量相等. 试证明：$H:R=\sqrt{3}:2$.

证明：设 $\begin{cases}x=r\cos\theta\\ y=r\sin\theta\\ z=z\end{cases}$.

$\Omega=\{(x,y,z)\mid x^2+y^2\le R^2,\ |z|\le H\}$

$=\{(r,\theta,z)\mid 0\le r\le R,\ -H\le z\le H\}$.

$d\sigma=r\,dr\,d\theta\,dz$.

$$I_x=\iiint_\Omega(y^2+z^2)\rho_0\,d\sigma=\rho_0\int_0^{2\pi}d\theta\int_0^R dr\cdot\int_{-H}^{H}(r^2\sin^2\theta+z^2)\cdot r\,dz$$

$$=\rho_0\int_0^{2\pi}d\theta\int_0^R r\,dr\int_{-H}^{H}(r^2\sin^2\theta+z^2)\,dz$$

$$=\rho_0\int_0^{2\pi}d\theta\cdot\int_0^R r\,dr\cdot\left(r^2\sin^2\theta\cdot z+\frac{1}{3}z^3\right)\Big|_{-H}^{H}$$

$$=\rho_0\int_0^{2\pi}d\theta\cdot\int_0^R\cdot\left(\frac{2}{3}H^3+2Hr^2\sin^2\theta\right)r\,dr$$

$$=\rho_0\int_0^{2\pi}\left(\frac{1}{3}R^2H^3+\frac{1}{2}HR^4\sin^2\theta\right)d\theta=\rho_0\left(\frac{2\pi}{3}R^2H^3+\frac{\pi}{2}HR^4\right)$$

$$=\pi\rho_0R^2H\left(\frac{2}{3}H^2+\frac{1}{2}R^2\right).$$

$$I_z=\iiint_\Omega(x^2+y^2)\cdot\rho_0\,d\sigma=\rho_0\int_0^{2\pi}d\theta\int_0^R dr\cdot\int_{-H}^{H}r^2\cdot r\,dz$$

$$=2\pi\rho_0\cdot\int_0^R r^3\,dr\cdot 2H=\pi\rho_0HR^4.$$

$\because I_x=I_z$. $\therefore \pi\rho_0R^2H\left(\frac{2}{3}H^2+\frac{1}{2}R^2\right)=\pi\rho_0HR^4$.

即：$\frac{2}{3}H^2+\frac{1}{2}R^2=R^2\Rightarrow\cdots\frac{2}{3}H^2=\frac{1}{2}R^2\Rightarrow H:R=\sqrt{3}:2$.

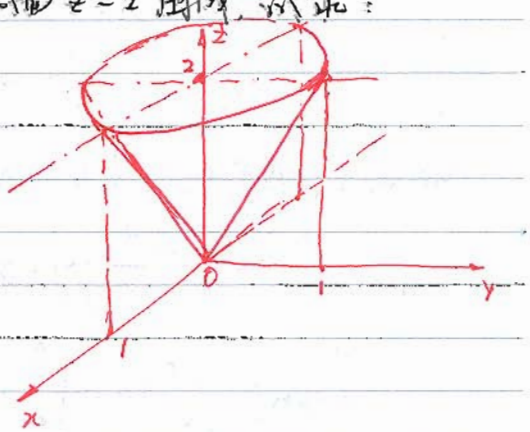
7. 一个质量为 $M$ 的均匀物体由曲面 $z=2\sqrt{x^2+y^2}$ 和平面 $z=2$ 围成，试求：

(1) 质心坐标；

(2) 关于中心轴的转动惯量；

(3) 关于质直径的转动惯量；

(4) 对坐标原点处质量为 $m$ 的质点的引力.




解：(1) 由对称性可知，质心 $(0,0,\bar z)$.

设 $\begin{cases}x=r\cos\theta\\ y=r\sin\theta\\ z=z\end{cases}$. $d\sigma=r\,dr\,d\theta\,dz$.

$$M=\iiint_\Omega\rho_0\,d\sigma=\rho_0\int_0^{2\pi}d\theta\int_0^{\frac{z}{2}}r\,dr\int_0^2 dz=2\pi\rho_0\int_0^2\frac{1}{2}\cdot\frac{z^2}{4}\,dz=\frac{\pi\rho_0}{4}\int_0^2 z^2\,dz=\frac{2\pi\rho_0}{3}.$$

$$\bar{z}=\frac{1}{M}\iiint_{\Omega} z\rho_0\,d\sigma=\frac{\rho_0}{M}\int_0^{2\pi}d\theta\int_0^{\frac{z}{2}}r\,dr\int_0^{2}z\,dz=\frac{2\pi\rho_0}{M}\int_0^{2}z\cdot\frac{1}{8}z^2dz$$

$$=\frac{\pi\rho_0}{M}\cdot\frac{1}{4}\int_0^{2}z^3dz=\frac{\pi\rho_0}{16M}\cdot z^4\Big|_0^2=\frac{\pi\rho_0}{M}=\pi\rho_0\cdot\frac{3}{2\pi\rho_0}=\frac{3}{2}.$$

$\therefore$ 质心坐标 $(0,0,\frac{3}{2})$.

(2) 关于中心轴之轴的转动惯量:

$$I_z=\iiint_{\Omega}(x^2+y^2)\rho_0\,d\sigma=\rho_0\int_0^{2\pi}d\theta\int_0^{\frac{z}{2}}r\,dr\int_0^{2}r^2dz$$

$$=2\pi\rho_0\int_0^{\frac{2}{2}}r^3dr\cdot\int_0^{2}dz=2\pi\rho_0\int_0^{2}\cdot\frac{1}{4}\cdot\frac{z^4}{16}dz=\frac{\pi\rho_0}{32}\int_0^{2}z^4dz$$

$$=\frac{\pi\rho_0}{32}\cdot\frac{1}{5}z^5\Big|_0^2=\frac{\pi\rho_0}{32}\cdot\frac{1}{5}\cdot 32=\frac{\pi\rho_0}{5}.$$

(3) 关于直线 $\begin{cases}x=0\\z=2\end{cases}$ 的转动惯量:

$$I_1=\iiint_{\Omega}\left[x^2+(z-2)^2\right]\rho_0\,d\sigma=\rho_0\int_0^{2\pi}d\theta\int_0^{\frac{z}{2}}r\,dr\int_0^{2}\left[r^2\cos^2\theta+(z-2)^2\right]dz$$

$$=\rho_0\int_0^{2\pi}d\theta\int_0^{\frac{z}{2}}\left[r^3\cos^2\theta+(z-2)^2\cdot r\right]dr\cdot\int_0^{2}dz$$

$$=\rho_0\int_0^{2\pi}d\theta\cdot\int_0^{2}\left[\frac{1}{4}\cdot\left(\frac{z}{2}\right)^4\cos^2\theta+\frac{1}{2}(z-2)^2\cdot\left(\frac{z}{2}\right)^2\right]dz$$

$$=\rho_0\int_0^{2\pi}d\theta\cdot\int_0^{2}\left[\frac{1}{64}z^4\cos^2\theta+\frac{1}{8}(z^4-4z^3+4z^2)\right]dz$$

$$=\rho_0\int_0^{2\pi}d\theta\cdot\left[\frac{1}{64}\cos^2\theta\cdot\frac{1}{5}z^5+\frac{1}{8}\left(\frac{1}{5}z^5-z^4+\frac{4}{3}z^3\right)\right]\Big|_0^2$$

$$=\rho_0\int_0^{2\pi}d\theta\cdot\left(\frac{1}{10}\cos^2\theta+\frac{2}{15}\right)=\rho_0\cdot\frac{11}{60}\cdot 2\pi=\frac{11}{30}\pi\rho_0.$$

(4) 对中心原点处质量为 $m$ 质点的引力:

$$F_z=\iiint_{\Omega}Gm\cdot\frac{\rho_0\,d\sigma}{(x^2+y^2+z^2)}\cdot\frac{z}{\sqrt{x^2+y^2+z^2}}=Gm\rho_0\iiint_{\Omega}\frac{z\,d\sigma}{(x^2+y^2+z^2)^{3/2}}$$

$$=Gm\rho_0\int_0^{2\pi}d\theta\cdot\int_0^{\frac{z}{2}}r\,dr\cdot\int_0^{2}\frac{z}{(r^2+z^2)^{3/2}}dz=2\pi Gm\rho_0\int_0^{2}dz\cdot\int_0^{\frac{z}{2}}\frac{r\,z\,dr}{(r^2+z^2)^{3/2}}$$

$$= 2\pi G m \rho_0 \int_0^2 dz \cdot z \cdot \left(-\frac{1}{\sqrt{r^2+z^2}}\right)\Big|_0^{\frac{z}{2}} = 2\pi G m \rho_0 \int_0^2 z\left(\frac{1}{z} - \frac{2}{z\sqrt{5}}\right) dz$$

$$= 4\pi G m \rho_0 \left(1-\frac{2}{\sqrt{5}}\right) = 4\left(1-\frac{2}{\sqrt{5}}\right)\pi G m \rho_0.$$

## §4. 广义重积分

习题. P.107.

1. 讨论下列广义积分的敛散性:

(1) $\iint\limits_{x^2+y^2\le 1} \frac{dx\,dy}{(1-x^2-y^2)^p}$.

解: $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$ $d\sigma = dx\,dy = r\,dr\,d\theta$.

$D = \{(x,y) \mid x^2+y^2 \le 1\} = \{(r,\theta) \mid 0\le r\le 1,\ 0\le\theta\le 2\pi\}$.

$$\iint\limits_{x^2+y^2\le 1} \frac{dx\,dy}{(1-x^2-y^2)^p} = \int_0^{2\pi} d\theta \int_0^1 \frac{r\,dr}{(1-r^2)^p} = -\pi \int_0^1 \frac{d(1-r^2)}{(1-r^2)^p} \quad (\text{令 } 1-r^2 = t)$$

$$= -\pi \int_0^1 \frac{dt}{t^p} = -\pi \lim_{\varepsilon\to 0^+} \int_\varepsilon^1 \frac{dt}{t^p} = \pi \lim_{\varepsilon\to 0^+} \int_\varepsilon^1 t^{-p}\,dt$$

$$= \begin{cases} +\infty & p\ge 1 \\ \frac{\pi}{1-p} & p<1. \end{cases} \quad \therefore \begin{array}{l} \text{当 } p\ge 1 \text{ 时, 原积分发散.} \\ p<1 \text{ 时, 原积分收敛于 } \frac{\pi}{1-p}. \end{array}$$

(2) $\iiint\limits_{x^2+y^2+z^2\le 1} \frac{dx\,dy\,dz}{(x^2+y^2+z^2)^p}$.

解: $\begin{cases} x = r\sin\varphi\cos\theta \\ y = r\sin\varphi\sin\theta \\ z = r\cos\varphi \end{cases}$ $d\sigma = dx\,dy\,dz = r^2\sin\varphi\,d\varphi\,d\theta\,dr$.

$D = \{(x,y,z) \mid x^2+y^2+z^2\le 1\} = \{(r,\theta,\varphi) \mid 0\le r\le 1,\ 0\le\theta\le 2\pi,\ 0\le\varphi\le\pi\}$.

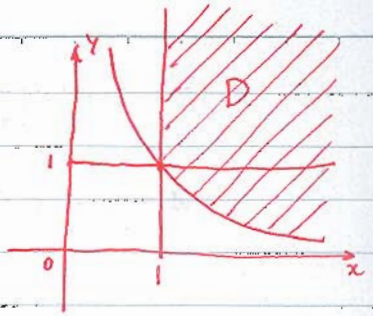
$$\iiint\limits_{x^2+y^2+z^2\le 1} \frac{dx\,dy\,dz}{(x^2+y^2+z^2)^p} = \int_0^{2\pi} d\theta \int_0^\pi d\varphi \int_0^1 \frac{r^2\sin\varphi\,dr}{r^{2p}} = 2\pi \int_0^\pi \sin\varphi\,d\varphi \int_0^1 \frac{dr}{r^{2(p-1)}}$$

$$= 4\pi \int_0^1 \frac{dr}{r^{2(p-1)}} = 4\pi \lim_{\varepsilon\to 0^+} \int_\varepsilon^1 r^{-2(p-1)}\,dr$$

$$= \lim_{\varepsilon\to 0^+} \begin{cases} \frac{4\pi(1-\varepsilon^{3-2p})}{3-2p} & p\ne\frac{3}{2} \\ -4\pi\ln\varepsilon & p=\frac{3}{2} \end{cases} = \begin{cases} +\infty & p\ge\frac{3}{2} \\ \frac{4\pi}{3-2p} & p<\frac{3}{2}. \end{cases}$$

$\therefore$ 当 $p\ge\frac{3}{2}$ 时, 原积分发散. 当 $p<\frac{3}{2}$ 时, 原积分收敛于 $\frac{4\pi}{3-2p}$.

2. 计算下列广义积分：




(1) $\iint_D \frac{dxdy}{x^p y^q}$，其中 $D=\{(x,y)\mid xy\geq 1,\ x\geq 1$ 且 $p>q>1\}$

解：$\iint_D \frac{dxdy}{x^p y^q}=\int_0^1 dy\int_{\frac{1}{y}}^{+\infty}\frac{dx}{x^p y^q}+\int_1^{+\infty}dy\int_1^{+\infty}\frac{dx}{x^p y^q}$

$=\int_0^1 \frac{dy}{y^q}\int_{\frac{1}{y}}^{+\infty}\frac{dx}{x^p}+\int_1^{+\infty}\frac{dy}{y^q}\int_1^{+\infty}\frac{dx}{x^p}$

$=\int_0^1 \frac{dy}{y^q}\cdot\frac{1}{1-p}x^{1-p}\Big|_{\frac{1}{y}}^{+\infty}+\frac{1}{1-q}y^{1-q}\Big|_1^{+\infty}\cdot\frac{1}{1-p}x^{1-p}\Big|_1^{+\infty}$

$=\frac{1}{p-1}\cdot\int_0^1\frac{dy}{y^{1-p+q}}+\frac{1}{q-1}\cdot\frac{1}{p-1}=\frac{1}{(p-1)(p-q)}y^{p-q}\Big|_0^1+\frac{1}{(p-1)(q-1)}$

$=\frac{1}{(p-1)(p-q)}+\frac{1}{(p-1)(q-1)}=\frac{1}{(p-q)(q-1)}$

或：$\iint_D \frac{dxdy}{x^p y^q}=\int_1^{+\infty}\frac{dx}{x^p}\int_{\frac{1}{x}}^{+\infty}\frac{dy}{y^q}=\int_1^{+\infty}\frac{dx}{x^p}\cdot\frac{1}{q-1}\cdot\frac{1}{x^{1-q}}$

$=\int_1^{+\infty}\frac{dx}{(q-1)x^{p-q+1}}=\frac{1}{q-1}\cdot x^{-(p-q)}\Big|_1^{+\infty}\cdot\frac{-1}{p-q}=\frac{1}{(p-q)(q-1)}$

(2) $\iint_{\frac{x^2}{a^2}+\frac{y^2}{b^2}\geq 1} e^{-\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)}dxdy$

解：$\begin{cases}x=ar\cos\theta\\ y=br\sin\theta\end{cases}$ $d\sigma=dxdy=abr\,dr\,d\theta$. $D=\left\{(x,y)\mid \frac{x^2}{a^2}+\frac{y^2}{b^2}\geq 1\right\}$

$=\{(r,\theta)\mid r\geq 1,\ 0\leq\theta\leq 2\pi\}$

$\iint_{\frac{x^2}{a^2}+\frac{y^2}{b^2}\geq 1} e^{-\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)}dxdy=\int_0^{2\pi}ab\,d\theta\int_1^{+\infty}e^{-r^2}r\,dr=\pi ab\int_1^{+\infty}e^{-r^2}dr^2$

$=\pi ab\,(-e^{-r^2})\Big|_1^{+\infty}=\frac{\pi ab}{e}$

(3) $\iiint_{R^3} e^{-(x^2+y^2+z^2)}dxdydz$

解：$\begin{cases}x=r\sin\varphi\cos\theta\\ y=r\sin\varphi\sin\theta\\ z=r\cos\varphi\end{cases}$ $d\sigma=dxdydz=r^2\sin\varphi\,dr\,d\varphi\,d\theta$.

$\Omega=R^3=\left\{(r,\varphi,\theta)\mid \begin{array}{l}0\leq r\leq+\infty\\ 0\leq\theta\leq 2\pi\\ 0\leq\varphi\leq\pi\end{array}\right\}$.

$$\iiint_{R^3} e^{-(x^2+y^2+z^2)}\,dx\,dy\,dz = \int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_0^{+\infty} e^{-r^2}\cdot r^2\sin\varphi\,dr$$

$$= \int_0^{2\pi} d\theta \int_0^{\pi} \sin\varphi\,d\varphi \cdot \int_0^{+\infty} r^2 e^{-r^2}\,dr$$

$$= 2\pi\cdot 2\cdot\frac{1}{2}\int_0^{+\infty} r e^{-r^2}\,dr^2 = -2\pi\int_0^{+\infty} r\,d e^{-r^2}$$

$$= -2\pi\, r e^{-r^2}\Big|_0^{+\infty} + 2\pi\int_0^{+\infty} e^{-r^2}\,dr = 2\pi\cdot\int_0^{+\infty} e^{-r^2}\,dr$$

$$= 2\pi\cdot\frac{\sqrt{\pi}}{2} = \pi\sqrt{\pi}. \quad \left(\text{利用 Possion 积分: } \int_0^{+\infty} e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}\right).$$

或: $$\iiint_{R^3} e^{-(x^2+y^2+z^2)}\,dx\,dy\,dz = \int_{-\infty}^{+\infty} e^{-x^2}dx\cdot\int_{-\infty}^{+\infty} e^{-y^2}dy\cdot\int_{-\infty}^{+\infty} e^{-z^2}dz$$

$$= \left(\int_{-\infty}^{+\infty} e^{-x^2}dx\right)^3 = \left(2\cdot\int_0^{+\infty} e^{-x^2}dx\right)^3 = \left(2\cdot\frac{\sqrt{\pi}}{2}\right)^3 = \pi\sqrt{\pi}.$$

3. 判别广义积分 $I = \iint_{R^2} \frac{dx\,dy}{(1+x^2)(1+y^2)}$ 是绝对收敛、条件收敛, 或发散.

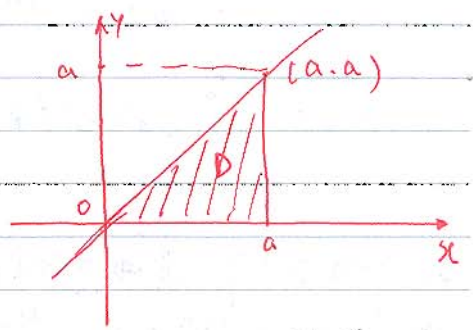
解: $$I = \iint_{R^2} \frac{dx\,dy}{(1+x^2)(1+y^2)} = \int_{-\infty}^{+\infty}\frac{dx}{1+x^2}\cdot\int_{-\infty}^{+\infty}\frac{dy}{1+y^2}$$

$$\because \int_{-\infty}^{+\infty}\frac{dx}{1+x^2} = \arctan x\Big|_{-\infty}^{+\infty} = \frac{\pi}{2}-\left(-\frac{\pi}{2}\right) = \pi.$$

$\therefore$ $I$ 收敛. 且有: $I = \left(\int_{-\infty}^{+\infty}\frac{dx}{1+x^2}\right)^2 = \pi^2$.

4. 设一元函数 $f$ 在 $[0,a]$ 上连续, 证明:

$$\iint_{0\le y\le x\le a} \frac{f(y)}{\sqrt{(a-x)(x-y)}}\,dx\,dy = \pi\int_0^a f(x)\,dx.$$




证明: $$\iint_{0\le y\le x\le a} \frac{f(y)}{\sqrt{(a-x)(x-y)}}\,dx\,dy = \int_0^a dy\int_y^a \frac{f(y)\,dx}{\sqrt{(a-x)(x-y)}}$$

$$= \int_0^a f(y)\,dy\cdot\int_y^a \frac{dx}{\sqrt{(a-x)(x-y)}} \quad \xleftarrow{\Sigma} \quad x = y\cos^2 t + a\sin^2 t. \quad \text{则: } \begin{matrix} x: y\to a, \\ t: 0\to\frac{\pi}{2}. \end{matrix}$$

$$\sqrt{(a-x)(x-y)} = \sqrt{(a-y\cos^2 t - a\sin^2 t)(y\cos^2 t + a\sin^2 t - y)}$$

$$= \sqrt{(a-y)^2\sin^2 t\cos^2 t} = (a-y)\sin t\cos t. \quad \left(\begin{matrix} 0\le y\le a \\ 0\le t\le\frac{\pi}{2}\end{matrix}\right).$$

$$= (a-y)\cdot\frac{1}{2}\sin 2t.$$

$dx = [2y\cos t(-\sin t) + 2a\sin t\cos t]dt = 2(a-y)\sin t\cos t\,dt$

$= (a-y)\sin 2t\,dt.$

$\therefore \iint_{0\le y\le x\le a} \frac{f(y)}{\sqrt{(a-x)(x-y)}}\,dx\,dy = \int_0^a f(y)\,dy \cdot \int_0^{\frac{\pi}{2}} \frac{(a-y)\sin 2t\,dt}{(a-y)\cdot\frac{1}{2}\sin 2t}$

$= 2\int_0^{\frac{\pi}{2}} dt \cdot \int_0^a f(y)\,dy = \pi\int_0^a f(x)\,dx.$

## §5. 两类曲线积分

习题 P116.

1. 求下列曲线的弧长.

(1) $x = 3t$, $y = 3t^2$, $z = 2t^3$ 自 $O(0,0,0)$ 到 $A(3,3,2)$;

解: $ds = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}\,dt = \sqrt{3^2 + (6t)^2 + (6t^2)^2}\,dt = \sqrt{9 + 36t^2 + 36t^4}\,dt$

从 $O(0,0,0)$ 到 $A(3,3,2)$, 即 $t: 0 \to 1$.

$\therefore L = \int_L ds = \int_0^1 \sqrt{9 + 36t^2 + 36t^4}\,dt = 3\int_0^1 (1 + 2t^2)\,dt$

$= 3\left(t + \frac{2}{3}t^3\right)\Big|_0^1 = 5.$

(2) $x = e^{-t}\cos t$, $y = e^{-t}\sin t$, $z = e^{-t}$ $(0 \le t < +\infty)$

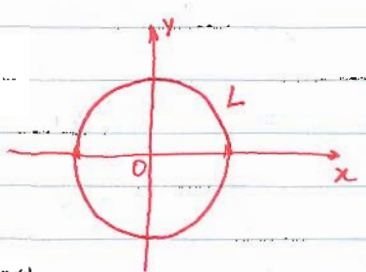
解: $L = \int_0^{+\infty} \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}\,dt = \int_0^{+\infty} \sqrt{(-e^{-t}\cos t - e^{-t}\sin t)^2 + (-e^{-t}\sin t + e^{-t}\cos t)^2 + (-e^{-t})^2}\,dt$

$= \int_0^{+\infty} \sqrt{e^{-2t}(1+\sin 2t) + e^{-2t}(1-\sin 2t) + e^{-2t}}\,dt$

$= \int_0^{+\infty} \sqrt{3}\cdot e^{-t}\,dt = -\sqrt{3}e^{-t}\Big|_0^{+\infty} = \sqrt{3}.$

2. 计算下列第一类曲线积分:

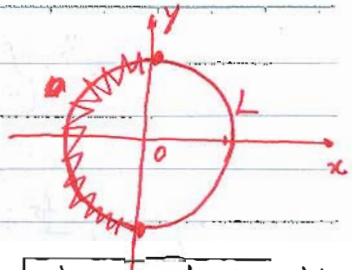
(1) $\int_L (x^2+y^2)^n\,ds$, 其中 $L$ 是圆心在原点, 半径为 $R$ 的圆周;




解: $L$: $\begin{cases} x = R\cos\theta \\ y = R\sin\theta \end{cases}$ $0 \le \theta \le 2\pi$

$\int_L (x^2+y^2)^n\,ds = \int_0^{2\pi} R^{2n}\cdot\sqrt{R^2\sin^2\theta + R^2\cos^2\theta}\,d\theta = \int_0^{2\pi} R^{2n+1}\,d\theta = 2\pi R^{2n+1}.$

(2) $\int_L |y|\,ds$，其中 $L$ 是右半圆周 $x=a\cos t$，$y=a\sin t$，$-\frac{\pi}{2}\le t\le\frac{\pi}{2}$.



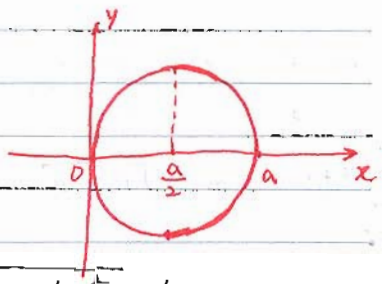

解：$L:\begin{cases}x=a\cos t\\ y=a\sin t\end{cases}$ $-\frac{\pi}{2}\le t\le\frac{\pi}{2}$.

$$\int_L |y|\,ds=\int_{-\frac{\pi}{2}}^{0}(-a\sin t)\sqrt{a^2\sin^2t+a^2\cos^2t}\,dt+\int_0^{\frac{\pi}{2}}(a\sin t)\sqrt{a^2\sin^2t+a^2\cos^2t}\,dt$$

$$=\int_0^{-\frac{\pi}{2}}(a^2\sin t)\,dt+\int_0^{\frac{\pi}{2}}a^2\sin t\,dt$$

$$=a^2(-\cos t)\Big|_0^{-\frac{\pi}{2}}+a^2(-\cos t)\Big|_0^{\frac{\pi}{2}}=2a^2.$$

(3) $\int_L\sqrt{x^2+y^2}\,ds$，其中 $L$ 是圆周 $x^2+y^2=ax$ $(a>0)$.




解：$L:\begin{cases}x=\frac{a}{2}(1+\cos\theta)\\ y=\frac{a}{2}\sin\theta\end{cases}$ $0\le\theta\le2\pi$ ($\theta$ 为参数)

$$\int_L\sqrt{x^2+y^2}\,ds=\int_0^{2\pi}a\cdot\sqrt{\left(\frac{1+\cos\theta}{2}\right)^2+\left(\frac{\sin\theta}{2}\right)^2}\cdot\sqrt{x'(\theta)^2+y'(\theta)^2}\,d\theta$$

$$=\frac{a}{2}\int_0^{2\pi}\sqrt{2+2\cos\theta}\cdot\frac{a}{2}\,d\theta=\int_0^{2\pi}\frac{a^2}{2}\sqrt{\frac{1+\cos\theta}{2}}\,d\theta$$

$$=\int_0^{2\pi}a^2\cdot\left|\cos\frac{\theta}{2}\right|d\left(\frac{\theta}{2}\right)=a^2\int_0^{\pi}\left|\cos\alpha\right|d\alpha$$

$$=a^2\int_0^{\frac{\pi}{2}}\cos\alpha\,d\alpha-a^2\int_{\frac{\pi}{2}}^{\pi}\cos\alpha\,d\alpha=2a^2.$$

或：$L:\ r=a\cos\theta$. $-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$ ($\theta$ 为极角)

(3)

$$\sqrt{x^2+y^2}=r=a\cos\theta.\qquad ds=\sqrt{r^2+r'(\theta)^2}\,d\theta=\sqrt{a^2\cos^2\theta+a^2\sin^2\theta}\,d\theta=a\,d\theta.$$

$$\int_L\sqrt{x^2+y^2}\,ds=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}a\cos\theta\cdot a\,d\theta=a^2\sin\theta\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}=2a^2.$$

(4) $\int_L(x^{4/3}+y^{4/3})\,ds$，其中 $L$ 为星形线 $x=a\cos^3t$，$y=a\sin^3t$ $(0\le t\le\frac{\pi}{2})$.

解：$\int_L(x^{4/3}+y^{4/3})\,ds=\int_0^{\frac{\pi}{2}}(a^{4/3}\cos^4t+a^{4/3}\sin^4t)\sqrt{[3a\cos^2t(-\sin t)]^2+[3a\sin^2t\cos t]^2}\,dt$

$$=a^{4/3}\int_0^{\frac{\pi}{2}}(\cos^4t+\sin^4t)\cdot3a\sin t\cos t\,dt=3a^{7/3}\int_0^{\frac{\pi}{2}}(\sin t\cos^5t+\sin^5t\cos t)\,dt$$

$$= 3a^{2/3}\cdot\left[\left(-\frac{1}{6}\cos^6 t\right)\Big|_0^{\frac{\pi}{2}} + \left(\frac{1}{6}\sin^6 t\right)\Big|_0^{\frac{\pi}{2}}\right] = 3a^{2/3}\cdot\frac{1}{3} = a^{2/3}.$$

(5) $\int_L xyz\,ds$, 其中曲线 $L$ 的参数方程为: $x=t,\ y=\frac{2\sqrt{2}}{3}t^{3/2},\ z=\frac{1}{2}t^2,\ (0\le t\le 1)$.

解: $ds = \sqrt{x'(t)^2+y'(t)^2+z'(t)^2}\,dt = \sqrt{1+2t+t^2}\,dt = |t+1|\,dt.$

$$\int_L xyz\,ds = \int_0^1 t\cdot\frac{2\sqrt{2}}{3}t^{3/2}\cdot\frac{1}{2}t^2\cdot(1+t)\,dt = \frac{\sqrt{2}}{3}\int_0^1\left(t^{\frac{9}{2}}+t^{\frac{11}{2}}\right)dt$$

$$= \frac{\sqrt{2}}{3}\left(\frac{2}{11}t^{\frac{11}{2}}+\frac{2}{13}t^{\frac{13}{2}}\right)\Big|_0^1 = \frac{16\sqrt{2}}{143}$$

(6) $\int_L x\sqrt{x^2-y^2}\,ds$, 其中 $L$ 是双纽线 $r^2=a^2\cos 2\theta$ 的右半支: $\left(-\frac{\pi}{4}\le\theta\le\frac{\pi}{4}\right)$.

解: $r = a\sqrt{\cos 2\theta}$. 化成极坐标: $\begin{cases} x = r\cos\theta = a\cos\theta\sqrt{\cos 2\theta} \\ y = r\sin\theta = a\sin\theta\sqrt{\cos 2\theta} \end{cases}$ ($\theta$ 为参数)

$$x'(\theta) = \frac{a(\sin\theta\cos 2\theta+\cos\theta\sin 2\theta)}{\sqrt{\cos 2\theta}} \qquad y'(\theta) = \frac{a(\cos\theta\cos 2\theta-\sin\theta\sin 2\theta)}{\sqrt{\cos 2\theta}}$$

$$x'(\theta)^2+y'(\theta)^2 = \frac{a^2}{\cos 2\theta}(\cos^2 2\theta+\sin^2 2\theta) = \frac{a^2}{\cos 2\theta}.$$

$$ds = \sqrt{x'(\theta)^2+y'(\theta)^2}\,d\theta = \frac{a}{\sqrt{\cos 2\theta}}\,d\theta.$$

$$\int_L x\sqrt{x^2-y^2}\,ds = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} a\cos\theta\sqrt{\cos 2\theta}\cdot\sqrt{a^2\cos 2\theta\cdot\cos 2\theta}\cdot\frac{a}{\sqrt{\cos 2\theta}}\,d\theta$$

$$= a^3\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\cos\theta\cos 2\theta\,d\theta = a^3\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}(1-2\sin^2\theta)\,d\sin\theta = \frac{2\sqrt{2}}{3}a^3.$$

3. 曲线 $y=\ln x$ 的线密度 $\rho(x,y)=x^2$, 试求曲线从 $x=\sqrt{3}$ 到 $x=\sqrt{15}$ 间的质量.

解: $M = \int_L \rho(x,y)\,ds = \int_{\sqrt{3}}^{\sqrt{15}} x^2\cdot\sqrt{1+y'(x)^2}\,dx = \int_{\sqrt{3}}^{\sqrt{15}} x^2\cdot\sqrt{1+\frac{1}{x^2}}\,dx = \int_{\sqrt{3}}^{\sqrt{15}} x\sqrt{1+x^2}\,dx$

$$= \frac{1}{2}\int_{\sqrt{3}}^{\sqrt{15}}(1+x^2)^{\frac{1}{2}}\,d(1+x^2) = \frac{1}{3}(1+x^2)^{\frac{3}{2}}\Big|_{\sqrt{3}}^{\sqrt{15}} = \frac{56}{3}.$$

4. 求摆线 $x=a(t-\sin t),\ y=a(1-\cos t)\ (0\le t\le\pi)$ 的质心, 其中密度为常值.

解: $M = \int_L \rho_0\,ds = \rho_0\int_0^{\pi}\sqrt{x'(t)^2+y'(t)^2}\,dt = \rho_0 a\int_0^{\pi}\sqrt{(1-\cos t)^2+(\sin t)^2}\,dt$

$= \rho_0 a \int_0^{\pi} \sqrt{2-2\cos t}\, dt = 4\rho_0 a \int_0^{\pi} \sin\frac{t}{2}\, d\frac{t}{2} = 4\rho_0 a$. 设质心 $(\bar{x}, \bar{y})$.

则: $\bar{x} = \frac{1}{M}\int_L x\rho_0 ds = \frac{1}{M}\int_0^{\pi} a(t-\sin t)\cdot \rho_0 a \sqrt{2-2\cos t}\, dt = -a\int_0^{\pi}(t-\sin t)\, d(\cos\frac{t}{2})$

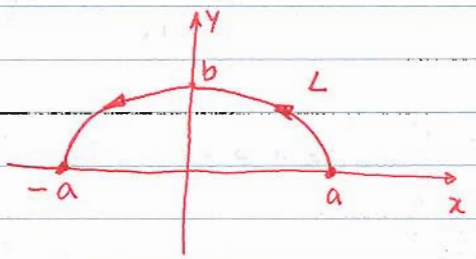
$= -a(t-\sin t)\cos\frac{t}{2}\Big|_0^{\pi} + a\int_0^{\pi}\cos\frac{t}{2}\cdot(1-\cos t)\, dt = 4a\int_0^{\pi}\sin^2\frac{t}{2}\cos\frac{t}{2}\, d\frac{t}{2} = \frac{4}{3}a$.

$\bar{y} = \frac{1}{M}\int_L y\rho_0 ds = \frac{1}{M}\int_0^{\pi} a(1-\cos t)\cdot\rho_0 a\sqrt{2-2\cos t}\, dt = a\int_0^{\pi}\sin^3\frac{t}{2}\, dt = \frac{4}{3}a$.

∴ 摆线的质心为 $(\frac{4}{3}a, \frac{4}{3}a)$.

5. 计算第二类曲线积分

(1) $\int_L y^2 dx + x^2 dy$, 其中 $L$ 是上半椭圆 $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $y\geq 0$, 方向从 $(-a,0)$ 到 $(-a,0)$.




解: $L$: $\begin{cases} x = a\cos t \\ y = b\sin t \end{cases}$ $t: 0 \to \pi$.

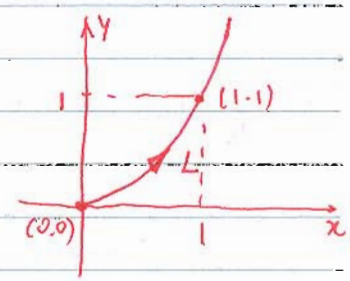
$\int_L y^2 dx + x^2 dy = \int_0^{\pi}\left[ b^2\sin^2 t\,(-a\sin t)\, dt \right.$

$\left. + a^2\cos^2 t\cdot(b\cos t)\, dt \right]$

$= ab\int_0^{\pi}\left[ -b(1-\cos^2 t)\, d\cos t + a(1-\sin^2 t)\, d\sin t \right]$

$= ab\left(b\cos t - \frac{1}{3}\cos^3 t\right)\Big|_0^{\pi} + ab\left(a\sin t - \frac{1}{3}\sin^3 t\right)\Big|_0^{\pi} = -\frac{4}{3}ab^2$.

(2) $\int_L (x-y)^2 dx + 2xy\, dy$, 其中 $L$ 为曲线 $y = x^5$, 方向从 $(0,0)$ 到 $(1,1)$.




解: $L$: $y = x^5$, $x: 0 \to 1$.

$\int_L (x-y)^2 dx + 2xy\, dy = \int_0^1 (x-x^5)^2 dx + 2x\cdot x^5\cdot 5x^4\, dx$

$= \int_0^1 (x^2 - 2x^6 + 11x^{10})\, dx = \frac{1}{3} - \frac{2}{7} + 1 = \frac{22}{21}$.

(3) $\int_L (y^2-z^2)dx + 2yz\, dy - x^2 dz$, 其中 $L$ 为 $x=t$, $y=t^2$, $z=t^3$, $t: 0 \to 1$.

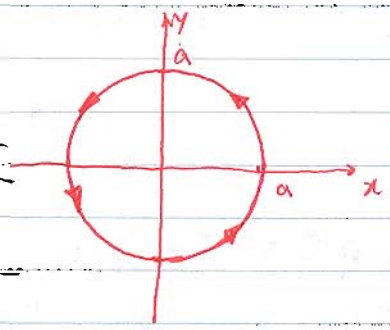
解: $L$: $\begin{cases} x = t \\ y = t^2 \\ z = t^3 \end{cases}$ $t: 0 \to 1$.

$$\int_L (y^2-z^2)dx+2yz\,dy-x^2dz=\int_0^1 (t^4-t^6)\cdot dt+2t^2\cdot t^3\cdot 2t\,dt-t^2\cdot 3t^2dt$$

$$=\int_0^1 (t^4-t^6+4t^6-3t^4)dt=\int_0^1 (3t^6-2t^4)dt=\frac{3}{7}-\frac{2}{5}=\frac{1}{35}.$$

(4) $\oint_L \frac{(x+y)\,dx-(x-y)\,dy}{x^2+y^2}$，其中 $L$ 为圆周 $x^2+y^2=a^2$，方向为逆时针方向.

解：$L:\begin{cases}x=a\cos t\\ y=a\sin t\end{cases}$ $t:0\to 2\pi$.




$$\oint_L \frac{(x+y)dx-(x-y)dy}{x^2+y^2}=\int_0^{2\pi}\frac{a(\cos t+\sin t)\cdot(-a\sin t)dt}{a^2}-\frac{a(\cos t-\sin t)\cdot a\cos t\,dt}{a^2}$$

$$=\frac{a}{a}\int_0^{2\pi}(-\cos t\sin t-\sin^2 t-\cos^2 t+\sin t\cos t)\,dt=\int_0^{2\pi}(-1)\,dt=-2\pi.$$

(5) $\oint_L x^2dx+z\,dy-y\,dz$，其中 $L$ 为 $x=k\theta$，$y=a\cos\theta$，$z=a\sin\theta$，$\theta:0\to 2\pi$.

解：$\oint_L x^2dx+z\,dy-y\,dz=\int_0^{2\pi}k^2\theta^2\cdot k\,d\theta+a\sin\theta\cdot(-a\sin\theta)d\theta-a\cos\theta\cdot a\cos\theta\,d\theta$

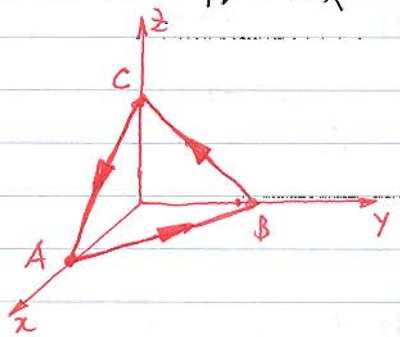
$$=\int_0^{2\pi}(k^3\theta^2-a^2\sin^2\theta-a^2\cos^2\theta)\,d\theta=\int_0^{2\pi}(k^3\theta^2-a^2)\,d\theta$$

$$=\left(\frac{k^3}{3}\theta^3-a^2\theta\right)\Big|_0^{2\pi}=\frac{k^3(2\pi)^3}{3}-a^2\cdot 2\pi.$$

(6) $\oint_L dx-dy+y\,dz$，其中 $L$ 为取逆时针方向的闭折线 $ABCA$，这里的 $A,B,C$ 依次为点 $(1,0,0)$，$(0,1,0)$，$(0,0,1)$.

解：$AB:\begin{cases}x+y=1\\ z=0\end{cases}$ $x:1\to 0$，$y=1-x$.　$BC:\begin{cases}y+z=1\\ x=0\end{cases}$ $y:1\to 0$，$z=1-y$.

$CA:\begin{cases}x+z=1\\ y=0\end{cases}$ $z:1\to 0$，$x=1-z$.

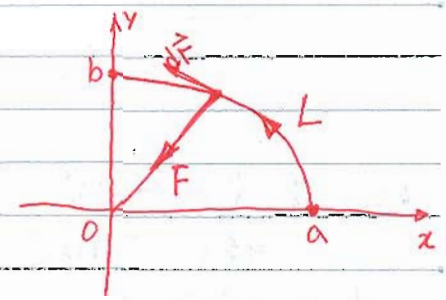



$$\oint_L dx-dy+y\,dz=\left(\int_{\widehat{AB}}+\int_{\widehat{BC}}+\int_{\widehat{CA}}\right)dx-dy+y\,dz$$

$$=\int_1^0(dx+dx)+\int_1^0(-dy-y\,dy)+\int_1^0(-dz)$$

$$= -2\int_0^1 dx + \int_0^1 (-1+y)dy + \int_0^1 -dz = -2 + (1+\frac{1}{2}) + 1 = \frac{1}{2}.$$

6. 弹力的方向指向坐标原点，力的大小和质点与原点的距离成正比，设此质点依逆时针方向描绘出椭圆 $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ 在第二象限的部分，试求弹力所做的功.




解: $W=\int_L \vec{F}\cdot d\vec{r}$. $L:\begin{cases}x=a\cos\theta \\ y=a\sin\theta\end{cases}$ $\theta: 0\to\frac{\pi}{2}$.

($\theta$ 为参数，并非极角)

$$\vec{F} = k\sqrt{x^2+y^2}\left(-\frac{x}{\sqrt{x^2+y^2}}\vec{i}-\frac{y}{\sqrt{x^2+y^2}}\vec{j}\right)$$

$$= -k(a\cos\theta\vec{i}+b\sin\theta\vec{j}). \qquad d\vec{r} = dx\vec{i}+dy\vec{j} = -a\sin\theta\,d\theta\,\vec{i}+b\cos\theta\,d\theta\,\vec{j}$$

$$\therefore W=\int_L \vec{F}\cdot d\vec{r} = \int_0^{\frac{\pi}{2}} -k(a\cos\theta\vec{i}+b\sin\theta\vec{j})\cdot(-a\sin\theta\,d\theta\,\vec{i}+b\cos\theta\,d\theta\,\vec{j})$$

$$= \int_0^{\frac{\pi}{2}} (ka^2\sin\theta\cos\theta - kb^2\sin\theta\cos\theta)\,d\theta = -k(a^2-b^2)\int_0^{\frac{\pi}{2}}\sin\theta\,d\sin\theta = \frac{k}{2}(a^2-b^2).$$

7. 有一力场，其力的大小与力的作用点到 $Oxy$ 平面的距离成反比，方向指向原点，试计算质点沿直线 $x=at$, $y=bt$, $z=ct$ ($c\neq 0$) 从点 $(a,b,c)$ 移动到点 $(2a,2b,2c)$ 时，该力场所作的功.

解: $\vec{F} = \frac{k}{|z|}\cdot\frac{-x\vec{i}-y\vec{j}-z\vec{k}}{\sqrt{x^2+y^2+z^2}} = \frac{k}{|c|\cdot t^2}\cdot\frac{at\vec{i}+bt\vec{j}+ct\vec{k}}{\sqrt{a^2+b^2+c^2}}$

$d\vec{r} = dx\vec{i}+dy\vec{j}+dz\vec{k} = a\,dt\,\vec{i}+b\,dt\,\vec{j}+c\,dt\,\vec{k}$. $L:\begin{cases}x=at\\ y=bt\\ z=ct\end{cases}$ $t: 1\to 2$.

$$W=\int_L \vec{F}\cdot d\vec{r} = \int_1^2 -\frac{k}{|c|\cdot t^2}\cdot\frac{1}{\sqrt{a^2+b^2+c^2}}(at\vec{i}+bt\vec{j}+ct\vec{k})\cdot(a\,dt\,\vec{i}+b\,dt\,\vec{j}+c\,dt\,\vec{k})$$

$$= \int_1^2 -\frac{k}{|c|\cdot t^2}\cdot\frac{(a^2t+b^2t+c^2t)}{\sqrt{a^2+b^2+c^2}}dt = -\frac{k}{|c|}\sqrt{a^2+b^2+c^2}\int_1^2\frac{1}{t}dt$$

$$= -\frac{k\ln 2}{|c|}\sqrt{a^2+b^2+c^2}$$

8. 设 $L$ 为曲线 $x=t$, $y=t^2$, $z=t^3$ 上相应于 $t: 0\to 1$ 的曲线弧，试把第二类曲线积分 $\int_L P\,dx+Q\,dy+R\,dz$ 化为第一类曲线积分.

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解: $d\vec{r} = \vec{\tau}\cdot ds = \dfrac{x'(t)\vec{i}+y'(t)\vec{j}+z'(t)\vec{k}}{\sqrt{x'(t)^2+y'(t)^2+z'(t)^2}}\cdot\sqrt{x'(t)^2+y'(t)^2+z'(t)^2}\,dt$

$= (x'(t)\vec{i}+y'(t)\vec{j}+z'(t)\vec{k})\,dt = (\vec{i}+2t\vec{j}+3t^2\vec{k})\,dt$

$\therefore \int_L P\,dx+Q\,dy+R\,dz = \int_L \vec{F}\cdot d\vec{r} = \int_L \vec{F}\cdot\vec{\tau}\,ds = \int_0^1 (P\vec{i}+Q\vec{j}+R\vec{k})\cdot(\vec{i}+2t\vec{j}+3t^2\vec{k})\,dt$

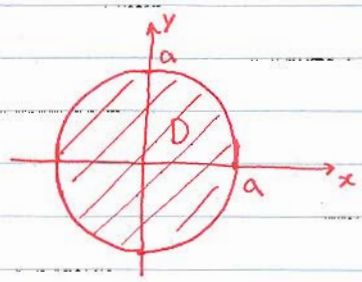
$= \int_0^1 (P+2tQ+3t^2R)\,dt.$

## §6 第一类曲面积分

### 习题 P.122.

1. 求下列曲面的面积:

(1) $z=axy$ 包含在圆柱面 $x^2+y^2=a^2$ $(a>0)$ 内的部分.




解: $S = \iint_{x^2+y^2\le a^2} \sqrt{1+f_x'^2+f_y'^2}\,dx\,dy$

$= \iint_{x^2+y^2\le a^2} \sqrt{1+a^2(x^2+y^2)}\,dx\,dy \xrightarrow{\text{令}} \begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases}$

$D = \{(x,y)\mid x^2+y^2\le a^2\} = \{(r,\theta)\mid 0\le r\le a,\ 0\le\theta\le 2\pi\}.$

$d\sigma = dx\,dy = r\,dr\,d\theta$

$= \int_0^{2\pi} d\theta\cdot\int_0^a \sqrt{1+a^2r^2}\cdot r\,dr$

$= 2\pi\cdot\frac{1}{2a^2}\int_0^a \sqrt{1+a^2r^2}\,d(1+a^2r^2) = \frac{\pi}{a^2}(1+a^2r^2)^{\frac{3}{2}}\Big|_0^a\cdot\frac{2}{3} = \frac{2\pi}{3a^2}\left[(1+a^4)^{\frac{3}{2}}-1\right].$

(2) 抛物面 $x^2+y^2=2az$ 包含在柱面 $(x^2+y^2)^2=2a^2xy$ $(a>0)$ 内的部分.

解: 抛物面 $x^2+y^2=2az$. $z_x' = \frac{x}{a}$. $z_y' = \frac{y}{a}$.

$D = \{(x,y)\mid (x^2+y^2)^2\le 2a^2xy\} = \{(r,\theta)\mid r^4\le 2a^2r^2\sin\theta\cos\theta\} = \left\{(r,\theta)\mid r^2\le a^2\sin 2\theta,\ 0\le\theta\le\frac{\pi}{2}\text{ 或 }\pi\le\theta\le\frac{5\pi}{2}\right\}$

$S = \iint_D \sqrt{1+z_x'^2+z_y'^2}\,dx\,dy = \iint_D \sqrt{1+\frac{1}{a^2}(x^2+y^2)}\,dx\,dy$

$= 2\int_0^{\frac{\pi}{2}} d\theta\int_0^{a\sqrt{\sin 2\theta}} \sqrt{1+\frac{r^2}{a^2}}\,r\,dr = a^2\int_0^{\frac{\pi}{2}} d\theta\cdot\int_0^{a\sqrt{\sin 2\theta}} \sqrt{1+\frac{r^2}{a^2}}\,d\left(1+\frac{r^2}{a^2}\right)$

$= a^2\int_0^{\frac{\pi}{2}} d\theta\cdot\frac{2}{3}\left(1+\frac{r^2}{a^2}\right)^{\frac{3}{2}}\Big|_0^{a\sqrt{\sin 2\theta}} = \frac{2}{3}a^2\int_0^{\frac{\pi}{2}}\left[(1+\sin 2\theta)^{\frac{3}{2}}-1\right]d\theta$

$= \frac{2a^2}{3}\int_0^{\frac{\pi}{2}}\left[(\sin\theta+\cos\theta)^3-1\right]d\theta = \frac{2a^2}{3}\int_0^{\frac{\pi}{2}}(\sin^3\theta+3\sin^2\theta\cos\theta+3\sin\theta\cos^2\theta+\cos^3\theta-1)\,d\theta$

$$= \frac{2a^2}{3}\left(\frac{1}{3}\cos^3\theta - \cos\theta + \sin^3\theta - \cos^3\theta + \sin\theta - \frac{1}{3}\sin^3\theta - \theta\right)\Big|_0^{\frac{\pi}{2}}$$

$$= \frac{2a^2}{3}\left(1+1-\frac{1}{3}-\frac{\pi}{2}-\frac{1}{3}+1+1\right) = \left(\frac{20}{9}-\frac{\pi}{3}\right)a^2.$$

(3) 求曲面 $x=(b+a\cos\varphi)\cos\theta$, $y=(b+a\cos\varphi)\sin\theta$, $z=a\sin\varphi$. $(0\le\varphi\le 2\pi,\ 0\le\theta\le 2\pi)$

其中 $0<a<b$.

解: $\vec{r}'_\varphi = x'_\varphi\vec{i} + y'_\varphi\vec{j} + z'_\varphi\vec{k} = -a\sin\varphi\cos\theta\,\vec{i} - a\sin\varphi\sin\theta\,\vec{j} + a\cos\varphi\,\vec{k}$

$\vec{r}'_\theta = x'_\theta\vec{i} + y'_\theta\vec{j} + z'_\theta\vec{k} = -(b+a\cos\varphi)\sin\theta\,\vec{i} + (b+a\cos\varphi)\cos\theta\,\vec{j} + 0\cdot\vec{k}$

$$\vec{r}'_\varphi \times \vec{r}'_\theta = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -a\sin\varphi\cos\theta & -a\sin\varphi\sin\theta & a\cos\varphi \\ -(b+a\cos\varphi)\sin\theta & (b+a\cos\varphi)\cos\theta & 0 \end{vmatrix}$$

$$= a(b+a\cos\varphi)\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -\sin\varphi\cos\theta & -\sin\varphi\sin\theta & \cos\varphi \\ -\sin\theta & \cos\theta & 0 \end{vmatrix}$$

$$= a(b+a\cos\varphi)\left(-\cos\varphi\cos\theta\,\vec{i} + \cos\varphi\sin\theta\,\vec{j} - \sin\varphi\,\vec{k}\right)$$

$$\left\|\vec{r}'_\varphi \times \vec{r}'_\theta\right\| = a(b+a\cos\varphi)$$

$$S = \iint_D \left\|\vec{r}'_\varphi \times \vec{r}'_\theta\right\| d\varphi\, d\theta = \int_0^{2\pi} d\theta \int_0^{2\pi} a(b+a\cos\varphi)\, d\varphi = 2\pi a\int_0^{2\pi}(b+a\cos\varphi)\, d\varphi$$

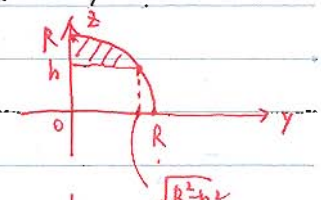
$$= 4\pi^2 ab.$$

2. 计算下列第一类曲面积分:

(1) $\iint_\Sigma \frac{1}{z}\, dS$, 其中 $\Sigma$ 是球面 $x^2+y^2+z^2=R^2$ 被平面 $z=h$ $(0<h<R)$ 截去的顶部.

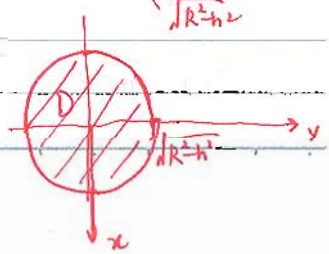
解: $z=\sqrt{R^2-x^2-y^2}$. $z'^2_x + z'^2_y = \frac{x^2+y^2}{R^2-x^2-y^2}$. $\sqrt{1+z'^2_x+z'^2_y} = \frac{R}{\sqrt{R^2-x^2-y^2}}$

$$dS = \sqrt{1+z'^2_x+z'^2_y}\, dx\,dy = \frac{R\,dx\,dy}{\sqrt{R^2-x^2-y^2}}$$




$$\iint_\Sigma \frac{1}{z}\, dS = \iint_D \frac{1}{\sqrt{R^2-x^2-y^2}} \cdot \frac{R\,dx\,dy}{\sqrt{R^2-x^2-y^2}} = \iint_D \frac{R}{R^2-x^2-y^2}\, dx\,dy.$$

$$D = \left\{(x,y)\,\middle|\, x^2+y^2 \le R^2-h^2\right\} = \left\{(r,\theta)\,\middle|\, \begin{matrix} 0\le r\le\sqrt{R^2-h^2} \\ 0\le\theta\le 2\pi \end{matrix}\right\}$$



$$\iint_{\Sigma}\frac{1}{z}dS=\int_0^{2\pi}d\theta\int_0^{\sqrt{R^2-h^2}}\frac{R}{R^2-r^2}r\,dr=-\pi R\int_0^{\sqrt{R^2-h^2}}\frac{1}{R^2-r^2}d(R^2-r^2)$$

$$=-\pi R\ln(R^2-r^2)\Big|_0^{\sqrt{R^2-h^2}}=2\pi R\ln\frac{R}{h}.$$

(2) $\iint_{\Sigma}(x+y+z)dS$，其中$\Sigma$是方程球面$x^2+y^2+z^2=R^2$，$y\le 0$.

解1：由对称性：$\iint_{\Sigma}(x+y+z)dS=\iint_{\Sigma}y\,dS=-\iint_{\Sigma_{zx}}\sqrt{R^2-x^2-z^2}\cdot\frac{R\,dx\,dz}{\sqrt{R^2-x^2-z^2}}$

$$=-R\iint_{\Sigma_{zx}}dx\,dz=-\pi R^3.$$

解2：$\begin{cases}x=R\sin\varphi\cos\theta\\ y=R\sin\varphi\sin\theta\\ z=R\cos\varphi\end{cases}$ $0\le\varphi\le\pi$，$\pi\le\theta\le 2\pi$.　$dS=R^2\sin\varphi\,d\varphi d\theta$.

$$\iint_{\Sigma}(x+y+z)dS=\int_{\pi}^{2\pi}d\theta\int_0^{\pi}(R\sin\varphi\cos\theta+R\sin\varphi\sin\theta+R\cos\varphi)\cdot R^2\sin\varphi\,d\varphi$$

$$=R^3\int_{\pi}^{2\pi}d\theta\int_0^{\pi}\left[(\cos\theta+\sin\theta)\sin^2\varphi+\frac{1}{2}\sin 2\varphi\right]d\varphi$$

$$=R^3\left[\int_{\pi}^{2\pi}(\sin\theta+\cos\theta)\,d\theta\cdot\int_0^{\pi}\sin^2\varphi\,d\varphi+\pi\int_0^{\pi}\sin 2\varphi\,d\varphi\right]$$

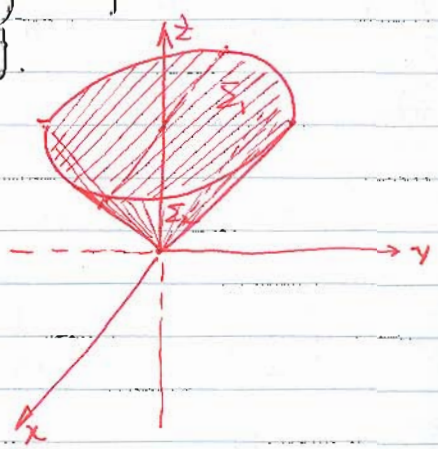
$$=R^3\left[(-\cos\theta)\Big|_{\pi}^{2\pi}\cdot\frac{\pi}{2}-\frac{\pi}{2}\cos 2\varphi\Big|_0^{\pi}\right]=-\pi R^3.$$

(3) $\iint_{\Sigma}(x^2+y^2)dS$，其中$\Sigma$是由锥面$\left\{(x,y,z)\,\middle|\,\sqrt{x^2+y^2}\le z\le 1\right\}$与边界.

解：$\Sigma=\Sigma_1+\Sigma_2$，

$\Sigma_1=\left\{(x,y,z)\,\middle|\,z=\sqrt{x^2+y^2}\quad 0\le z\le 1\right\}$.

$\Sigma_2=\left\{(x,y,z)\,\middle|\,z=1\quad x^2+y^2\le 1\right\}$.




$$\iint_{\Sigma}(x^2+y^2)dS=\left(\iint_{\Sigma_1}+\iint_{\Sigma_2}\right)(x^2+y^2)dS.$$

$\Sigma_1$：$dS=\sqrt{1+z_x'^2+z_y'^2}\,dx\,dy=\sqrt{2}\,dx\,dy$.

$\Sigma_2$：$dS=\sqrt{1+z_x'^2+z_y'^2}\,dx\,dy=dx\,dy$.

令 $\begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases}$ $D=\{(x,y)\mid x^2+y^2\le 1\}=\left\{(r,\theta)\mid \begin{matrix}0\le r\le 1\\ 0\le\theta\le 2\pi\end{matrix}\right\}$. $d\sigma=r\,dr\,d\theta$.

$$\iint_\Sigma (x^2+y^2)\,dS=\iint_D (1+\sqrt{2})\,dx\,dy=(1+\sqrt{2})\int_0^{2\pi}d\theta\int_0^1 r^3\,dr=\frac{1+\sqrt{2}}{2}\pi.$$

(4) $\iint_\Sigma (xy+yz+zx)\,dS$，其中 $\Sigma$ 为锥面 $z=\sqrt{x^2+y^2}$ 被柱面 $x^2+y^2=2ax$ 所截出的部分，$a>0$.

解: $dS=\sqrt{1+z_x'^2+z_y'^2}\,dx\,dy=\sqrt{2}\,dx\,dy$ $D=\{(x,y)\mid x^2+y^2\le 2ax\}$

$=\left\{(r,\theta)\mid \begin{matrix}0\le r\le 2a\cos\theta\\ -\frac{\pi}{2}\le\theta\le\frac{\pi}{2}\end{matrix}\right\}$.

$$\iint_\Sigma (xy+yz+zx)\,dS=\iint_D\left[xy+(x+y)\sqrt{x^2+y^2}\right]\cdot\sqrt{2}\,dx\,dy \qquad d\sigma=dx\,dy=r\,dr\,d\theta.$$

$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}d\theta\int_0^{2a\cos\theta}\left[r^2\sin\theta\cos\theta+(r\cos\theta+r\sin\theta)\cdot r\right]\sqrt{2}\,r\,dr$$

$$=-\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(-\sin\theta\cos\theta+\sin\theta+\cos\theta)\,d\theta\cdot\sqrt{2}\int_0^{2a\cos\theta}r^3\,dr$$

$$=4\sqrt{2}a^4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(-\sin\theta\cos^5\theta+\sin\theta\cos^4\theta+\cos^5\theta)\,d\theta=4\sqrt{2}a^4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^5\theta\,d\theta=\frac{64}{15}\sqrt{2}a^4.$$

(5) $\iint_\Sigma \frac{1}{x^2+y^2+z^2}\,dS$，其中 $\Sigma$ 是圆柱面 $x^2+y^2=a^2$ 介于平面 $z=0$ 与 $z=H$ 间的部分.

解: 柱坐标 $\begin{cases} x=a\cos\theta \\ y=a\sin\theta \\ z=z \end{cases}$ $\vec{r}_\theta'=x_\theta'\vec{i}+y_\theta'\vec{j}+z_\theta'\vec{k}=-a\sin\theta\,\vec{i}+a\cos\theta\,\vec{j}$

$\vec{r}_z'=x_z'\vec{i}+y_z'\vec{j}+z_z'\vec{k}=\vec{k}$.

$$\vec{r}_\theta'\times\vec{r}_z'=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\ -a\sin\theta&a\cos\theta&0\\ 0&0&1\end{vmatrix}=a\cos\theta\,\vec{i}-a\sin\theta\,\vec{j}$$

$\|\vec{r}_\theta'\times\vec{r}_z'\|=a$. $dS=a\,d\theta\,dz$. $D_{\theta z}=\left\{(\theta,z)\mid \begin{matrix}0\le\theta\le 2\pi\\ 0\le z\le H\end{matrix}\right\}$.

$$\iint_\Sigma \frac{1}{x^2+y^2+z^2}\,dS=\iint_{D_{\theta z}}\frac{1}{a^2+z^2}\,a\,d\theta\,dz=\int_0^{2\pi}d\theta\int_0^H\frac{a}{a^2+z^2}\,dz=2\pi\arctan\frac{H}{a}.$$

(6) $\iint_\Sigma x\,dS$，其中 $\Sigma$ 是螺旋面 $x=u\cos v$，$y=u\sin v$，$z=av$ 在 $D_{uv}=\left\{(u,v)\mid \begin{matrix}0\le u\le r\\ 0\le v\le 2\pi\end{matrix}\right\}$ 上的部分.

解: $\begin{cases} x = u\cos v \\ y = u\sin v \\ z = av \end{cases}$ $\vec{r}'_u = \cos v\,\vec{i} + \sin v\,\vec{j} + 0\cdot\vec{k}$ $D_{uv} = \left\{(u,v) \,\middle|\, \begin{matrix} 0 \le u \le r \\ 0 \le v \le 2\pi \end{matrix}\right\}$

$\vec{r}'_v = -u\sin v\,\vec{i} + u\cos v\,\vec{j} + a\cdot\vec{k}$

$$\vec{r}'_u \times \vec{r}'_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos v & \sin v & 0 \\ -u\sin v & u\cos v & a \end{vmatrix} = a\sin v\,\vec{i} - a\cos v\,\vec{j} + u\vec{k}$$

$$\|\vec{r}'_u \times \vec{r}'_v\| = \sqrt{a^2+u^2} \qquad dS = \|\vec{r}'_u \times \vec{r}'_v\|\,du\,dv = \sqrt{a^2+u^2}\,du\,dv$$

$$\iint_\Sigma x\,dS = \iint_{D_{uv}} u\cos v\cdot\sqrt{a^2+u^2}\,du\,dv = \int_0^{2\pi} dv \int_0^r u\cos v\sqrt{a^2+u^2}\,du$$

$$= \int_0^{2\pi} \cos v\,dv \cdot \int_0^r u\sqrt{a^2+u^2}\,du = 0$$

3. 求抛物面壳 $z = \frac{1}{2}(x^2+y^2)$ 在 $z\in[0,1]$ 部分的质量，已知其面密度 $\rho(x,y,z) = z$.

解: $dS = \sqrt{1+z'^2_x+z'^2_y}\,dx\,dy = \sqrt{1+x^2+y^2}\,dx\,dy$ $D_{xy} = \left\{(x,y) \,\middle|\, 0 \le x^2+y^2 \le 2\right\}$

$$M = \iint_\Sigma z\,dS = \iint_{D_{xy}} \frac{1}{2}(x^2+y^2)\cdot\sqrt{1+x^2+y^2}\,dx\,dy \qquad = \left\{(r,\theta) \,\middle|\, \begin{matrix} 0 \le r \le \sqrt{2} \\ 0 \le \theta \le 2\pi \end{matrix}\right\}$$

$$= \frac{1}{2}\int_0^{2\pi} d\theta \cdot \int_0^{\sqrt{2}} r^2\cdot\sqrt{1+r^2}\,r\,dr = \frac{\pi}{2}\int_0^{\sqrt{2}} r^2\sqrt{1+r^2}\,dr^2$$

$$= \frac{\pi}{2}\int_0^2 t\sqrt{1+t}\,dt = \frac{\pi}{2}\int_0^2 \left[(1+t)^{\frac{3}{2}} - (1+t)^{\frac{1}{2}}\right] d(1+t) = \frac{1}{15}(12\sqrt{3}+2)\pi.$$

4. 求质量均匀分布的上半球壳 $x^2+y^2+z^2 = a^2$, $z \ge 0$ 的质心坐标.

解: 设面密度 $\rho_0$. 质心 $(\bar{x},\bar{y},\bar{z})$. 由对称性可知: $\bar{x}=0$, $\bar{y}=0$, $\bar{z} = \frac{1}{M}\iint_\Sigma z\rho_0\,dS$.

$z = \sqrt{a^2-x^2-y^2}$. $dS = \sqrt{1+z'^2_x+z'^2_y}\,dx\,dy = \dfrac{a\,dx\,dy}{\sqrt{a^2-x^2-y^2}}$. $D = \left\{(x,y) \,\middle|\, x^2+y^2 \le a^2\right\}$

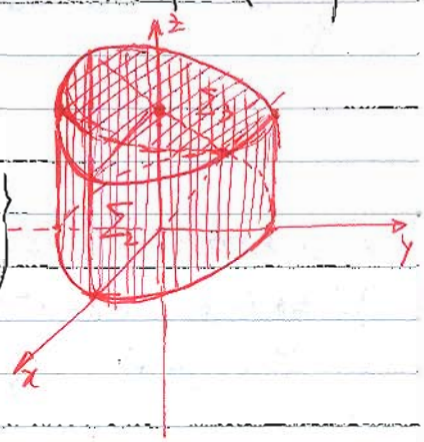
$$M = \iint_\Sigma \rho_0\,dS = \rho_0\iint_D \frac{a\,dx\,dy}{\sqrt{a^2-x^2-y^2}} = \rho_0\int_0^{2\pi} d\theta\int_0^a \frac{a}{\sqrt{a^2-r^2}}\,r\,dr \qquad = \left\{(r,\theta) \,\middle|\, \begin{matrix} 0 \le r \le a \\ 0 \le \theta \le 2\pi \end{matrix}\right\}$$

$$= 2\pi\rho_0 a^2.$$

$$\bar{z} = \frac{1}{M}\iint_\Sigma z\rho_0\,dS = \frac{\rho_0}{2\pi\rho_0 a^2}\iint_D \sqrt{a^2-x^2-y^2}\cdot\frac{a}{\sqrt{a^2-x^2-y^2}}\,dx\,dy = \frac{1}{2\pi a}\iint_D dx\,dy$$

$$= \frac{1}{2\pi a}\cdot\pi a^2 = \frac{a}{2}. \qquad \therefore \text{质心 } \left(0,0,\frac{a}{2}\right).$$

5. 计算 $\oint_\Sigma z\,dS$，其中 $\Sigma$ 是由圆柱面 $x^2+y^2=1$，平面 $z=0$ 和 $z=x+1$ 所围成空间区域的边界曲面.




解: $\Sigma=\Sigma_1+\Sigma_2+\Sigma_3$.  $\Sigma_1=\{(x,y,z)\mid z=0,\ x^2+y^2\le 1\}$

$\Sigma_2=\{(x,y,z)\mid x^2+y^2=1,\ 0\le z\le x+1\}=\{(r,\theta,z)\mid r=1,\ 0\le\theta\le 2\pi,\ 0\le z\le\cos\theta+1\}$

$\Sigma_3=\{(x,y,z)\mid z=x+1,\ x^2+y^2\le 1\}=\{(r,\theta,z)\mid 0\le r\le 1,\ 0\le\theta\le 2\pi,\ z=r\cos\theta+1\}$

$$\oint_\Sigma z\,dS=\left(\iint_{\Sigma_1}+\iint_{\Sigma_2}+\iint_{\Sigma_3}\right)z\,dS.\qquad \iint_{\Sigma_1} z\,dS=0$$

$$\iint_{\Sigma_2} z\,dS=\iint_{D_{\theta z}} z\,\|\vec r'_\theta\times\vec r'_z\|\,d\theta dz.\qquad \begin{cases}\vec r_\theta=x'_\theta\vec i+y'_\theta\vec j+z'_\theta\vec k\\ \vec r_z=x'_z\vec i+y'_z\vec j+z'_z\vec k\end{cases}\qquad \begin{cases}x=\cos\theta\\ y=\sin\theta\\ z=z\end{cases}$$

$$\vec r'_\theta\times\vec r'_z=\begin{vmatrix}\vec i&\vec j&\vec k\\ -\sin\theta&\cos\theta&0\\ 0&0&1\end{vmatrix}=\cos\theta\,\vec i+\sin\theta\,\vec j,\qquad \|\vec r'_\theta\times\vec r'_z\|\equiv 1.$$

$$\therefore\iint_{\Sigma_2} z\,dS=\iint_{D_{\theta z}} z\cdot d\theta dz=\int_0^{2\pi}d\theta\int_0^{\cos\theta+1} z\,dz=\int_0^{2\pi}\frac{1}{2}(1+\cos\theta)^2\,d\theta=\frac{3}{2}\pi.$$

$$\iint_{\Sigma_3} z\,dS=\iint_{\Sigma_3}(x+1)\,dS.\qquad dS=\sqrt{1+z'^2_x+z'^2_y}\,dxdy=\sqrt{2}\,dxdy$$

$$=\iint_{D_{xy}}(x+1)\cdot\sqrt{2}\,dxdy=\sqrt{2}\int_0^{2\pi}d\theta\int_0^1(r\cos\theta+1)\,r\,dr$$

$$=\sqrt{2}\cdot\int_0^{2\pi}d\theta\cdot\left(\frac{1}{3}\cos\theta+\frac{1}{2}\right)=\sqrt{2}\,\pi.$$

$$\therefore\oint_\Sigma z\,dS=0+\frac{3}{2}\pi+\sqrt{2}\pi=\left(\frac{3}{2}+\sqrt{2}\right)\pi.$$

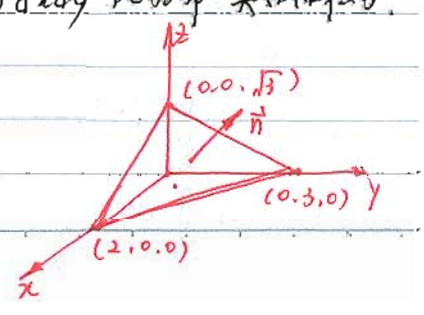
## §7. 第二类曲面积分

习题. P. 130

1. 把第二类曲面积分 $\iint_\Sigma P(x,y,z)\,dydz+Q(x,y,z)\,dzdx+R(x,y,z)\,dxdy$ 化为第一类曲面积分.

其中:

i) $\Sigma$ 是平面 $3x+2y+2\sqrt{3}z=6$ 在第一卦限部分的上侧;



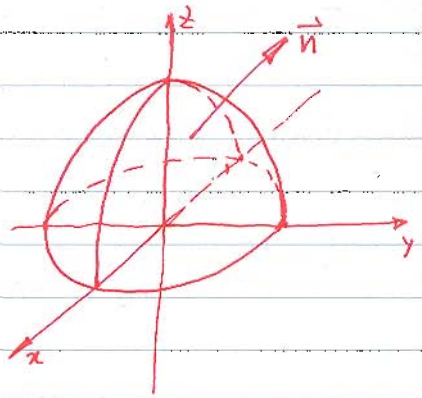

解: 平面 $3x+2y+2\sqrt{3}z=6$ 在第一卦限部分的上侧, 则

$$\vec n=(3,2,2\sqrt{3})\cdot\frac{1}{5}=\left(\frac{3}{5},\frac{2}{5},\frac{2\sqrt{3}}{5}\right).$$

$$\iint_\Sigma P\,dydz + Q\,dzdx + R\,dxdy = \iint_\Sigma \left[ P\cos(\vec{n},x) + Q\cos(\vec{n},y) + R\cos(\vec{n},z) \right] dS$$

$$= \iint_\Sigma \left( \frac{3}{5}P + \frac{2}{5}Q + \frac{2\sqrt{3}}{5}R \right) dS.$$

(2) $\Sigma$ 是抛物面 $z = 8 - x^2 - y^2$ 在 $Oxy$ 平面上方部分的上侧.




解: 抛物面 $z = 8 - x^2 - y^2$ 在 $Oxy$ 平面上方部分的上侧,

$$\vec{n} = (-f'_x, -f'_y, 1) \Big/ \sqrt{1 + f'^2_x + f'^2_y}$$

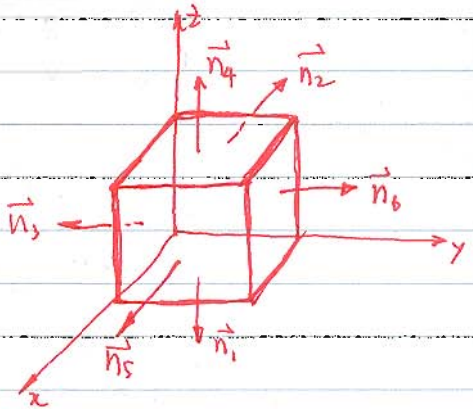
$$= \frac{1}{\sqrt{1+4x^2+4y^2}} (2x, 2y, 1)$$

$$\therefore \iint_\Sigma P\,dydz + Q\,dzdx + R\,dxdy$$

$$= \iint_\Sigma \vec{F}\cdot\vec{n}\,dS = \iint_\Sigma \frac{2xP + 2yQ + R}{\sqrt{1+4x^2+4y^2}}\,dS.$$

2. 计算第二类曲面积分:

(1) $\iint_\Sigma x\,dydz + y\,dzdx + z\,dxdy$, 其中 $\Sigma$ 是由平面 $x=0, y=0, z=0, x=1, y=1, z=1$ 所围立方体表面的外侧.




解: $\vec{F} = (x, y, z)$. $\Sigma = \Sigma_1 + \Sigma_2 + \Sigma_3 + \Sigma_4 + \Sigma_5 + \Sigma_6$

$\Sigma_1$: $z=0$, $\vec{n} = (0, 0, -1)$, $D_1 = \{(x,y) \mid 0 \le x, y \le 1\}$.

$\Sigma_2$: $x=0$, $\vec{n} = (-1, 0, 0)$, $D_2 = \{(y,z) \mid 0 \le y, z \le 1\}$.

$\Sigma_3$: $y=0$, $\vec{n} = (0, -1, 0)$, $D_3 = \{(x,z) \mid 0 \le x, z \le 1\}$.

$\Sigma_4$: $z=1$, $\vec{n} = (0, 0, 1)$, $D_4 = \{(x,y) \mid 0 \le x, y \le 1\}$.

$\Sigma_5$: $x=1$, $\vec{n} = (1, 0, 0)$, $D_5 = \{(y,z) \mid 0 \le y, z \le 1\}$.

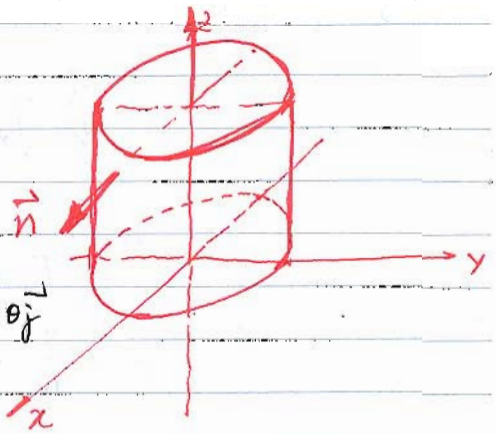
$\Sigma_6$: $y=1$, $\vec{n} = (0, 1, 0)$, $D_6 = \{(x,z) \mid 0 \le x, z \le 1\}$.

$$\therefore \iint_\Sigma x\,dydz + y\,dzdx + z\,dxdy = \iint_\Sigma \vec{F}\cdot\vec{n}\,dS$$

$$= \iint_{\Sigma_1} (-z)\,dS + \iint_{\Sigma_2} (-x)\,dS + \iint_{\Sigma_3} (-y)\,dS + \iint_{\Sigma_4} z\,dS + \iint_{\Sigma_5} x\,dS + \iint_{\Sigma_6} y\,dS$$

$$= \iint_{\Sigma_4} dS + \iint_{\Sigma_5} dS + \iint_{\Sigma_6} dS = 3$$

(2) $\iint_{\Sigma} x^2dydz + y^2dzdx + z^2dxdy$，其中$\Sigma$是柱面$x^2+y^2=1$被平面$z=0$及$z=3$所截部分的外侧.




解: $\vec{F} = (x^2, y^2, z^2)$.

$\Sigma$采用柱坐标: $\Sigma = \left\{ (\theta, z) \middle| \begin{matrix} r=1,\ 0 \le \theta \le 2\pi \\ 0 \le z \le 3 \end{matrix} \right\}$

$$\begin{cases} x = \cos\theta \\ y = \sin\theta \\ z = z \end{cases} \qquad \vec{r}'_\theta \times \vec{r}'_z = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = \cos\theta\vec{i} + \sin\theta\vec{j}$$

$\Sigma$的外侧: $\vec{n} = \dfrac{\vec{r}'_\theta \times \vec{r}'_z}{\|\vec{r}'_\theta \times \vec{r}'_z\|} = (\cos\theta, \sin\theta, 0)$.

$E = x'^2_\theta + y'^2_\theta + z'^2_\theta = \sin^2\theta + \cos^2\theta = 1$. $G = x'^2_z + y'^2_z + z'^2_z = 1$.

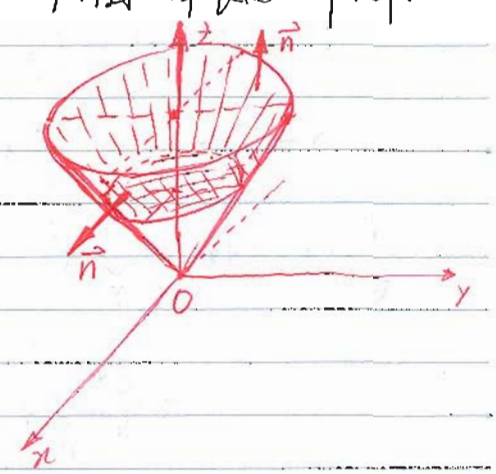
$F = x'_\theta x'_z + y'_\theta y'_z + z'_\theta z'_z = 0$. $\therefore dS = \sqrt{EG - F^2}\, d\theta dz = d\theta dz$.

$$\therefore \iint_{\Sigma} x^2dydz + y^2dzdx + z^2dxdy = \iint_{\Sigma} \vec{F}\cdot\vec{n}\, dS = \iint_{\Sigma} (x^2, y^2, z^2)\cdot(\cos\theta, \sin\theta, 0)\cdot d\theta dz$$

$$= \iint_{\Sigma} (x^2\cos\theta + y^2\sin\theta)\, d\theta dz = \iint_{\Sigma} (\cos^3\theta + \sin^3\theta)\, d\theta dz = \int_0^3 dz \int_0^{2\pi} (\cos^3\theta + \sin^3\theta)\, d\theta$$

$$= 3\cdot\int_0^{2\pi} (\cos\theta + \sin\theta)(1 - \sin\theta\cos\theta)\, d\theta = 0.$$

(3) $\iint_{\Sigma} \dfrac{e^z dxdy}{\sqrt{x^2+y^2}}$，其中$\Sigma$是锥面$z=\sqrt{x^2+y^2}$及平面$z=1$、$z=2$所围立体表面的外侧.




解: $\vec{F} = \left(0, 0, \dfrac{e^z}{\sqrt{x^2+y^2}}\right)$ $\Sigma = \Sigma_1 + \Sigma_2 + \Sigma_3$.

$\Sigma_1$: $z = \sqrt{x^2+y^2}$，外侧 $\vec{n}_1 = \dfrac{1}{\sqrt{2}}\left(\dfrac{x}{\sqrt{x^2+y^2}}, \dfrac{y}{\sqrt{x^2+y^2}}, -1\right)$.

$dS = \sqrt{1 + f'^2_x + f'^2_y}\, dxdy = \sqrt{2}\, dxdy$

$D_1 = \left\{ (x,y) \middle| 1 \le x^2+y^2 \le 4 \right\}$.

$\Sigma_2$: $z=1$. 下侧 $\vec{n}_2 = (0,0,-1)$. $dS = dxdy$.

$D_2 = \left\{ (x,y) \middle| 0 \le x^2+y^2 \le 1 \right\}$.

$\Sigma_3$: $z=2$. 上侧 $\vec{n}_3 = (0,0,1)$. $dS = dxdy$. $D_3 = \left\{ (x,y) \middle| 0 \le x^2+y^2 \le 4 \right\}$.

$$\iint_{\Sigma}\frac{e^z\,dx\,dy}{\sqrt{x^2+y^2}}=\left(\iint_{\Sigma_1}+\iint_{\Sigma_2}+\iint_{\Sigma_3}\right)\vec{F}\cdot\vec{n}\,dS$$

$$=\iint_{\Sigma_1}\frac{-e^z}{\sqrt{x^2+y^2}}\cdot\frac{1}{\sqrt{2}}\,dS+\iint_{\Sigma_2}\frac{-e^z}{\sqrt{x^2+y^2}}\,dS+\iint_{\Sigma_3}\frac{e^z}{\sqrt{x^2+y^2}}\,dS$$

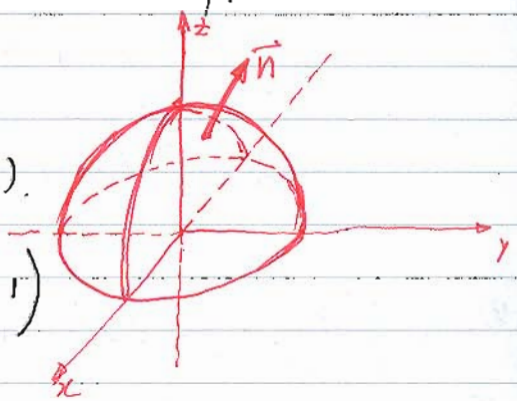
$$=-\iint_{D_1}\frac{e^z}{\sqrt{x^2+y^2}}\,dx\,dy-\iint_{D_2}\frac{e^z}{\sqrt{x^2+y^2}}\,dx\,dy+\iint_{D_3}\frac{e^z}{\sqrt{x^2+y^2}}\,dx\,dy$$

$$=-\iint_{D_1}\frac{e^{\sqrt{x^2+y^2}}}{\sqrt{x^2+y^2}}\,dx\,dy-\iint_{D_2}\frac{e}{\sqrt{x^2+y^2}}\,dx\,dy+\iint_{D_3}\frac{e^1}{\sqrt{x^2+y^2}}\,dx\,dy$$

$$=-\int_0^{2\pi}d\theta\cdot\int_1^2\frac{e^r}{r}\cdot r\,dr-\int_0^{2\pi}d\theta\cdot\int_0^1\frac{e}{r}\cdot r\,dr+\int_0^{2\pi}d\theta\cdot\int_0^2\frac{e^1}{r}\cdot r\,dr$$

$$=-2\pi(e^2-e)-2\pi e+4\pi e^2=2\pi e^2.$$

(4) $\iint_{\Sigma}yz\,dz\,dx$, 其中$\Sigma$是$b$半椭球面: $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$, $z\geq 0$ 的上侧.




解(I): $\vec{F}=(0,yz,0)$. $\Sigma$: $z=c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}$

$\vec{n}=(-f'_x,-f'_y,1)/\|\vec{n}\|$ (上侧).

即: $\vec{n}=\frac{1}{\sqrt{1+f'^2_x+f'^2_y}}\left(\frac{cx}{a^2\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}},\frac{cy}{b^2\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}},1\right)$

$dS=\sqrt{1+f'^2_x+f'^2_y}\,dx\,dy$.

$$D=\left\{(x,y)\,\middle|\,\frac{x^2}{a^2}+\frac{y^2}{b^2}\leq 1\right\}=\left\{(r,\theta)\,\middle|\,\begin{matrix}0\leq r\leq 1\\ 0\leq\theta\leq 2\pi\end{matrix}\right\}.\quad\begin{cases}x=ar\cos\theta\\ y=br\sin\theta\end{cases}$$

$$\iint_{\Sigma}yz\,dz\,dx=\iint_{\Sigma}\vec{F}\cdot\vec{n}\,dS=\iint_{D}\frac{cy^2z}{b^2\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}}\,dx\,dy=\iint_{D}\frac{c^2y^2}{b^2}\,dx\,dy$$

$$=\frac{c^2}{b^2}\iint_{D}y^2\,dx\,dy=\frac{c^2}{b^2}\int_0^{2\pi}d\theta\cdot\int_0^1 b^2r^2\sin^2\theta\cdot ab\,r\,dr$$

$$=c^2\int_0^{2\pi}\sin^2\theta\,d\theta\cdot\int_0^1 abr^3\,dr=abc^2\cdot\frac{1}{2}\cdot 2\pi\cdot\frac{1}{4}=\frac{\pi abc^2}{4}.$$

解(II): $\vec{F}=(0,yz,0)$. $\Sigma$: $\begin{cases}x=a\sin\varphi\cos\theta\\ y=b\sin\varphi\sin\theta\\ z=c\cos\varphi\end{cases}$ $\begin{matrix}0\leq\theta\leq 2\pi\\ 0\leq\varphi\leq\frac{\pi}{2}\end{matrix}$.

$$\vec{n}=\frac{1}{\sqrt{EG-F^2}}\left(\begin{vmatrix} y'_\varphi & y'_\theta \\ z'_\varphi & z'_\theta \end{vmatrix},\begin{vmatrix} z'_\varphi & z'_\theta \\ x'_\varphi & x'_\theta \end{vmatrix},\begin{vmatrix} x'_\varphi & x'_\theta \\ y'_\varphi & y'_\theta \end{vmatrix}\right) \quad (\text{上侧}).$$

$$=\frac{1}{\sqrt{EG-F^2}}(bc\sin^2\varphi\cos\theta,\ ac\sin^2\varphi\sin\theta,\ ab\sin\varphi\cos\varphi). \quad 0\le\varphi\le\frac{\pi}{2}.$$

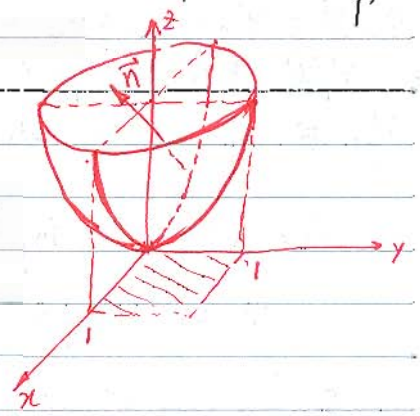
$$dS=\sqrt{EG-F^2}\,d\varphi d\theta.$$

$$\therefore \iint_\Sigma yz\,dzdx=\iint_\Sigma \vec{F}\cdot\vec{n}\,dS=\iint_D bc\sin\varphi\sin\theta\cdot\cos\varphi\cdot ac\sin^2\varphi\sin\theta\,d\varphi d\theta$$

$$=abc^2\int_0^{2\pi}\sin^2\theta\,d\theta\cdot\int_0^{\frac{\pi}{2}}\sin^3\varphi\cos\varphi\,d\varphi=abc^2\cdot\frac{1}{2}\cdot 2\pi\cdot\frac{1}{4}=\frac{\pi abc^2}{4}.$$

3. 已知向量场 $\vec{F}$ 和有向曲面 $\Sigma$，求下列积分 $\iint_\Sigma \vec{F}\cdot\vec{n}\,dS$.

(1) $\vec{F}=e^y\vec{i}+ye^x\vec{j}+x^2y\vec{k}$，其中 $\Sigma$ 是抛物面 $z=x^2+y^2$，$0\le x\le 1$，$0\le y\le 1$，取上侧；




解：$\vec{F}=(e^y,\ ye^x,\ x^2y)$. $\Sigma$：$z=x^2+y^2$，$0\le x\le 1$，$0\le y\le 1$.

$\Sigma$ 上侧 $\vec{n}=\frac{1}{\sqrt{f'^2_x+f'^2_y+1}}(-f'_x,-f'_y,1)$

$$=\frac{1}{\sqrt{f'^2_x+f'^2_y+1}}(-2x,-2y,1).$$

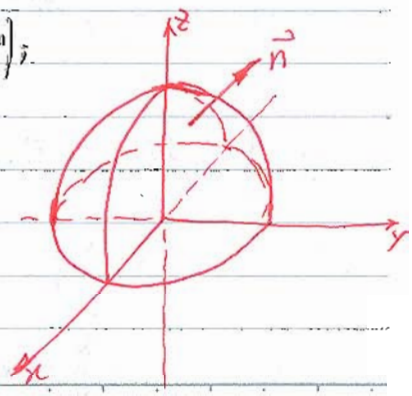
$$dS=\sqrt{f'^2_x+f'^2_y+1}\,dxdy. \quad D=\{(x,y)\mid 0\le x\le 1,\ 0\le y\le 1\}.$$

$$\iint_\Sigma \vec{F}\cdot\vec{dS}=\iint_\Sigma \vec{F}\cdot\vec{n}\,dS=\iint_D(-2xe^y-2y^2e^x+x^2y)\,dxdy$$

$$=\int_0^1 dy\int_0^1(-2xe^y-2y^2e^x+x^2y)\,dx=\int_0^1(-e^y-2ey^2+\frac{1}{3}y+2y^2)\,dy$$

$$=\frac{11}{6}-\frac{5}{3}e.$$

(2) $\vec{F}=-y\vec{i}-x\vec{j}+z\vec{k}$，其中 $\Sigma$ 是半球面 $z=\sqrt{1-x^2-y^2}$，取上侧；




解：$\vec{F}=(-y,-x,z)$. $\Sigma$：$z=\sqrt{1-x^2-y^2}$.

$\Sigma$ 上侧 $\vec{n}=\frac{1}{\sqrt{1+f'^2_x+f'^2_y}}(-f'_x,-f'_y,1)$

$$=\frac{1}{\sqrt{1+f'^2_x+f'^2_y}}\left(\frac{x}{\sqrt{1-x^2-y^2}},\frac{y}{\sqrt{1-x^2-y^2}},1\right).$$

Date

$dS = \sqrt{1+f_x'^2+f_y'^2}\,dxdy$. $D=\{(x,y)\mid 0\le x^2+y^2\le 16\}$

$=\{(r,\theta)\mid 0\le r\le 4,\ 0\le\theta\le 2\pi\}$. $\begin{cases}x=r\cos\theta\\ y=r\sin\theta\end{cases}$

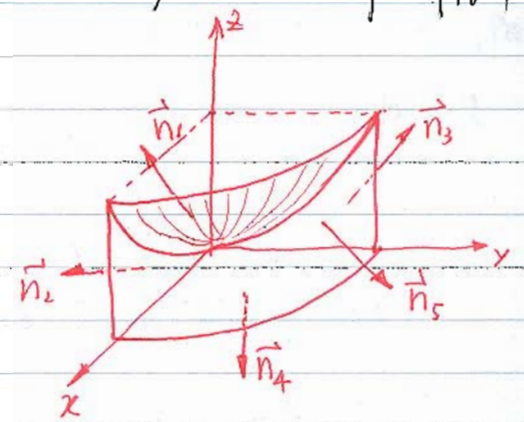
$$\iint_\Sigma \vec{F}\cdot d\vec{S}=\iint_\Sigma \vec{F}\cdot\vec{n}\,dS=\iint_D(-y,-z,3z)\cdot\left(\frac{x}{\sqrt{16-x^2-y^2}},\frac{y}{\sqrt{16-x^2-y^2}},1\right)dxdy$$

$$=\iint_D\left[\frac{-2xy}{\sqrt{16-x^2-y^2}}+3z\right]dxdy=\iint_D\left[\frac{-2xy}{\sqrt{16-x^2-y^2}}+3\sqrt{16-x^2-y^2}\right]dxdy$$

$$=\int_0^{2\pi}d\theta\cdot\int_0^4\left[\frac{-2r^3\sin\theta\cos\theta}{\sqrt{16-r^2}}+3\sqrt{16-r^2}\right]\cdot r\,dr$$

$$=2\pi\cdot 3\int_0^4\sqrt{16-r^2}\cdot r\,dr=128\pi.$$

(3) $\vec{F}=y^2z\vec{i}+z^2x\vec{j}+x^2y\vec{k}$, 其中$\Sigma$是由$z=x^2+y^2$, $x^2+y^2=1$, $x=0$, $y=0$, $z=0$在第一卦限中所围立体表面的外侧;




解: $\vec{F}=(-y^2z,\ z^2x,\ x^2y)$. $\Sigma=\Sigma_1+\Sigma_2+\Sigma_3+\Sigma_4+\Sigma_5$.

$\Sigma_1$: $z=x^2+y^2$. 上侧 $\vec{n}_1=\frac{1}{\sqrt{EG-F^2}}(-2x,-2y,1)$.

$D_1=\{(x,y)\mid 0\le x^2+y^2\le 1,\ 0\le x,y\le 1\}$. $dS=\sqrt{EG-F^2}\,dxdy$.

$\Sigma_2$: $y=0$. $\vec{n}_2=(0,-1,0)$ (左侧)

$D_2=\{(x,z)\mid 0\le z\le x^2,\ 0\le x\le 1\}$.

$\Sigma_3$: $x=0$. $\vec{n}_3=(-1,0,0)$. (后侧). $D_3=\{(y,z)\mid 0\le z\le y^2,\ 0\le y\le 1\}$.

$\Sigma_4$: $z=0$. $\vec{n}_4=(0,0,-1)$. (下侧). $D_4=\{(x,y)\mid x^2+y^2\le 1,\ 0\le x,y\le 1\}$.

$\Sigma_5$: $x^2+y^2=1$. 参数形式: $\begin{cases}x=\cos\theta\\ y=\sin\theta\\ z=z\end{cases}$ $\vec{n}_5=(\cos\theta,\sin\theta,0)$ (后侧).

$D_5=\{(\theta,z)\mid 0\le\theta\le\frac{\pi}{2},\ 0\le z\le 1\}$ (在第一卦限内)

$$\therefore \iint_\Sigma \vec{F}\cdot d\vec{S}=\iint_\Sigma \vec{F}\cdot\vec{n}\,dS=\iint_{\Sigma_1}\vec{F}\cdot\vec{n}_1\,dS+\iint_{\Sigma_2}\vec{F}\cdot\vec{n}_2\,dS+\iint_{\Sigma_3}\vec{F}\cdot\vec{n}_3\,dS+\iint_{\Sigma_4}\vec{F}\cdot\vec{n}_4\,dS+\iint_{\Sigma_5}\vec{F}\cdot\vec{n}_5\,dS$$

$$=\iint_{D_1}\left[-2x^2y(x^2+y^2)-2xy(x^2+y^2)^2+x^2y\right]dxdy+\iint_{D_2}(-z^2x)\,dxdz+\iint_{D_3}(-y^2z)\,dydz$$

$$+\iint_{D_4}(-x^2y)\,dxdy+\iint_{D_5}(y^2z\cos\theta+z^2x\sin\theta)\,d\theta dz$$

$$= \int_0^{\frac{\pi}{2}} d\theta \cdot \int_0^1 \left[ -2r\cos\theta \cdot r^2\sin^2\theta \cdot r^2 - 2r\cos\theta\cdot\sin\theta\cdot r^4 + r^2\cos^2\theta \cdot r\sin\theta \right] r\,dr$$

$$- \int_0^1 dx \int_0^{x^2} z^2 x\,dz - \int_0^1 dy \int_0^{y^2} y^2 z\,dz = \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r^2\cos^2\theta \cdot r\sin\theta \cdot r\,dr$$

$$+ \int_0^{\frac{\pi}{2}} d\theta \int_0^1 \left(-\sin^2\theta\cos\theta\cdot z + \cos\theta\sin\theta\cdot z^2\right) dz$$

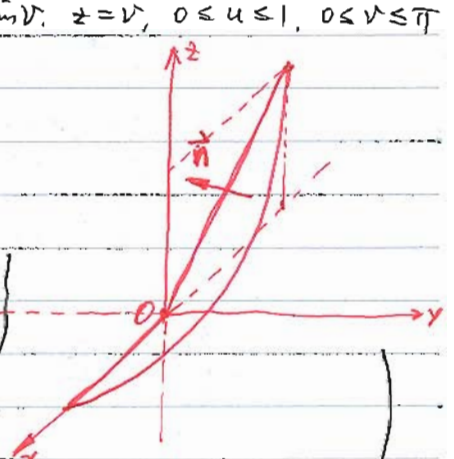
$$= \left(-\frac{2}{21} - \frac{1}{8} + \frac{1}{15}\right) - \frac{1}{24} - \frac{1}{14} - \frac{1}{15} + \left(\frac{1}{6} + \frac{1}{6}\right) = 0.$$

(事实上，本题中 $\operatorname{div}\vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 0$. 即 $\vec{F}$ 为无源场，由 Gauss 公式，

对任何光滑闭曲面 $\Sigma$: $\iint_\Sigma \vec{F}\cdot\vec{n}\,dS = 0$. 即穿出 $\Sigma$ 的通量为 0. )

(4) $\vec{F} = y\vec{i} + x\vec{j} + z^2\vec{k}$, 其中 $\Sigma$ 是螺旋面 $x = u\cos v$, $y = u\sin v$, $z = v$, $0 \le u \le 1$, $0 \le v \le \pi$ 的上侧.

解: $\vec{F} = (y, x, z^2)$. $\Sigma$: $\begin{cases} x = u\cos v & 0 \le u \le 1 \\ y = u\sin v & 0 \le v \le \pi \\ z = v \end{cases}$




$\Sigma$ 的上侧 $\vec{n} = \frac{1}{\sqrt{EG-F^2}}(\sin v, -\cos v, u)$. $D = \left\{(u,v) \,\middle|\, \begin{matrix} 0 \le u \le 1 \\ 0 \le v \le \pi \end{matrix}\right\}$

$$\left( \begin{vmatrix} y'_u & y'_v \\ z'_u & z'_v \end{vmatrix} = \begin{vmatrix} \sin v & u\cos v \\ 0 & 1 \end{vmatrix} = \sin v \right.$$

$$\left. \begin{vmatrix} z'_u & z'_v \\ x'_u & x'_v \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ \cos v & -u\sin v \end{vmatrix} = -\cos v. \quad \begin{vmatrix} x'_u & x'_v \\ y'_u & y'_v \end{vmatrix} = \begin{vmatrix} \cos v & -u\sin v \\ \sin v & u\cos v \end{vmatrix} = u. \right)$$

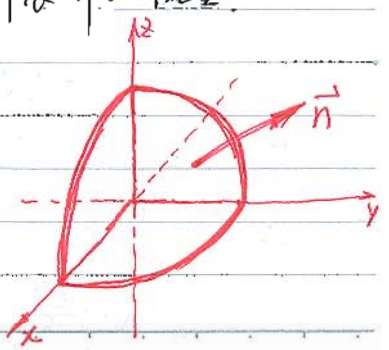
$$\iint_\Sigma \vec{F}\cdot d\vec{S} = \iint_\Sigma \vec{F}\cdot\vec{n}\,dS = \iint_D \left(y\sin v - x\cos v + z^2 u\right) du\,dv = \iint_D \left(-u\sin^2 v - u\cos^2 v + uv^2\right) du\,dv$$

$$= \int_0^\pi dv \int_0^1 \left(\sin^2 v - \cos^2 v + v^2\right) u\,du = \frac{\pi^3}{3}\cdot\frac{1}{2} = \frac{\pi^3}{6}.$$

4. 求流速场 $\vec{V} = xy\vec{i} + yz\vec{j} + zx\vec{k}$ 由内侧穿过单位球面在第一卦限部分的流量.

解: $Q = -\iint_\Sigma xy\,dy\,dz + yz\,dz\,dx + zx\,dx\,dy$

$\Sigma$: $x^2 + y^2 + z^2 = 1$, $0 \le x, y, z \le 1$. 参数式: $\begin{cases} x = \sin\varphi\cos\theta & 0 \le \varphi \le \frac{\pi}{2} \\ y = \sin\varphi\sin\theta & 0 \le \theta \le \frac{\pi}{2} \\ z = \cos\varphi \end{cases}$.



$$Q=\iint_D\left(\sin^2\varphi\sin\theta\cos\theta\cdot\begin{vmatrix}y'_\varphi & y'_\theta\\ z'_\varphi & z'_\theta\end{vmatrix}+\sin\varphi\sin\theta\cos\varphi\cdot\begin{vmatrix}z'_\varphi & z'_\theta\\ x'_\varphi & x'_\theta\end{vmatrix}+\sin\varphi\cos\theta\cos\varphi\begin{vmatrix}x'_\varphi & x'_\theta\\ y'_\varphi & y'_\theta\end{vmatrix}\right)d\varphi d\theta$$

$$=\iint_D\left(\sin^2\varphi\sin\theta\cos\theta\cdot\sin^2\varphi\cos\theta+\sin\varphi\cos\varphi\cdot\sin\theta\cdot\sin^2\varphi\sin\theta+\sin\varphi\cos\varphi\cos\theta\cdot\sin\varphi\cos\varphi\right)d\varphi d\theta$$

$$=\int_0^{\frac{\pi}{2}}d\theta\cdot\int_0^{\frac{\pi}{2}}\left(\sin^4\varphi\sin\theta\cos^2\theta+\sin^3\varphi\cos\varphi\sin^2\theta+\sin^2\varphi\cos^2\varphi\cos\theta\right)d\varphi=\frac{3}{16}\pi.$$

解II: $\Sigma$ 在三个坐标面上的投影区域分别为 $D_1, D_2, D_3$.

$$Q=\iint_\Sigma\vec{V}\cdot d\vec{S}=-\iint_{D_1}y\sqrt{1-y^2-z^2}\,dy\,dz+\iint_{D_2}z\sqrt{1-x^2-z^2}\,dx\,dz+\iint_{D_3}x\sqrt{1-x^2-y^2}\,dx\,dy$$

$$=3\iint_D x\sqrt{1-x^2-y^2}\,dx\,dy.\qquad D=\left\{(x,y)\,\middle|\,\begin{matrix}0\le x^2+y^2\le 1\\ 0\le x-y\le 1\end{matrix}\right\}=\left\{(r,\theta)\,\middle|\,\begin{matrix}0\le r\le 1\\ 0\le\theta\le\frac{\pi}{2}\end{matrix}\right\}$$

$$=3\int_0^{\frac{\pi}{2}}d\theta\cdot\int_0^1 r\cos\theta\cdot\sqrt{1-r^2}\cdot r\,dr=3\cdot\int_0^{\frac{\pi}{2}}\cos\theta\,d\theta\cdot\int_0^1 r^2\sqrt{1-r^2}\,dr$$

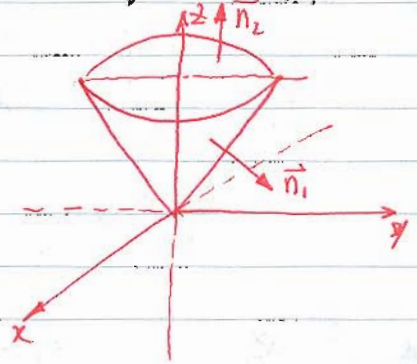
$$=3\cdot\int_0^1 r^2\sqrt{1-r^2}\,dr\xrightarrow{r=\sin t}3\cdot\int_0^{\frac{\pi}{2}}\sin^2t\cdot\cos t\cdot\cos t\,dt$$

$$=3\cdot\int_0^{\frac{\pi}{2}}\sin^2t\cdot\cos^2t\,dt=\frac{3}{4}\int_0^{\frac{\pi}{2}}\sin^2 2t\,dt=\frac{3}{4}\cdot\int_0^{\frac{\pi}{2}}\frac{1}{2}(1-\cos 4t)\,dt$$

$$=\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}=\frac{3}{16}\pi.$$

5. 求向量场 $\vec{F}=x\vec{i}+y\vec{j}+z\vec{k}$ 由内向外穿过锥体 $\sqrt{x^2+y^2}\le z\le h$ 表面的通量.

解: $\vec{F}=(x,y,z)$. $\Sigma=\Sigma_1+\Sigma_2$.




$\Sigma_1$: $z=\sqrt{x^2+y^2}$.

外侧 $\vec{n}_1=\frac{1}{\sqrt{EG-F^2}}(z'_x,z'_y,-1)$

$=\frac{1}{\sqrt{EG-F^2}}\left(\frac{x}{\sqrt{x^2+y^2}},\frac{y}{\sqrt{x^2+y^2}},-1\right)$

$dS=\sqrt{EG-F^2}\,dx\,dy$. $D_1=\left\{(x,y)\,\middle|\,0\le x^2+y^2\le h^2\right\}=\left\{(r,\theta)\,\middle|\,\begin{matrix}0\le r\le h\\ 0\le\theta\le 2\pi\end{matrix}\right\}=D$

$\Sigma_2$: $z=h$, 上侧 $\vec{n}_2=(0,0,1)$. $D_2=D_1=D$ $dS=dx\,dy$.

$$\therefore Q=\iint_\Sigma\vec{F}\cdot d\vec{S}=\iint_{\Sigma_1}\vec{F}\cdot\vec{n}_1\,dS+\iint_{\Sigma_2}\vec{F}\cdot\vec{n}_2\,dS$$

$$=\iint_{D_1}\left(\frac{x^2}{\sqrt{x^2+y^2}}+\frac{y^2}{\sqrt{x^2+y^2}}-z\right)dx\,dy+\iint_{D_2}z\,dx\,dy=\iint_D\sqrt{x^2+y^2}\,dx\,dy$$

$= \int_0^{2\pi} d\theta \int_0^h r \cdot r\,dr = 2\pi \cdot \frac{1}{3} r^3 \Big|_0^h = \frac{2}{3}\pi h^3$

$= \iint_{D_1} \left(\sqrt{x^2+y^2} - \sqrt{x^2+y^2}\right) dx\,dy + \iint_{D_2} h \cdot dx\,dy = h \iint_{D_2} dx\,dy = h \cdot \pi h^2 = \pi h^3$.

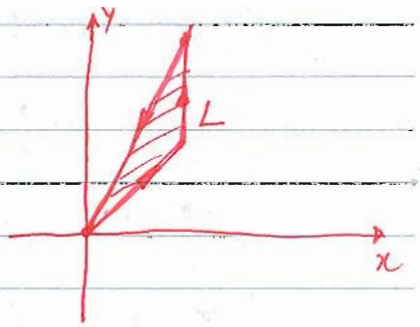
(事实上，由Gauss公式：$Q = \oint_\Sigma \vec{F} \cdot d\vec{S} = \iiint_\Omega \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) d\sigma = 3 \iiint_\Omega d\sigma = 3 \cdot \frac{1}{3}\pi h^3 = \pi h^3$)

## §8. Green公式和Stokes公式.

习题. P.141.

1. 利用Green公式计算下列曲线积分：

(1) $\oint_L (x^2y - 2y)\,dx + \left(\frac{x^3}{3} - x\right) dy$，其中$L$是由直线$y = x$，$y = 2x$和$x = 1$构成的三角形闭边，取逆时针为正向.



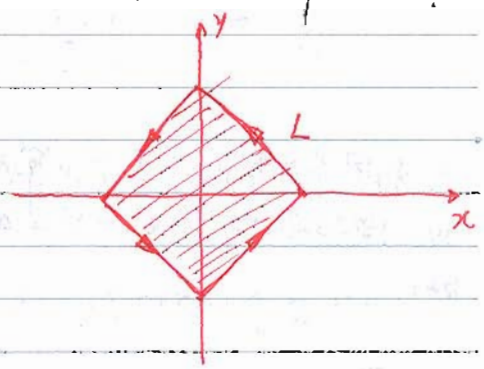

解：$P = x^2y - 2y$. $Q = \frac{x^3}{3} - x$.

$\frac{\partial P}{\partial y} = x^2 - 2$. $\frac{\partial Q}{\partial x} = x^2 - 1$.

$\oint_L (x^2y - 2y)\,dx + \left(\frac{x^3}{3} - x\right) dy = \oint_L P\,dx + Q\,dy$

$= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \iint_D (x^2 - 1 - x^2 + 2)\,dx\,dy = \iint_D dx\,dy = \frac{1}{2}$.

(2) $\oint_L (x+y)\,dx + (x-y)\,dy$，其中$L$是由方程$|x| + |y| = 1$所给出的闭曲线，取逆时针为正向.

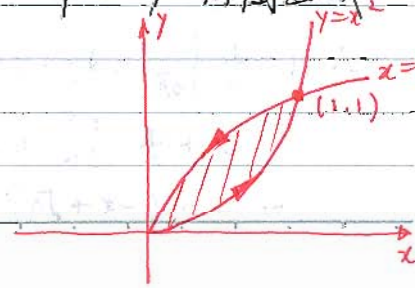



解：$P = x + y$. $Q = x - y$. $\frac{\partial P}{\partial y} = 1$. $\frac{\partial Q}{\partial x} = 1$.

$\oint_L (x+y)\,dx + (x-y)\,dy = \oint_L P\,dx + Q\,dy$

$= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = 0$.

(3) $\oint_L (-y^2 + e^{e^x})\,dx + \arctan y\,dy$. 其中$L$是由两条抛物线$y = x^2$和$x = y^2$所围成的区域的边界，取逆时针为正向.




解：$P = -y^2 + e^{e^x}$. $Q = \arctan y$. $\frac{\partial P}{\partial y} = -2y$. $\frac{\partial Q}{\partial x} = 0$.

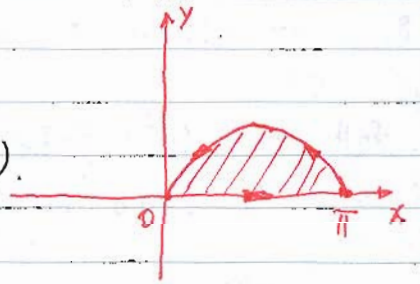
$\oint_L (-y^2 + e^{e^x})\,dx + \arctan y\,dy = \oint_L P\,dx + Q\,dy$

$$= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx dy = \iint_D 2y\, dx dy = 2\int_0^1 dx \int_{x^2}^{\sqrt{x}} y\, dy$$

$$= \int_0^1 (x - x^4)\, dx = \frac{1}{2} - \frac{1}{5} = \frac{3}{10}.$$

(4) $\oint_L e^x[(1-\cos y)dx - (y-\sin y)dy]$，其中 $L$ 是区域 $D = \left\{(x,y) \middle| \begin{matrix} 0 \le y \le \sin x \\ 0 \le x \le \pi \end{matrix}\right\}$ 的边界，

取逆时针为正向.




解: $P = e^x(1-\cos y)$. $Q = -e^x(y-\sin y)$

$\frac{\partial P}{\partial y} = e^x \sin y$. $\frac{\partial Q}{\partial x} = -e^x(y-\sin y) = e^x(\sin y - y)$.

$$\oint_L e^x[(1-\cos y)dx - (y-\sin y)dy] = \oint_L P\,dx + Q\,dy$$

$$= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx dy = \iint_D (-e^x y)\, dx dy = -\int_0^\pi e^x dx \int_0^{\sin x} y\, dy$$

$$= -\int_0^\pi e^x \cdot \frac{1}{2}\sin^2 x\, dx = \frac{1}{4}\int_0^\pi e^x \cos 2x\, dx - \frac{1}{4}\int_0^\pi e^x dx = \frac{1}{4}I - \frac{1}{4}(e^\pi - 1).$$

$$I = \int_0^\pi e^x \cos 2x\, dx = -\int_0^\pi \cos 2x\, de^x = -e^x \cos 2x\Big|_0^\pi + 2\int_0^\pi e^x \sin 2x\, dx$$

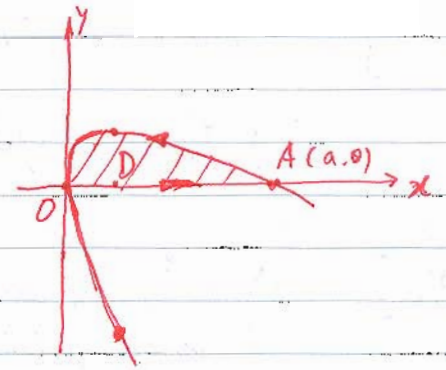
$$= (e^\pi - 1) + 2\int_0^\pi \sin 2x\, de^x = (e^\pi - 1) - 2\int_0^\pi e^x \cdot 2\cos 2x\, dx = (e^\pi - 1) - 4I.$$

$$\therefore I = \frac{1}{5}(e^\pi - 1)$$

$$\therefore \text{原式} = \frac{1}{4}\cdot\frac{1}{5}(e^\pi - 1) - \frac{1}{4}(e^\pi - 1) = \frac{1}{5}(1 - e^\pi).$$

2. 利用第二类曲线积分计算面积:

(1) 由抛物线 $(x+y)^2 = ax$ 和 $x$ 轴围成的平面图形.




解: $A = \frac{1}{2}\oint_{\partial D} x\,dy - y\,dx = \frac{1}{2}\left(\int_{\overline{OA}} + \int_{\widehat{AO}}\right)(x\,dy - y\,dx)$

$\overline{OA}$: $y = 0$. $x$: $0 \mapsto a$.

$\widehat{AO}$: $y = -x + \sqrt{ax}$. $x$: $a \mapsto 0$.

$$\therefore A = \frac{1}{2}\int_{\overline{OA}} x\,dy - y\,dx + \frac{1}{2}\int_{\widehat{AO}} x\,dy - y\,dx$$

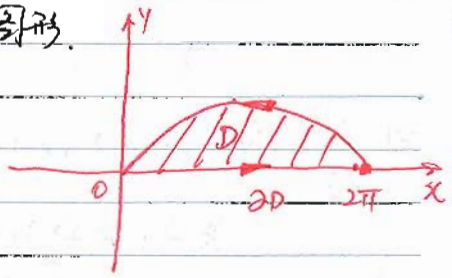
$$= 0 + \frac{1}{2}\int_a^0 x \cdot d(-x + \sqrt{ax}) - (-x + \sqrt{ax})\,dx$$

$$= \frac{1}{2}x(-x + \sqrt{ax})\Big|_a^0 - \frac{1}{2}\int_a^0 (-x + \sqrt{ax})\,dx - \frac{1}{2}\int_a^0 (-x + \sqrt{ax})\,dx$$

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$$= \int_0^a (-x+\sqrt{ax})\,dx = -\frac{1}{2}x^2 + \frac{2}{3}\sqrt{a}\,x^{\frac{3}{2}}\Big|_0^a = -\frac{1}{2}a^2 + \frac{2}{3}a^2 = \frac{1}{6}a^2.$$

(2) 由旋轮线段 $\begin{cases} x = a(t-\sin t) \\ y = a(1-\cos t) \end{cases}$ $t\in[0,2\pi]$ 与 $x$ 轴围成的平面图形.




解: $A = \frac{1}{2}\oint_{\partial D} x\,dy - y\,dx = \frac{1}{2}\left(\int_{\overline{OA}} + \int_{\widehat{AO}}\right)(x\,dy - y\,dx)$.

$\overline{OA}$: $y=0$, $x: 0 \mapsto 2\pi$.

$\widehat{AO}$: $\begin{cases} x = a(t-\sin t) \\ y = a(1-\cos t) \end{cases}$ $t: 2\pi \mapsto 0$

$$\therefore A = \frac{1}{2}\oint_{\partial D} x\,dy - y\,dx = \frac{1}{2}\int_{\overline{OA}} x\,dy - y\,dx + \frac{1}{2}\int_{\widehat{AO}} x\,dy - y\,dx$$

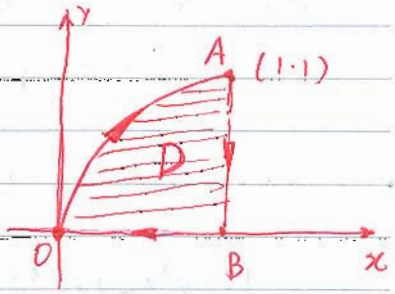
$$= 0 + \frac{1}{2}\int_{2\pi}^{0} a(t-\sin t)\,d[a(1-\cos t)] - a(1-\cos t)\,d[a(t-\sin t)]$$

$$= \frac{1}{2}\int_{2\pi}^{0} a^2\left[(t-\sin t)\cdot\sin t - (1-\cos t)^2\right]dt$$

$$= \frac{a^2}{2}\int_{2\pi}^{0} (t\sin t + 2\cos t - 2)\,dt = \frac{a^2}{2}\int_{2\pi}^{0} (-t\,d\cos t) - 2\,dt = 2\pi a^2.$$

3. 利用 Green 公式计算下列第二类曲线积分:

(1) $\int_L (x^2-y)\,dx - (x+\sin^2 y)\,dy$, 其中 $L$ 是图形 $y = \sqrt{2x-x^2}$ 上由 $(0,0)$ 到 $(1,1)$ 的一段弧.




解: $\left(\int_L + \int_{\overline{AB}} + \int_{\overline{BO}}\right)(x^2-y)\,dx - (x+\sin^2 y)\,dy$

$$= -\iint_D \left\{\frac{\partial}{\partial x}\left[-(x+\sin^2 y)\right] - \frac{\partial}{\partial y}\left[x^2-y\right]\right\}dx\,dy$$

$$= -\iint_D (-1+1)\,dx\,dy = 0.$$

$\overline{AB}$: $x=1$, $y: 1 \mapsto 0$.

$$\int_{\overline{AB}} (x^2-y)\,dx - (x+\sin^2 y)\,dy = \int_1^0 \left[-(1+\sin^2 y)\right]dy = \int_0^1 \left(1 + \frac{1}{2} - \frac{1}{2}\cos 2y\right)dy$$

$$= \frac{3}{2} - \frac{1}{4}\sin 2.$$

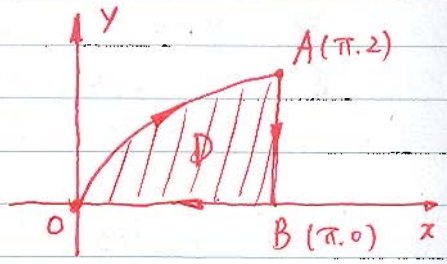
$\overline{BO}$: $y=0$, $x: 1 \mapsto 0$

$$\int_{\overline{BO}} (x^2-y)\,dx - (x+\sin^2 y)\,dy = \int_1^0 x^2\,dx = -\frac{1}{3}.$$

$$\therefore \int_L (x^2-y)\,dx-(x+\sin^2 y)\,dy = -\left(\int_{\overline{AB}}+\int_{\overline{BO}}\right)(x^2-y)\,dx-(x+\sin^2 y)\,dy$$

$$= -\left(\frac{3}{2}-\frac{1}{4}\sin 2\right)-\left(-\frac{1}{3}\right) = -\frac{7}{6}+\frac{1}{4}\sin 2.$$

(2) $\int_L (x^2y-3xe^x)\,dx+\left(\frac{1}{3}x^3-y\sin y\right)dy$，其中$L$是沿摆线 $\begin{cases} x=t-\sin t \\ y=1-\cos t \end{cases}$ 从点 $(0,0)$ 到点 $A(\pi,2)$ 的一段弧.




解：$\left(\int_L+\int_{\overline{AB}}+\int_{\overline{BO}}\right)(x^2y-3xe^x)\,dx+\left(\frac{1}{3}x^3-y\sin y\right)dy$

$$= -\iint_D\left[\frac{\partial}{\partial x}\left(\frac{1}{3}x^3-y\sin y\right)-\frac{\partial}{\partial y}\left(x^2y-3xe^x\right)\right]dx\,dy$$

$$= -\iint_D (x^2-x^2)\,dx\,dy = 0.$$

$\overline{AB}$：$x=\pi$，$y$：$2\mapsto 0$.

$$\int_{\overline{AB}}(x^2y-3xe^x)\,dx+\left(\frac{1}{3}x^3-y\sin y\right)dy = \int_2^0\left(\frac{1}{3}\pi^3-y\sin y\right)dy$$

$$= -\frac{2}{3}\pi^3+\int_0^2 y\sin y\,dy = -\frac{2}{3}\pi^3-y\cos y\Big|_0^2+\int_0^2\cos y\,dy$$

$$= -\frac{2}{3}\pi^3-2\cos 2+\sin 2.$$

$\overline{BO}$：$y=0$，$x$：$\pi\mapsto 0$.

$$\int_{\overline{BO}}(x^2y-3xe^x)\,dx+\left(\frac{1}{3}x^3-y\sin y\right)dy = \int_\pi^0(-3xe^x)\,dx = 3\int_0^\pi x\,de^x$$

$$= 3xe^x\Big|_0^\pi-3\int_0^\pi e^x dx = 3\pi e^\pi-3e^x\Big|_0^\pi = 3e^\pi(\pi-1)+3.$$

$$\therefore \int_L (x^2y-3xe^x)\,dx+\left(\frac{1}{3}x^3-y\sin y\right)dy = -\left(\int_{\overline{AB}}+\int_{\overline{BO}}\right)(x^2y-3xe^x)\,dx+\left(\frac{1}{3}x^3-y\sin y\right)dy$$

$$= -\left(-\frac{2}{3}\pi^3-2\cos 2+\sin 2\right)-\left[-3e^\pi(\pi-1)+3\right]$$

$$= \frac{2}{3}\pi^3+2\cos 2-\sin 2-3(\pi-1)e^\pi-3.$$

4. 利用 Stokes 公式计算第二类曲线积分.

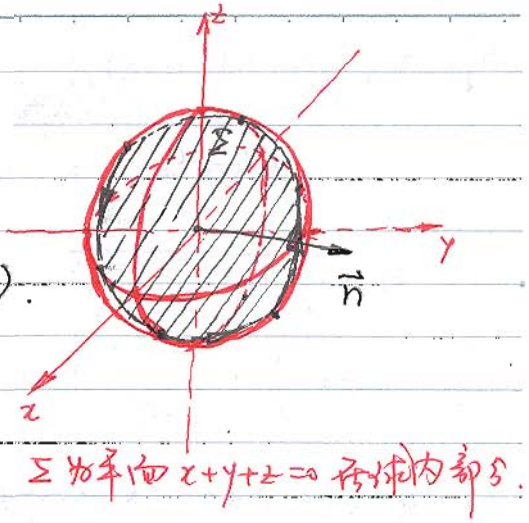
(1) $\oint_L y\,dx+z\,dy+x\,dz$，其中$L$是圆周 $\begin{cases} x^2+y^2+z^2=a^2 \\ x+y+z=0 \end{cases}$，顶视为逆时针方向

解: $\oint_L y\,dx + z\,dy + x\,dz = \iint_\Sigma \begin{vmatrix} dy\,dz & dz\,dx & dx\,dy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix}$

$= \iint_\Sigma (-dy\,dz - dz\,dx - dx\,dy)$ $\Sigma: \vec{n} = \frac{1}{\sqrt{3}}(1,1,1)$.

$= \iint_\Sigma (-1,-1,-1)\cdot\frac{1}{\sqrt{3}}(1,1,1)\,dS$

$= -\sqrt{3}\iint_\Sigma dS = -\sqrt{3}\,\pi a^2$.




Σ为平面 $x+y+z=0$ 在球内部分.

(2) $\oint_L (y^2+z^2)dx + (z^2+x^2)dy + (x^2+y^2)dz$, 其中L是上半球面 $z=\sqrt{R^2-x^2-y^2}$ 与圆柱面 $x^2+y^2=2rx$ ($R>2r>0$) 的交线, 顶视为逆时针走向

解: $\Sigma: \vec{n} = (-z'_x, -z'_y, 1)\Big/\sqrt{1+z'^2_x+z'^2_y}$. $z=\sqrt{R^2-x^2-y^2}$.

$= \left(\frac{x}{R}, \frac{y}{R}, \frac{z}{R}\right)$.

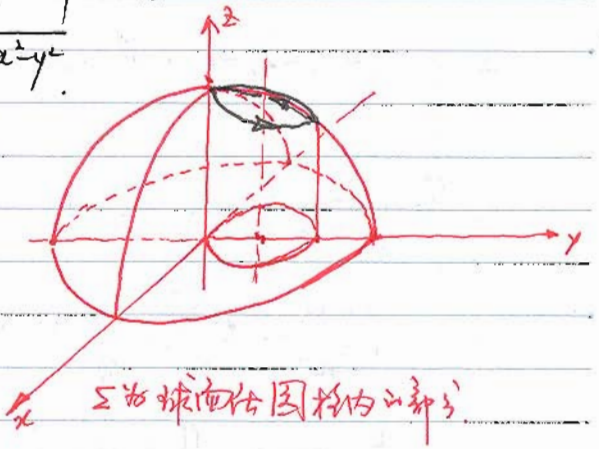
$\oint_L (y^2+z^2)dx + (z^2+x^2)dy + (x^2+y^2)dz$

$= \iint_\Sigma \begin{vmatrix} dy\,dz & dz\,dx & dx\,dy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2+z^2 & z^2+x^2 & x^2+y^2 \end{vmatrix}$

$= \iint_\Sigma (2y-2z)\,dy\,dz + (2z-2x)\,dz\,dx + (2x-2y)\,dx\,dy$

$= \iint_\Sigma (2y-2z, 2z-2x, 2x-2y)\cdot\left(\frac{x}{R}, \frac{y}{R}, \frac{z}{R}\right)dS$

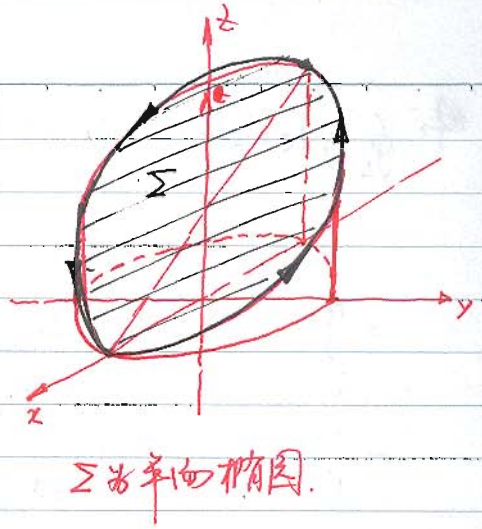
$= \iint_\Sigma \frac{2}{R}(xy - xz + zy - xy + xz - yz)\,dS = \iint_\Sigma \frac{2}{R}\cdot 0\,dS = 0$.




Σ为球面位圆柱内部分.

(3) $\oint_L (y-z)dx + (z-x)dy + (x-y)dz$, 其中L为椭圆 $\begin{cases} x^2+y^2=a^2 \\ bx+az=ab \end{cases}$ $(a>0, b>0)$,

从x轴正向看去, 取逆时针走向.



Σ为平面椭圆.

解: $\Sigma: \vec{n} = (-b, 0, -a) \Big/ \sqrt{a^2+b^2} = \frac{1}{\sqrt{a^2+b^2}}(-b, 0, -a)$.

(平面方程: $z = b - \frac{b}{a}x$).

$$dS = \sqrt{1 + z_x'^2 + z_y'^2}\, dx\,dy$$

$$= \sqrt{1 + \frac{b^2}{a^2}}\, dx\,dy = \frac{1}{a}\sqrt{a^2+b^2}\, dx\,dy.$$

$$\oint_L (y-z)dx + (z-x)dy + (x-y)dz$$

$$= \iint_\Sigma \begin{vmatrix} dy\,dz & dz\,dx & dx\,dy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y-z & z-x & x-y \end{vmatrix} = \iint_\Sigma (-2\,dy\,dz - 2\,dz\,dx - 2\,dx\,dy)$$

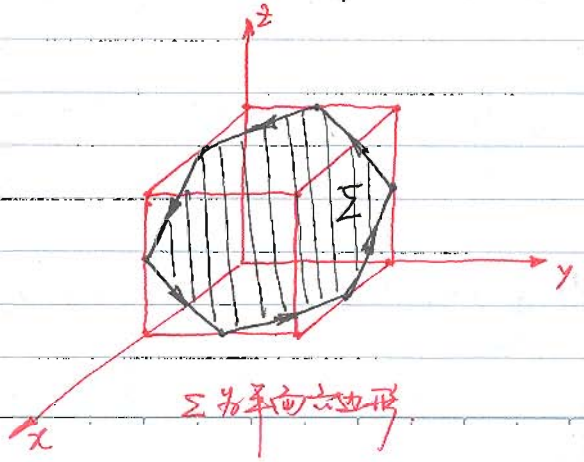
$$= \iint_\Sigma (-2,-2,-2)\cdot \vec{n}\, dS = \iint_\Sigma (-2,-2,-2)\cdot \frac{1}{\sqrt{a^2+b^2}}(-b, 0, -a)\, dS$$

$$= \iint_\Sigma \frac{1}{\sqrt{a^2+b^2}}(2b - 2a)\, dS = -\frac{2(a+b)}{\sqrt{a^2+b^2}} \iint_\Sigma dS = -\frac{2(a+b)}{\sqrt{a^2+b^2}} \iint_D \frac{1}{a}\sqrt{a^2+b^2}\, dx\,dy$$

$$= -\frac{2(a+b)}{\sqrt{a^2+b^2}} \cdot \frac{\sqrt{a^2+b^2}}{a} \iint_D dx\,dy. \qquad D = \{(x,y) \mid x^2+y^2 \le a^2\}$$

$$= -\frac{2}{a}(a+b) \cdot \iint_D dx\,dy = -\frac{2}{a}(a+b)\cdot \pi a^2 = -2\pi a(a+b)$$

(4) $\oint_L (y^2-z^2)dx + (z^2-x^2)dy + (x^2-y^2)dz$, 其中L是用平面 $x+y+z=\frac{3}{2}$ 截立方体 $\begin{cases} 0 \le x \le 1 \\ 0 \le y \le 1 \\ 0 \le z \le 1 \end{cases}$ 的表面所得的截线, 取逆时针方向.



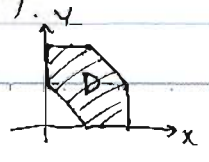
Σ为平面六边形

解: $\Sigma: x+y+z=\frac{3}{2}$, $\begin{matrix} 0 \le x \le 1 \\ 0 \le y \le 1 \\ 0 \le z \le 1 \end{matrix}$ $z = \frac{3}{2} - x - y$.

$$\vec{n} = \frac{1}{\sqrt{3}}(1,1,1).$$

$$dS = \sqrt{1 + z_x'^2 + z_y'^2} = \sqrt{3}\, dx\,dy$$

$$D = \left\{(x,y) \;\middle|\; \begin{matrix} \frac{1}{2} \le x+y \le \frac{3}{2} \\ 0 \le x, y \le 1 \end{matrix}\right\}$$



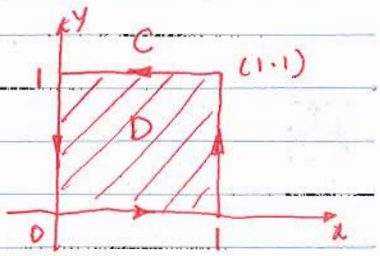
$$\oint_L (y^2-z^2)dx+(z^2-x^2)dy+(x^2-y^2)dz = \iint_\Sigma \begin{vmatrix} \cos(\vec{n},\vec{x}) & \cos(\vec{n},\vec{y}) & \cos(\vec{n},\vec{z}) \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} dS$$

$$= \iint_\Sigma \begin{vmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2-z^2 & z^2-x^2 & x^2-y^2 \end{vmatrix} dS = \frac{1}{\sqrt{3}} \iint_\Sigma (-2y-2z-2z-2x-2x-2y)\,dS$$

$$= -\frac{4}{\sqrt{3}} \iint_\Sigma (x+y+z)\,dS = -\frac{4}{\sqrt{3}} \iint_D \frac{3}{2}\sqrt{3}\,dxdy = -6\iint_D dxdy$$

$$= -6\left(1-\frac{1}{2}\cdot 2\cdot\frac{1}{2}\cdot\frac{1}{2}\right) = -6\cdot\frac{3}{4} = -\frac{9}{2}.$$

5. 设 $f(x)$ 是 $[0,1]$ 上的连续正值一元函数，平面区域 $D=\{(x,y)\mid 0\le x\le 1,\ 0\le y\le 1\}$，$C$ 为 $D$ 的正向边界，证明：$\oint_C x f(y)\,dy - \frac{y}{f(x)}dx \ge 2$.




证明：$\oint_C x f(y)\,dy - \frac{y}{f(x)}dx$

$$= \iint_D \left[\frac{\partial}{\partial x}(x f(y)) - \frac{\partial}{\partial y}\left(-\frac{y}{f(x)}\right)\right] dxdy$$

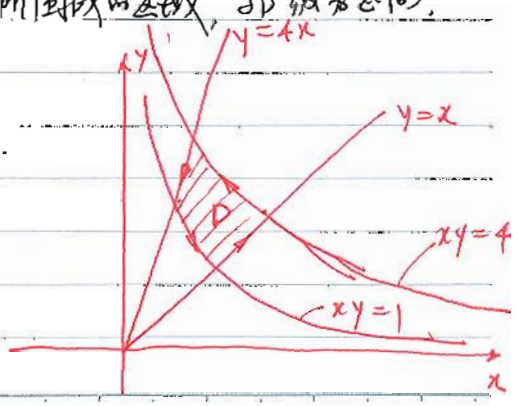
$$= \iint_D \left[f(y)+\frac{1}{f(x)}\right] dxdy = \iint_D f(y)\,dxdy + \iint_D \frac{1}{f(x)}dxdy$$

$$= \int_0^1 dx\cdot\int_0^1 f(y)\,dy + \int_0^1 dy\cdot\int_0^1 \frac{1}{f(x)}dx = \int_0^1 f(y)\,dy + \int_0^1 \frac{1}{f(x)}dx$$

$$= \int_0^1 f(x)\,dx + \int_0^1 \frac{1}{f(x)}dx = \int_0^1 \left[f(x)+\frac{1}{f(x)}\right] dx \ge \int_0^1 2\,dx = 2. \quad \text{※}$$

6. 设 $D$ 为两条直线 $y=x$，$y=4x$ 和两条双曲线 $xy=1$，$xy=4$ 所围成的区域，$\partial D$ 取为正向，$F(u)$ 是具有连续导数的一元函数，记 $f(u)=F'(u)$. 证明：

$$\int_{\partial D} \frac{F(xy)}{y}dy = \ln 2 \int_1^4 f(u)\,du.$$




证明：令 $P=0$，$Q=\frac{F(xy)}{y}$，则：

$$\int_{\partial D} \frac{F(xy)}{y}dy = \iint_D \frac{\partial}{\partial x}\left(\frac{F(xy)}{y}\right) dxdy$$

$$= \iint_D \frac{1}{y} \cdot F'(xy) \cdot y \, dx\,dy = \iint_D F'(xy)\, dx\,dy. \quad \text{作变换：} \begin{cases} u = xy \\ v = \dfrac{y}{x} \end{cases}$$

$$\left|\begin{matrix} x'_u & x'_v \\ y'_u & y'_v \end{matrix}\right| = \frac{1}{\left|\begin{matrix} u'_x & u'_y \\ v'_x & v'_y \end{matrix}\right|} = \frac{1}{\left|\begin{matrix} y & x \\ -\frac{y}{x^2} & \frac{1}{x} \end{matrix}\right|} = \frac{1}{\frac{2y}{x}} = \frac{1}{2} \cdot \frac{x}{y} = \frac{1}{2v}.$$

即：$dx\,dy = \left|\begin{matrix} x'_u & x'_v \\ y'_u & y'_v \end{matrix}\right| du\,dv = \frac{1}{2v} du\,dv$. $D = \left\{ (u,v) \,\middle|\, \begin{matrix} 1 \le u \le 4 \\ 1 \le v \le 4 \end{matrix} \right\}$.

$$\therefore \int_{\partial D} \frac{F(xy)}{y}\, dy = \iint_D F'(xy)\, dx\,dy = \iint_D F'(u) \cdot \frac{1}{2v}\, du\,dv$$

$$= \int_1^4 F'(u)\, du \cdot \int_1^4 \frac{1}{2v}\, dv = \int_1^4 F'(u)\, du \cdot \frac{1}{2} \ln 4$$

$$= \ln 2 \cdot \int_1^4 f(u)\, du.$$

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# 第三篇 多元函数微积分

## 第七章 多元函数微分学

## 第八章 多元函数积分学

### §9. 旋度和无旋场

习题 P.151.

1. 求下列向量场的旋度.

(1) $\vec{F} = (3x^2z+2y)\vec{i} + (y^2-3xz)\vec{j} + 3xyz\vec{k}$.

解: $\operatorname{rot}\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2z+2y & y^2-3xz & 3xyz \end{vmatrix} = (3xz+3x)\vec{i} + (3x^2-3yz)\vec{j} + (-3z-2)\vec{k}$

$= (3xz+3x,\ 3x^2-3yz,\ -3z-2)$.

(2) $\vec{F} = y^2z\vec{i} + z^2x\vec{j} + x^2y\vec{k}$.

解: $\operatorname{rot}\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2z & z^2x & x^2y \end{vmatrix} = (x^2-2xz)\vec{i} + (y^2-2xy)\vec{j} + (z^2-2yz)\vec{k}$

$= (x^2-2xz,\ y^2-2xy,\ z^2-2yz)$.

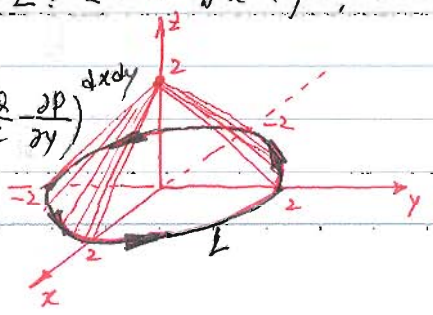
(3) $\vec{F} = P(x)\vec{i} + Q(y)\vec{j} + R(x)\vec{k}$.

解: $\operatorname{rot}\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P(x) & Q(y) & R(x) \end{vmatrix} = 0\cdot\vec{i} + [-R'(x)]\vec{j} + 0\cdot\vec{k} = (0, -R'(x), 0)$.

2. 求向量场 $\vec{F} = (x-z)\vec{i} + (x^3+yz)\vec{j} + 3xy^2\vec{k}$ 沿圆周 $L: z = 2-\sqrt{x^2+y^2}$, $z=0$ 的边界, 顶视 $L$ 为逆时针方向

解: $\Gamma = \oint_L \vec{F}\cdot d\vec{r} = \iint_\Sigma \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)dydz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)dxdz + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)dxdy$

$\Sigma: \begin{cases} \sqrt{x^2+y^2} \le 2 \\ z=0 \end{cases}$ $\vec{n} = (0,0,1)$.



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$$\therefore P=\oint_L \vec{F}\cdot d\vec{r}=\iint_\Sigma \operatorname{rot}\vec{F}\cdot\vec{n}\,dS$$

$$=\iint_\Sigma\left(\begin{vmatrix}\frac{\partial}{\partial y}&\frac{\partial}{\partial z}\\ Q&R\end{vmatrix},\begin{vmatrix}\frac{\partial}{\partial z}&\frac{\partial}{\partial x}\\ R&P\end{vmatrix},\begin{vmatrix}\frac{\partial}{\partial x}&\frac{\partial}{\partial y}\\ P&Q\end{vmatrix}\right)\cdot(0,0,1)\,dS$$

$$=\iint_\Sigma\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dS=\iint_{x^2+y^2\le 4}3x^2\cdot dxdy=\iint_{\substack{0\le r\le 2\\ 0\le\theta\le 2\pi}}3r^2\cos^2\theta\cdot r\,dr\,d\theta$$

$$=\int_0^{2\pi}\cos^2\theta\,d\theta\int_0^2 3r^3\,dr=\frac{1}{2}\cdot 2\pi\cdot\frac{3}{4}\cdot 2^4=12\pi.$$

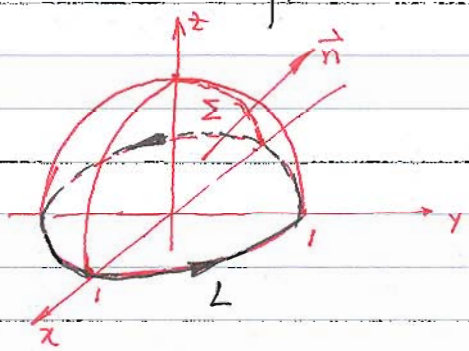
3. 利用 Stokes 公式把 $\iint_\Sigma \operatorname{rot}\vec{F}\cdot d\vec{S}$ 化为曲线积分，并计算积分值.

(1) $\vec{F}=xyz\,\vec{i}+x\,\vec{j}+e^{xy}\cos z\,\vec{k}$，$\Sigma$ 是半椭面 $x^2+y^2+z^2=1$，$z\ge 0$ 的上侧.

解: $\oint_L \vec{F}\cdot d\vec{r}=\iint_\Sigma \operatorname{rot}\vec{F}\cdot d\vec{S}=\iint_\Sigma \operatorname{rot}\vec{F}\cdot\vec{n}\,dS$

$\Sigma$: $x^2+y^2+z^2=1$，$z\ge 0$ 的上侧，$z=\sqrt{1-x^2-y^2}$.

$L$: $\begin{cases}x=\cos\theta\\ y=\sin\theta\\ z=0.\end{cases}$ $\theta: 0\mapsto 2\pi$.




(I): $\vec{n}=\frac{1}{\sqrt{z_x'^2+z_y'^2+1}}(-z_x',-z_y',1)=(x,y,z)$.

$$\operatorname{rot}\vec{F}=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\ \frac{\partial}{\partial x}&\frac{\partial}{\partial y}&\frac{\partial}{\partial z}\\ xyz&x&e^{xy}\cos z\end{vmatrix}=xe^{xy}\cos z\,\vec{i}+(xy-ye^{xy}\cos z)\vec{j}+(1-xz)\vec{k}$$

$$\iint_\Sigma \operatorname{rot}\vec{F}\cdot d\vec{S}=\iint_\Sigma(xe^{xy}\cos z,\ xy-ye^{xy}\cos z,\ 1-xz)\cdot(x,y,z)\,dS$$

$$=\iint_\Sigma\left(x^2e^{xy}\cos z+xy^2-y^2e^{xy}\cos z+z-xz^2\right)dS \qquad D=\left\{(r,\theta)\,\middle|\,\begin{matrix}0\le r\le 1\\ 0\le\theta\le\frac{\pi}{2}\end{matrix}\right\}$$

$$=\iint_D\left[(x^2-y^2)e^{xy}\cos\sqrt{1-x^2-y^2}+xy^2-x(1-x^2-y^2)+\sqrt{1-x^2-y^2}\right]\cdot\frac{1}{\sqrt{1-x^2-y^2}}dxdy$$

$$=\iint_D\frac{1}{\sqrt{1-r^2}}\left[r^2\cos 2\theta\cdot e^{\frac{1}{2}r^2\sin 2\theta}\cdot\cos\sqrt{1-r^2}+r^3\sin^2\theta\cos\theta-r\sqrt{1-r^2}\cos\theta+\sqrt{1-r^2}\right]r\,dr\,d\theta$$

$$=\int_0^{2\pi}d\theta\cdot\int_0^1\frac{r}{\sqrt{1-r^2}}\left[r^2\cos\sqrt{1-r^2}\cos 2\theta\cdot e^{\frac{r^2}{2}\sin 2\theta}+r^3\sin^2\theta\cos\theta-r\sqrt{1-r^2}\cos\theta\right]dr$$

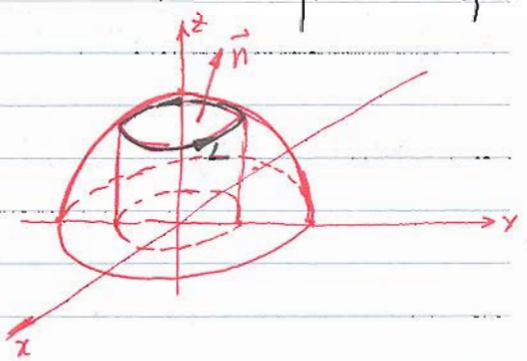
$$+\int_0^{2\pi} d\theta \int_0^1 r\,dr = 0 + 2\pi\cdot\frac{1}{2}r^2\Big|_0^1 = \pi.$$

(II): $\iint_\Sigma \mathrm{rot}\,\vec{F}\cdot\vec{dS} = \oint_L \vec{F}\cdot d\vec{r} = \oint_L xyz\,dx + x\,dy + e^{xy}\cos z\,dz.$

$(L: z=0)$ $= \oint_L x\,dy = \int_0^{2\pi} \cos\theta\cdot d\sin\theta = \int_0^{2\pi} \cos^2\theta\,d\theta$

$$= \int_0^{2\pi} \frac{1}{2}(1+\cos 2\theta)\,d\theta = \frac{1}{2}\cdot 2\pi = \pi.$$

(2) $\vec{F} = yz\vec{i} + xz\vec{j} + xy\vec{k}$, $\Sigma$ 是 $x^2+y^2+z^2=4$ 包含于柱面 $x^2+y^2=1$ 内部且 $z>0$ 的部分，取上侧.




解: $\Sigma: \begin{cases} x^2+y^2+z^2=4 \\ z>0 \\ x^2+y^2\le 1 \end{cases}$ $L: \begin{cases} x^2+y^2=1 \\ z=\sqrt{3} \end{cases} \Rightarrow \begin{cases} x=\cos\theta \\ y=\sin\theta \\ z=\sqrt{3} \end{cases}$ $\theta: 0\mapsto 2\pi.$

$$\iint_\Sigma \mathrm{rot}\,\vec{F}\cdot\vec{dS} = \oint_L \vec{F}\cdot d\vec{r}$$

$$= \oint_L yz\,dx + xz\,dy + xy\,dz$$

$$= \int_0^{2\pi}\left(\sqrt{3}\sin\theta\,d\cos\theta + \sqrt{3}\cos\theta\,d\sin\theta\right) = \int_0^{2\pi}\sqrt{3}\,(\cos^2\theta-\sin^2\theta)\,d\theta$$

$$= \sqrt{3}\int_0^{2\pi}\cos 2\theta\,d\theta = 0.$$

4. 证明下列向量场为势场，并求其势函数.

(1) $\vec{F} = y\cos(xy)\vec{i} + x\cos(xy)\vec{j} + \sin z\vec{k}$

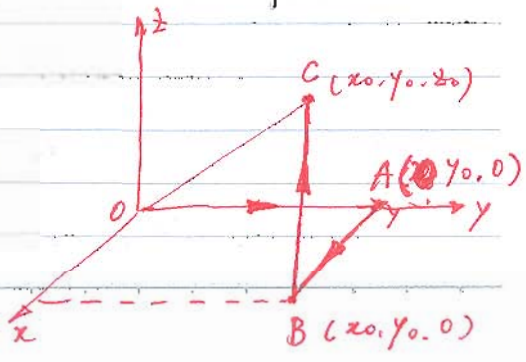
解: $\mathrm{rot}\,\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y\cos(xy) & x\cos(xy) & \sin z \end{vmatrix} = 0\cdot\vec{i} + 0\cdot\vec{j} + [\cos(xy) - xy\sin(xy) - \cos(xy) + xy\sin(xy)]\vec{k}$

$= \vec{0}$. $\therefore \vec{F}$ 为无旋场，即为势场.

设 $\mathrm{grad}\,U(x,y,z) = \vec{F}$.

$$U(x_0,y_0,z_0) = \int_{(0,0,0)}^{(x_0,y_0,z_0)} P\,dx + Q\,dy + R\,dz$$

$$= \left(\int_{\overline{OA}} + \int_{\overline{AB}} + \int_{\overline{BC}}\right) P\,dx + Q\,dy + R\,dz.$$



No. 4

Date

$\overline{OA}$: $\begin{cases} x=0 \\ z=0 \end{cases}$ $y: 0 \mapsto y_0$. $\overline{AB}$: $\begin{cases} y=y_0 \\ z=0 \end{cases}$ $x: 0 \mapsto x_0$. $\overline{BC}$: $\begin{cases} x=x_0 \\ y=y_0 \end{cases}$ $z: 0 \mapsto z_0$.

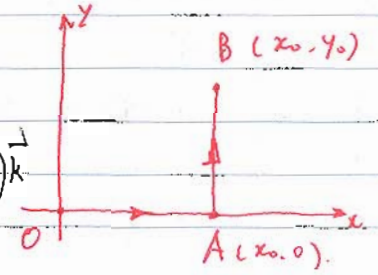
$$u(x_0,y_0,z_0) = \int_0^{y_0} x\cos(xy)\,dy + \int_0^{x_0} y\cos(xy)\,dx + \int_0^{z_0} \sin z\,dz$$

$$= \int_0^{y_0} 0\cdot\cos 0\,dy + \int_0^{x_0} y_0\cos(xy_0)\,dx + \int_0^{z_0} \sin z\,dz$$

$$= \sin xy_0\Big|_0^{x_0} - \cos z\Big|_0^{z_0} = \sin x_0y_0 - \cos z_0 + 1.$$

$\therefore\ u(x,y,z) = \sin xy - \cos z + C$

(2) $\vec{F} = (2x\cos y - y^2\sin x)\vec{i} + (2y\cos x - x^2\sin y)\vec{j}$.




解: $\mathrm{rot}\,\vec{F} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\vec{k} = \left(-2y\sin x - 2x\sin y + 2x\sin y + 2y\sin x\right)\vec{k}$

$= \vec{0}$. $\therefore \vec{F}$ 为无旋场 即势场.

设 $\mathrm{grad}\,u(x,y,0) = \vec{F}$.

$$u(x_0,y_0) = \int_{(0,0)}^{(x_0,y_0)} P\,dx + Q\,dy = \int_{(0,0)}^{(x_0,0)} P(x,0)\,dx + \int_{(x_0,0)}^{(x_0,y_0)} Q(x_0,y)\,dy$$

$$= \int_0^{x_0} 2x\,dx + \int_0^{y_0} (2y\cos x_0 - x_0^2\sin y)\,dy$$

$$= x_0^2 + y_0^2\cos x_0 + x_0^2\cos y_0 - x_0^2 = x_0^2\cos y_0 + y_0^2\cos x_0.$$

$\therefore\ u(x,y) = x^2\cos y + y^2\cos x + C$.

5. 验证下列微分形式均是全微分, 求其一个原函数

(1) $2xy\,dx + x^2\,dy$

解: $P = 2xy$. $Q = x^2$. $\frac{\partial P}{\partial y} = 2x = \frac{\partial Q}{\partial x}$ 记 $\vec{F} = P\vec{i} + Q\vec{j}$. 则 $\mathrm{rot}\,\vec{F} = \vec{0}$.

即存在 $u(x,y)$ 满足 $\mathrm{grad}\,u(x,y) = \vec{F} = P\vec{i} + Q\vec{j}$. 且 $du = P\,dx + Q\,dy = \vec{F}\cdot d\vec{r}$.

$\therefore$ 上述微分形式为全微分.

$$u(x_0,y_0) = \int_{(0,0)}^{(x_0,0)} P(x_0,0)\,dx + \int_{(x_0,0)}^{(x_0,y_0)} Q(x_0,y)\,dy$$

$$= \int_0^{x_0} 0\cdot dx + \int_0^{y_0} x_0^2\,dy = x_0^2y_0. \quad \therefore\ u(x,y) = x^2y + C$$

(2) $(3x^2y+8xy^2)dx+(x^3+8x^2y+12ye^y)dy$.

解: $P=3x^2y+8xy^2$. $Q=x^3+8x^2y+12ye^y$. $\frac{\partial P}{\partial y}=3x^2+16xy$. $\frac{\partial Q}{\partial x}=3x^2+16xy$.

$\therefore \frac{\partial P}{\partial y}=\frac{\partial Q}{\partial x}$. 与路径无关为全微分.

$$u(x_0,y_0)=\int_{(0,0)}^{(x_0,y_0)}P\,dx+Q\,dy=\int_{(0,0)}^{(x_0,0)}P(x,0)dx+\int_{(x_0,0)}^{(x_0,y_0)}Q(x_0,y)dy$$

$$=\int_0^{x_0}0\cdot dx+\int_0^{y_0}(x_0^3+8x_0^2y+12ye^y)dy$$

$$=x_0^3y_0+4x_0^2y_0^2+12y_0e^{y_0}-12e^{y_0}+12$$

$\therefore u(x,y)=x^3y+4x^2y^2+12ye^y-12e^y+C$.

$=x^2y(x+4y)+12e^y(y-1)+C$.

b. 验证下列曲线积分与路径无关，并计算积分值.

(1) $\int_{(1,1,1)}^{(2,3,4)}x^3dx+y^3dy+z^3dz$.

解: 设 $\vec{F}=x^3\vec{i}+y^3\vec{j}+z^3\vec{k}$. $\operatorname{rot}\vec{F}=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\ \frac{\partial}{\partial x}&\frac{\partial}{\partial y}&\frac{\partial}{\partial z}\\ x^3&y^3&z^3\end{vmatrix}=\vec{0}$.

$\therefore \vec{F}$ 为保守场. 存在 $u(x,y,z)$ 满足 $\int_A^B P\,dx+Q\,dy+R\,dz=u(B)-u(A)$.

$$u(x_0,y_0,z_0)=\int_{(0,0,0)}^{(x_0,0,0)}P(x,0,0)dx+\int_{(x_0,0,0)}^{(x_0,y_0,0)}Q(x_0,y,0)dy+\int_{(x_0,y_0,0)}^{(x_0,y_0,z_0)}R(x_0,y_0,z)dz$$

$$=\int_0^{x_0}x^3dx+\int_0^{y_0}y^3dy+\int_0^{z_0}z^3dz=\frac{1}{4}(x_0^4+y_0^4+z_0^4).$$

$\therefore u(x,y,z)=\frac{1}{4}(x^4+y^4+z^4)$.

$\therefore \int_{(1,1,1)}^{(2,3,4)}x^3dx+y^3dy+z^3dz=u(2,3,4)-u(1,1,1)=\frac{1}{4}(2^4+3^4+4^4-3)=\frac{175}{2}$.

(2) $\int_{(0,0,0)}^{(1,2,2)}yz(2x+y+z)dx+zx(x+2y+z)dy+xy(x+y+2z)dz$

解: 设 $\vec{F}=yz(2x+y+z)\vec{i}+zx(x+2y+z)\vec{j}+xy(x+y+2z)\vec{k}=P\vec{i}+Q\vec{j}+R\vec{k}$.

$$\operatorname{rot}\vec{F}=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\ \frac{\partial}{\partial x}&\frac{\partial}{\partial y}&\frac{\partial}{\partial z}\\ yz(2x+y+z)&zx(x+2y+z)&xy(x+y+2z)\end{vmatrix}$$

$= (x^2+2xy+2xz-x^2-2xy-2xz)\vec{i} + (2xy+y^2+2yz-2xy-y^2-2yz)\vec{j}$

$+ (2xz+2yz+z^2-2xz-2yz-z^2)\vec{k} = \vec{0}$.

∴ $\vec{F}$ 为保守场. 存在 $U(x,y,z)$, 满足 $\int_A^B P\,dx+Q\,dy+R\,dz = U(B)-U(A)$.

$U(x_0,y_0,z_0) = \int_0^{x_0} P(x,0,0)\,dx + \int_0^{y_0} Q(x_0,y,0)\,dy + \int_0^{z_0} R(x_0,y_0,z)\,dz$

$= \int_0^{x_0} 0\cdot dx + \int_0^{y_0} 0\cdot dy + \int_0^{z_0} x_0y_0(x_0+y_0+2z)\,dz$

$= x_0y_0z_0(x_0+y_0+z_0)$. ∴ $U(x,y,z) = xyz(x+y+z)+C$.

∴ $\int_{(0,0,0)}^{(2,2,2)} yz(2x+y+z)\,dx + zx(x+2y+z)\,dy + xy(x+y+2z)\,dz$

$= U(2,2,2) - U(0,0,0) = 2\times2\times2(2+2+2) - 0 = 48$.

(3) $\int_{(0,0,0)}^{(3,2,\frac{\pi}{3})} (y+\sin z)\,dx + x\,dy + x\cos z\,dz$.

解: 设 $\vec{F} = (y+\sin z)\vec{i} + x\vec{j} + x\cos z\vec{k} = P\vec{i} + Q\vec{j} + R\vec{k}$.

$\text{rot}\,\vec{F} = (\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z})\vec{i} + (\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x})\vec{j} + (\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y})\vec{k}$

$= (0-0)\vec{i} + (\cos z - \cos z)\vec{j} + (1-1)\vec{k} = \vec{0}$

∴ $\vec{F}$ 为保守场. 存在 $U(x,y,z)$, $\int_A^B P\,dx+Q\,dy+R\,dz = U(B)-U(A)$.

$U(x_0,y_0,z_0) = \int_0^{x_0} P(x,0,0)\,dx + \int_0^{y_0} Q(x_0,y,0)\,dy + \int_0^{z_0} R(x_0,y_0,z)\,dz$

$= \int_0^{x_0} 0\cdot dx + \int_0^{y_0} x_0\,dy + \int_0^{z_0} x_0\cos z\,dz$

$= x_0y_0 + x_0\sin z_0$. ∴ $U(x,y,z) = xy + x\sin z$.

∴ $\int_{(0,0,0)}^{(3,2,\frac{\pi}{3})} (y+\sin z)\,dx + x\,dy + x\cos z\,dz = U(3,2,\frac{\pi}{3}) - U(0,0,0)$

$= 3\times2 + 3\times\sin\frac{\pi}{3} - 0 = 6 + \frac{3\sqrt{3}}{2}$.

## §10. Gauss 公式和散度

习题. P:162.

1. 利用 Gauss 公式计算下列曲面积分:

(1) $\iint_\Sigma x^3\,dy\,dz + y^3\,dz\,dx + z^3\,dx\,dy$, 其中 $\Sigma$ 是球面 $x^2+y^2+z^2=4$ 的外侧.



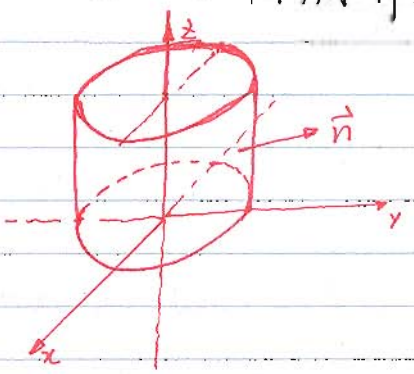
解: $\Sigma: \begin{cases} x = 2\sin\varphi\cos\theta \\ y = 2\sin\varphi\sin\theta \\ z = 2\cos\varphi \end{cases}$ $\Omega = \left\{ (r,\varphi,\theta) \,\middle|\, \begin{matrix} 0\le r\le 2 \\ 0\le\theta\le 2\pi \\ 0\le\varphi\le\pi \end{matrix} \right\}$

$$\iint_\Sigma x^3\,dydz + y^3\,dzdx + z^3\,dxdy = \iiint_\Omega \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dxdydz$$

$$= \iiint_\Omega 3(x^2+y^2+z^2)\,dxdydz = 3\int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_0^2 r^2 \cdot r^2\sin\varphi\, dr$$

$$= 3\cdot 2\pi \cdot \int_0^{\pi} \sin\varphi \cdot \int_0^2 r^4\,dr = 3\cdot 2\pi\cdot 2\cdot\frac{1}{5}\cdot 2^5 = \frac{384}{5}\pi.$$

(2) $\iint_\Sigma (x-y)\,dxdy + x(y-z)\,dydz$, 其中$\Sigma$为柱面 $x^2+y^2=1$ 及 $z=0$, $z=1$ 所围空间区域表面, 是向取外侧.




解: $\iint_\Sigma (x-y)\,dxdy + x(y-z)\,dydz$

$$= \iiint_\Omega \left[\frac{\partial}{\partial x}(x-y) + \frac{\partial}{\partial y}\cdot 0 + \frac{\partial}{\partial z}(x(y-z))\right] dxdydz$$

$$= \iiint_\Omega (y-z)\,dxdydz. \qquad \begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases}$$

$$= \int_0^1 dz \int_0^{2\pi} d\theta \int_0^1 (r\sin\theta - z)\, r\,dr$$

$$= \int_0^1 dz \cdot \int_0^{2\pi} d\theta \cdot \int_0^1 (r^2\sin\theta - zr)\,dr = \int_0^1 dz \cdot \int_0^{2\pi} \left(\frac{1}{3}\sin\theta - \frac{1}{2}z\right) d\theta$$

$$= \int_0^1 \left(-\frac{1}{3}\cos\theta - \frac{1}{2}z\cdot\theta\right)\Big|_0^{2\pi} \cdot dz = -\int_0^1 \pi z\,dz = -\frac{\pi}{2}.$$

(3) $\iint_\Sigma a^2b^2z^2\,dydz + b^2c^2x^2y\,dzdx + c^2a^2y^2z\,dxdy$, 其中$\Sigma$是由椭球面上半椭球面 $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$, $z\ge 0$, 是向取下侧.




解: 设 $\Sigma_1: \begin{cases} \frac{x^2}{a^2}+\frac{y^2}{b^2}\le 1 \\ z=0 \end{cases}$ 方向取上侧: $\vec{n}_1 = (0,0,1)$.

则 $\Sigma\cup\Sigma_1$ 上方向取内侧, 由积分可利用 Gauss 积分公式:

$$-\iint_{\Sigma\cup\Sigma_1} a^2b^2z^2x\,dydz+b^2c^2x^2y\,dzdx+c^2a^2y^2z\,dxdy.$$

$$=\iiint_{\Omega}\left[\frac{\partial}{\partial x}(a^2b^2z^2x)+\frac{\partial}{\partial y}(b^2c^2x^2y)+\frac{\partial}{\partial z}(c^2a^2y^2z)\right]dxdydz$$

$$=\iiint_{\Omega} a^2b^2c^2\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\right)dxdydz.$$

$$\Omega=\left\{(r,\varphi,\theta)\,\middle|\,\begin{matrix}0\le r\le 1\\ 0\le\theta\le 2\pi\\ 0\le\varphi\le\frac{\pi}{2}\end{matrix}\right\}$$

$$=a^2b^2c^2\int_0^{2\pi}d\theta\cdot\int_0^{\frac{\pi}{2}}d\varphi\cdot\int_0^1 r^2\cdot abcr^2\sin\varphi\,dr$$

$$\begin{cases}x=ar\sin\varphi\cos\theta\\ y=br\sin\varphi\sin\theta\\ z=cr\cos\varphi\end{cases}$$

$$=2\pi a^2b^2c^2\int_0^{\frac{\pi}{2}}\sin\varphi\,d\varphi\cdot\int_0^1 r^4\,dr$$

$$d\sigma=dxdydz=abcr^2\sin\varphi\,drd\varphi d\theta$$

$$=\frac{2\pi}{5}a^3b^3c^3.$$

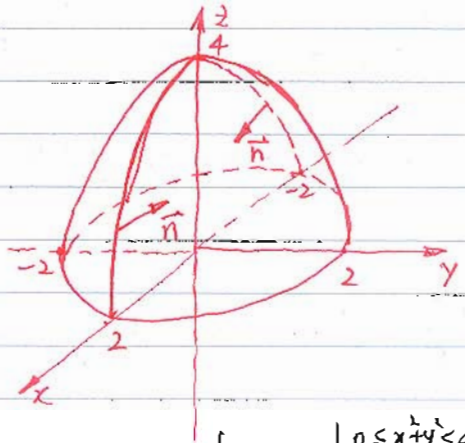
$$\iint_{\Sigma_1} a^2b^2z^2x\,dydz+b^2c^2x^2y\,dzdx+c^2a^2y^2z\,dxdy$$

$$=\iint_{\Sigma_1}(a^2b^2z^2x,\ b^2c^2x^2y,\ c^2a^2y^2z)\cdot(0,0,1)\,dS=\iint_{\Sigma_1}c^2a^2y^2z\cdot dS=0.$$

$$\therefore\iint_{\Sigma} a^2b^2z^2x\,dydz+b^2c^2x^2y\,dzdx+c^2a^2y^2z\,dxdy=-\iiint_{\Omega}(a^2b^2z^2+b^2c^2x^2+c^2a^2y^2)\,dxdydz$$

$$=-\frac{2\pi}{5}a^3b^3c^3.$$

(4) $\iint_{\Sigma} x^3dydz+2xz^2dzdx+3y^2zdxdy$，其中$\Sigma$是由抛物面$z=4-x^2-y^2$和$Oxy$坐标面所围立体表面的内侧.




解：$\iint_{\Sigma} x^3dydz+2xz^2dzdx+3y^2zdxdy$

$$=-\iiint_{\Omega}\left[\frac{\partial}{\partial x}(x^3)+\frac{\partial}{\partial y}(2xz^2)+\frac{\partial}{\partial z}(3y^2z)\right]dxdydz$$

$$=-\iiint_{\Omega}(3x^2+3y^2)\,dxdydz$$

$$=-3\iiint_{\Omega}(x^2+y^2)\,dxdydz.$$

采用柱坐标：$\begin{cases}x=r\cos\theta\\ y=r\sin\theta\\ z=z\end{cases}$

$$\Omega=\left\{(x,y,z)\,\middle|\,\begin{matrix}0\le x^2+y^2\le 4-z\\ 0\le z\le 4\end{matrix}\right\}$$

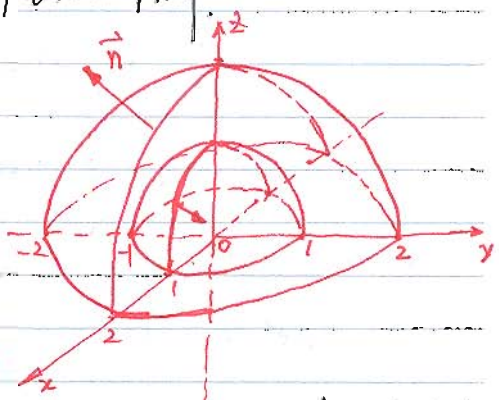
$$=\left\{(r,\theta,z)\,\middle|\,\begin{matrix}0\le r\le\sqrt{4-z}\\ 0\le z\le 4\\ 0\le\theta\le 2\pi\end{matrix}\right\}$$

$$=-3\int_0^{2\pi}d\theta\cdot\int_0^4 dz\cdot\int_0^{\sqrt{4-z}} r^2\cdot r\,dr$$

$$=-6\pi\int_0^4 dz\cdot\frac{1}{4}r^4\Big|_0^{\sqrt{4-z}}=-6\pi\int_0^4\frac{1}{4}(z^2-8z+16)\,dz$$

$$=-32\pi.$$

(5) $\iint_\Sigma (x^3+y\sin z)\,dydz+(y^3+z\sin x)\,dzdx+3z\,dxdy$，其中$\Sigma$是由两个半球面 $z=\sqrt{4-x^2-y^2}$，$z=\sqrt{1-x^2-y^2}$ 和平面 $z=0$ 所围立体表面的外侧.




解：$\iint_\Sigma (x^3+y\sin z)\,dydz+(y^3+z\sin x)\,dzdx+3z\,dxdy$

$=\iiint_\Omega \left[\frac{\partial}{\partial x}(x^3+y\sin z)+\frac{\partial}{\partial y}(y^3+z\sin x)+\frac{\partial}{\partial z}(3z)\right]dxdydz$

$=\iiint_\Omega (3x^2+3y^2+3)\,dxdydz=3\iiint_\Omega (x^2+y^2+1)\,dxdydz$

$=3\int_0^{2\pi}d\theta\cdot\int_0^{\frac{\pi}{2}}d\varphi\cdot\int_1^2(r^2\sin^2\varphi+1)\cdot r^2\sin\varphi\,dr$ 采用球坐标.

$=6\pi\cdot\int_0^{\frac{\pi}{2}}d\varphi\cdot\left(\frac{1}{5}r^5\sin^3\varphi+\frac{1}{3}r^3\sin\varphi\right)\Big|_{r=1}^{r=2}$

$=6\pi\cdot\int_0^{\frac{\pi}{2}}\left(\frac{31}{5}\sin^3\varphi+\frac{7}{3}\sin\varphi\right)d\varphi=6\pi\cdot\frac{97}{15}=\frac{194}{5}\pi.$

$\Omega=\left\{(x,y,z)\,\middle|\,\begin{matrix}1\le x^2+y^2+z^2\le 4\\ 0\le z\end{matrix}\right\}$

$=\left\{(r,\varphi,\theta)\,\middle|\,\begin{matrix}1\le r\le 2\\ 0\le\theta\le 2\pi\\ 0\le\varphi\le\frac{\pi}{2}\end{matrix}\right\}$

(6) $\iint_\Sigma (2x+2y+z^2)\,dS$，其中$\Sigma$是单位球面 $x^2+y^2+z^2=1$.

解：$\iint_\Sigma (2x+2y+z^2)\,dS=\iint_\Sigma (2x,2y,z^2)\cdot(-1,-1,-1)\,dS=\iint_\Sigma -2x\,dydz-2y\,dzdx-z^2\,dxdy$

$=\iiint_\Omega (-2-2-2z)\,dxdydz=-2\iiint_\Omega (2+z)\,dxdydz=-2\cdot\int_0^{2\pi}d\theta\int_0^{\pi}d\varphi\int_0^1(2+r\cos\varphi)r^2\sin\varphi\,dr$

$=4\pi\cdot\int_0^{\pi}\left(\frac{2}{3}\sin\varphi+\frac{1}{4}\sin\varphi\cos\varphi\right)d\varphi=4\pi\cdot\left(-\frac{2}{3}\cos\varphi\Big|_0^{\pi}+\frac{1}{8}\sin^2\varphi\Big|_0^{\pi}\right)=\frac{16}{3}\pi.$

2. 求 $\operatorname{div}\vec{F}$ 在指定点的值：

(1) $\vec{F}=x^2\vec{i}+y^2\vec{j}+z^2\vec{k}$，在点 $P(-1,1,2)$.

解：$\operatorname{div}\vec{F}=\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}=2x+2y+2z=2(x+y+z)\Big|_P=2\times 2=4.$

(2) $\vec{F}=xyz(x\vec{i}+y\vec{j}+z\vec{k})$，在点 $P(1,3,2)$.

解：$\operatorname{div}\vec{F}=\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}=2xyz+2xyz+2xyz=6xyz\Big|_P=6\times 1\times 3\times 2=36.$

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3. 设 $\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$ 是无源场.

$$X(x,y,z) = \int_{z_0}^{z} Q(x,y,\zeta)\,d\zeta - \int_{y_0}^{y} R(x,\eta,z_0)\,d\eta$$

$$Y(x,y,z) = -\int_{z_0}^{z} P(x,y,\zeta)\,d\zeta.$$

$$G(x,y,z) = X(x,y,z)\vec{i} + Y(x,y,z)\vec{j}.$$

证明: $\operatorname{rot}\vec{G} = \vec{F}$

证明:
$$\operatorname{rot}\vec{G} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ X & Y & 0 \end{vmatrix} = -\frac{\partial Y}{\partial z}\vec{i} + \frac{\partial X}{\partial z}\vec{j} + \left(\frac{\partial Y}{\partial x} - \frac{\partial X}{\partial y}\right)\vec{k}$$

$$= -\frac{\partial}{\partial z}\left(-\int_{z_0}^{z} P(x,y,\zeta)\,d\zeta\right)\vec{i} + \frac{\partial}{\partial z}\left(\int_{z_0}^{z} Q(x,y,\zeta)\,d\zeta - \int_{y_0}^{y} R(x,\eta,z_0)\,d\eta\right)\vec{j}$$

$$+\left[\frac{\partial}{\partial x}\left(-\int_{z_0}^{z} P(x,y,\zeta)\,d\zeta\right) - \frac{\partial}{\partial y}\left(\int_{z_0}^{z} Q(x,y,\zeta)\,d\zeta - \int_{y_0}^{y} R(x,\eta,z_0)\,d\eta\right)\right]\vec{k}$$

$$= P(x,y,z)\vec{i} + Q(x,y,z)\vec{j} + \left[-\int_{z_0}^{z}\left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right)(x,y,\zeta)\,d\zeta + R(x,y,z_0)\right]\vec{k}$$

$\because \vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$ 是无源场.

$\therefore \operatorname{div}\vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 0 \Rightarrow \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = -\frac{\partial R}{\partial z}.$

$$\therefore \operatorname{rot}\vec{G} = P(x,y,z)\vec{i} + Q(x,y,z)\vec{j} + \left[\int_{z_0}^{z} \frac{\partial R}{\partial z}(x,y,\zeta)\,d\zeta + R(x,y,z_0)\right]\vec{k}$$

$$= P\vec{i} + Q\vec{j} + \left[R(x,y,z) - R(x,y,z_0) + R(x,y,z_0)\right]\vec{k}$$

$$= P\vec{i} + Q\vec{j} + R\vec{k} = \vec{F} \quad ※$$

(任一无源场 $\vec{F}$ 必为某一向量场 $\vec{G}$ 的旋度场 $\operatorname{rot}\vec{G}$.
即可由 $P, Q, R$ 构造 $\vec{G}$.)

4. 求向量场 $\vec{F} = (x-y+z)\vec{i} + (y-z+x)\vec{j} + (z-x+y)\vec{k}$ 由内向外穿过椭球面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 的通量.

解: $$\iint_{\Sigma} \vec{F}\cdot d\vec{S} = \iint_{\Sigma} \vec{F}\cdot\vec{n}\,dS = \iiint_{\Omega} \operatorname{div}\vec{F}\cdot d\sigma = \iiint_{\Omega}\left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right)d\sigma$$

$$= 3\iiint_{\Omega} d\sigma = 3\cdot\frac{4}{3}\pi abc = 4\pi abc.$$

5. 设函数 $P(x,y,z)$、$Q(x,y,z)$、$R(x,y,z)$ 在 $R^3$ 上具有连续偏导数，且对于任意光滑曲面 $\Sigma$，成立 $\iint_{\Sigma} P\,dydz + Q\,dzdx + R\,dxdy = 0$.

证明在 $R^3$ 上，$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \equiv 0$.

证明：设某一点 $M$ 有 $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = K \neq 0$，不妨设 $K > 0$.

∵ $P$、$Q$、$R$ 在 $R^3$ 上具有连续偏导数，∴ 可取 $\varepsilon = \frac{K}{2}$，$\exists \delta > 0$.

在 $O(M,\delta)$ 内，有：$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} > \frac{K}{2} > 0$.

$$\iint_{\partial\Omega} P\,dydz + Q\,dzdx + R\,dxdy = \iiint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dxdydz, \quad \Omega = O(M,\delta)$$

$$> \frac{K}{2} \iiint_{\Omega} dxdydz > 0.$$

这与已知对于任意光滑曲面 $\Sigma$ 成立：

$$\iint_{\Sigma} P\,dydz + Q\,dzdx + R\,dxdy = 0 \text{ 矛盾.}$$

∴ 在 $R^3$ 上恒有：$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \equiv 0$.

6. 设 $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$，$\vec{a}$ 为常向量，验证：

(1) $\nabla \cdot (\vec{a} \times \vec{r}) = 0$.

(2) $\nabla \times (\vec{a} \times \vec{r}) = 2\vec{a}$.

(3) $\nabla \cdot ((\vec{r} \cdot \vec{r})\vec{a}) = 2\vec{r} \cdot \vec{a}$

证明：(1) $\nabla \cdot (\vec{a} \times \vec{r}) = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \cdot \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ x & y & z \end{vmatrix}$

$$= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \cdot (a_2 z - a_3 y,\ a_3 x - a_1 z,\ a_1 y - a_2 x)$$

$$= \frac{\partial}{\partial x}(a_2 z - a_3 y) + \frac{\partial}{\partial y}(a_3 x - a_1 z) + \frac{\partial}{\partial z}(a_1 y - a_2 x) = 0.$$

(2) $\nabla \times (\vec{a} \times \vec{r}) = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \times (a_2 z - a_3 y,\ a_3 x - a_1 z,\ a_1 y - a_2 x)$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_2 z - a_3 y & a_3 x - a_1 z & a_1 y - a_2 x \end{vmatrix}$$

$$= \left[\frac{\partial}{\partial y}(a_1 y - a_2 x) - \frac{\partial}{\partial z}(a_3 x - a_1 z)\right]\vec{i} + \left[\frac{\partial}{\partial z}(a_2 z - a_3 y) - \frac{\partial}{\partial x}(a_1 y - a_2 x)\right]\vec{j}$$

$$+ \left[\frac{\partial}{\partial x}(a_3 x - a_1 z) - \frac{\partial}{\partial y}(a_2 z - a_3 y)\right]\vec{k}$$

$$= 2a_1\vec{i} + 2a_2\vec{j} + 2a_3\vec{k} = 2\vec{a}.$$

(3) $\nabla\cdot\left((\vec{r}\cdot\vec{r})\vec{a}\right) = \nabla\cdot\left((x^2+y^2+z^2)\vec{a}\right) = \frac{\partial}{\partial x}\left[(x^2+y^2+z^2)a_1\right] + \frac{\partial}{\partial y}\left[(x^2+y^2+z^2)a_2\right]$

$$+ \frac{\partial}{\partial z}\left[(x^2+y^2+z^2)a_3\right] = 2a_1 x + 2a_2 y + 2a_3 z$$

$$= 2(x,y,z)\cdot(a_1,a_2,a_3) = 2\vec{r}\cdot\vec{a}.$$

7. 验证: $\nabla\cdot(g\nabla f) = \nabla g\cdot\nabla f + g\Delta f$, 其中 $g$ 和 $f$ 分别具有一阶和二阶连续偏导数.

证明: $\nabla\cdot(g\nabla f) = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)\cdot\left(g\frac{\partial f}{\partial x}, g\frac{\partial f}{\partial y}, g\frac{\partial f}{\partial z}\right)$

$$= \frac{\partial}{\partial x}\left[g\frac{\partial f}{\partial x}\right] + \frac{\partial}{\partial y}\left[g\frac{\partial f}{\partial y}\right] + \frac{\partial}{\partial z}\left[g\frac{\partial f}{\partial z}\right]$$

$$= \frac{\partial g}{\partial x}\cdot\frac{\partial f}{\partial x} + g\frac{\partial^2 f}{\partial x^2} + \frac{\partial g}{\partial y}\cdot\frac{\partial f}{\partial y} + g\frac{\partial^2 f}{\partial y^2} + \frac{\partial g}{\partial z}\cdot\frac{\partial f}{\partial z} + g\frac{\partial^2 f}{\partial z^2}$$

$$= \nabla g\cdot\nabla f + g\Delta f.$$

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